

Capstone Final Report

Team 2: Liminal Umbrella

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Design 388-1: MaDE Capstone Sequence
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Executive Summary

Liminal Umbrella targets the failure point of conventional umbrellas: the transition from rain protection to storage, with vehicle entry as the hardest case. Our research converged on three unmet needs: clean storage after use, smoother car and tight-space transitions, and reduced handling burden when users are multitasking.

We benchmarked three umbrella archetypes (Stick Reverse, Warped, Automatic) using a geometry-driven protocol (canopy dimensions, fold path, shaft motion, closed envelope, and vehicle interaction). All three failed vehicle clearance: door clearance “doesn’t fit” and entry motion flow “awkward.” Closure and wrapping averaged 7.64–8.75 seconds, with repeatable drivers such as strap position, bulk, and wrap geometry. Fold control effort ranged from “manual clean” under 5 pounds to “manual very difficult” around 15 pounds, exposing a trade space between controllability, stiffness, and user effort. These benchmarks established vehicle-entry usability, controlled collapse, and low-effort closure as primary requirements.

User testing reinforced the same constraint from the interaction side. First-time users defaulted to slider-based closure and searched for affordances (buttons, lock cues), but quickly recognized the value of lower-shaft actuation once demonstrated. Several users explicitly identified “reaching and pulling” as the dominant pain point in tight spaces and wanted a one-handed, single-motion close. Handle testing narrowed the interface: users favored compact, non-restrictive options (simple rubber and carabiner-style) over grips that constrain hand placement or become bulky near a car door. Container lid testing showed that a press-fit seal can feel secure, but opening force and secondary openings reduce trust and create leak paths during inversion; users asked for clearer grip features and a more decisive sealed-state cue.

These results support a close-and-contain system architecture: lower-shaft interaction to reduce reach, controlled collapse tuned for tight-space motion, and integrated containment to enable immediate post-use storage. Next steps are full-system validation in a vehicle-entry scenario with standardized metrics (closure time, closure effort, leak behavior, and seal confidence), followed by convergence on a manufacturable architecture that leverages standardized umbrella components and rigorous assembly-level quality control.

Differentiation Plan

Problem Exploration

The development of the Liminal Umbrella began with a broad exploration of consumer pain points across a variety of industries and everyday activities. Through brainstorming sessions, market research, preliminary interviews, and AI-assisted ideation, the team generated and evaluated over 60 potential problem spaces. These opportunities were systematically narrowed through research reports, journey mapping, stakeholder analysis, feasibility assessments, competitive analysis, and decision matrices, as shown in Appendix A eventually leading to a final focus on umbrella moisture management and vehicle-entry usability.

Umbrellas emerged as a compelling problem space because they are widely used, deeply integrated into daily routines, and consistently associated with frustration. Early research revealed recurring issues related to moisture management, awkward handling, durability, storage, and operation while carrying other belongings. Although umbrellas effectively provide protection from rain, users frequently experience difficulties during the transitions between outdoor and indoor environments.

Through interviews, surveys, observational studies, and market research, three major unmet needs were identified:

- **Clean Storage After Use:** Users need a convenient way to store wet umbrellas without dripping water onto surrounding surfaces.
- **Smooth Transitions in Cars and Tight Spaces:** Users need a way to move between outdoor and indoor environments without sacrificing rain protection or transferring water.
- **Reduced Handling Burden:** Users need a simpler and more ergonomic method of operating umbrellas while managing other belongings.

These findings suggested that the greatest opportunity for innovation was not improving rain protection itself, but improving what happens immediately before and after umbrella use. In particular, the team identified vehicle entry as a high-friction moment where users experience excessive water transfer, awkward movements, and increased cognitive load. This insight guided the project toward developing a solution focused on transition management, closure ergonomics, and moisture containment.

Research Reports

Following problem selection, the team conducted interviews, surveys, AEIOU observations, journey mapping, product dissections, stakeholder analyses, and patent investigations to better understand the umbrella user experience and identify opportunities for differentiation.

Interviews and Surveys

As shown in Appendix D and E, Interviews and surveys revealed that users generally view umbrellas as a necessary but inconvenient product. Participants consistently reported frustration with storing wet umbrellas, managing moisture when entering vehicles or buildings, and operating umbrellas while carrying bags, phones, groceries, or other belongings.

Many participants described entering a vehicle during rainfall as the most frustrating part of the umbrella experience. Users reported becoming wet while attempting to close the umbrella, maneuver through the vehicle doorway, and prevent water from dripping onto seats and interior surfaces. Several participants described improvised storage solutions such as placing umbrellas on passenger seats, in footwells, or inside plastic bags, none of which fully solved the problem.

Survey responses reinforced these findings and identified moisture containment, one-handed operation, improved storage, and ease of use as highly desirable features. Collectively, the interviews and surveys demonstrated that users struggle less with rain protection itself and more with the post-rain experience.

AEIOU Observations and Journey Mapping

Activity name: Umbrella Testing			Time of day: 8 pm Saturday, Nov 8 2025	
Testing 4 different umbrellas entering and exiting the car while raining.				
Activities	Environment	Objects	Interactions	Users
What are the modes people work in, and the specific activities and processes they go through?	Describe the context and characteristics of the space where activities are being observed.	What are the objects and devices people have in their environments and how do they relate to their activities?	What is the nature of routine and special interactions between people, between people and objects in their environment, and across distances?	Describe the people being observed. What are their values and biases, behaviors, needs and relationships?
- Participants walking to car parked in the rain - Entering the car - Storing Umbrella - Exiting the car - Participants walking to car parked closely next to another car in the rain - Entering the car - Storing Umbrella - Exiting the car	- Light rain, cold, windy - Uncomfortable/tight space between cars - Small parking lot outside of apartment building - Dark outside - Ground is wet with lots of debris e.g. leaves	- Umbrellas (opening, closing, storing) - Car door (opening, closing, getting wet) - Car seats (sitting down, storing umbrella, getting wet) - One participant was wearing a hood, which affected how much rain was getting on them	- Participants were being recorded from outside and inside the car of their routine which might've affected their actions - Small umbrellas were stored on the car seat while the stick umbrella was stored on the passenger seat floor.	- Close friends, understand the general premise of the task - Wearing coats - Do not want to get wet - Using another person's car - Car was left wet - The owner of the car asked why there were a lot of leaves on the floor the next morning.
User Experience				
Physical: Closing the umbrella takes strength and often requires reaching your arm out. Especially when car is next to another car and door angle is limited, participants had to reach up high above car roof or in some cases stand up to retract umbrella. Going into car seemed harder and less seamless than going out of car.				
Cognitive: Thinking about multiple things at once (entering into the car without getting wet, opening the door enough to get in but not hit the car next to it, thinking of how to efficiently bring umbrella into the car). Understanding the umbrella mechanism (ie. using the manual umbrella the first time takes longer than the second/third time).				
Social: Users don't want to look silly and try to perform the process elegantly (laughing off failures)				
Cultural: Participants vary in how much they care about getting themselves / the car wet.				
Emotional: Frustration in parts of the process e.g. closing the manual umbrella and getting wet at the same time				
Comments:				

Figure 1: AEIOU Observations



Figure 2: Umbrella-Car Journey Map

To validate interview findings through direct observation, the team conducted AEIOU studies (Figure 1 and Appendix H/K) and developed journey maps (Figure 2) focused on entering a vehicle during rainfall as shown in figures 1 and 2 and appendix H. These studies revealed that users repeatedly perform awkward body movements, reposition their umbrellas multiple times, and struggle to balance several simultaneous tasks.

The journey map revealed that the most difficult stage occurs during umbrella collapse and vehicle entry. Conventional umbrellas require users to reach upward toward the runner near the canopy and apply a significant pulling force to retract the umbrella. Because the canopy remains open during this process, users cannot fully enter the vehicle before collapsing the umbrella. Instead, they must remain partially outside the vehicle while reaching upward and pulling downward on the runner.

This combination of canopy geometry, constrained space, and required actuation force creates an awkward body position in which users are simultaneously exposed to rain, managing the vehicle door, balancing personal belongings, and attempting to prevent water from contacting the vehicle interior. The team observed repeated instances of users pausing, repositioning themselves, and adjusting their grip to complete the process successfully.

These observations revealed that the primary challenge is not moisture containment itself, but the interaction architecture of traditional umbrellas. Existing umbrellas force users to perform high-reach, high-force motions while operating within the constrained geometry of a vehicle doorway. As a result, users are exposed to rain at the exact moment they are attempting to transition into a protected environment.

Product Dissection and Benchmark Analysis

To better understand current solutions, the team conducted a detailed dissection of a reverse-folding umbrella, the Lanbrella, which is frequently marketed as a solution for vehicle entry and moisture management. The goal was to understand how existing products address moisture containment, how their mechanisms operate, and what limitations remain.

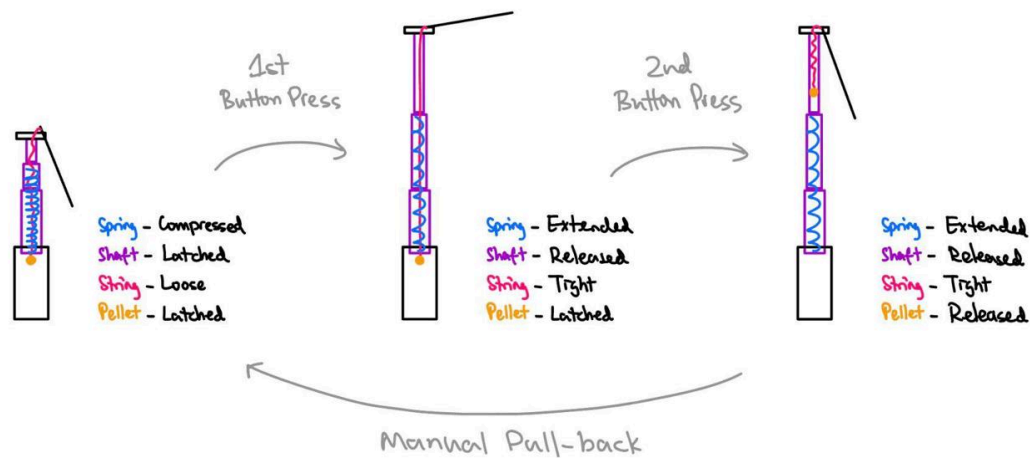


Figure 3: Umbrella Function Diagram

As shown in Appendix J/K, The dissection revealed that the umbrella relies on a spring-driven telescoping shaft and a cord-based linkage system to control canopy deployment and collapse. As illustrated in Figure 3, the first button press releases the compressed spring, extending the shaft and tensioning the cord to deploy the canopy. The second button press releases the cord mechanism, allowing the canopy to fold inward. To reset the system, the user must manually compress the shaft and re-engage the internal latching mechanism.

Although the reverse-folding architecture successfully keeps the wet canopy surface contained inside the umbrella, the dissection revealed several limitations. Users must still manually compress the shaft after closure, requiring significant force and an upward reach toward the upper portion of the umbrella. Furthermore, the large canopy geometry remains unchanged, meaning users must still collapse the umbrella while partially outside the vehicle. As a result, many of the ergonomic and transition-related challenges identified during observations and journey mapping remain unresolved.

The dissection also provided valuable insight into how runner position, shaft architecture, force transmission, and canopy geometry influence user interactions. These findings suggested that meaningful improvement would require rethinking the closure architecture itself rather than simply modifying canopy geometry or moisture-containment features.

Root Cause Analysis

OBSERVATION / SYMPTOM	WHY #1	WHY #2	WHY #3	WHY #4	ROOT CAUSE
Users get wet or drip water when entering vehicles.	Users collapse umbrella before fully entering car.	Car-door openings do not provide enough space to enter with the umbrella open.	Umbrella folding is difficult in tight spaces .	Umbrella geometry prioritizes maximum canopy span rather than maneuverability in confined spaces.	Umbrella mechanisms lack vehicle-entry adaptability.
Users are frustrated entering vehicle in the rain.	Rushing to get into their vehicles.	Users are getting wet when heading to important commitments.	Fumbling the umbrella as well as any other held items (bags, drinks, etc.).	Transition requires multiple coordinated actions (door, umbrella, etc.) at once.	Design lacks intuitive feedback and simplified motion flow.

Figure 4: Root Cause Analysis

To move beyond surface-level observations, the team conducted a root cause analysis using the 5 Whys methodology as shown in figure 4. While observations initially suggested that users simply became wet while entering vehicles, deeper analysis revealed two fundamental causes.

The first root cause was a lack of vehicle-entry adaptability. Existing umbrella architectures are designed for users standing in open spaces and do not accommodate the constrained movements required when entering a vehicle.

The second root cause was excessive reach, force, and cognitive effort. Existing umbrellas require users to perform awkward motions while coordinating multiple simultaneous tasks and receiving limited feedback about the state of the mechanism.

These findings reframed the challenge from a moisture-management problem to a vehicle-transition problem centered on ergonomics, motion flow, and user interaction.

Patent and Market Research

Umbrella Type	Patent #	Status	Key Claim
Inverted/Reverse Fold	US20040211451	Dead	Sliding/multi-collar system lets canopy invert.
	US7484518B2	Dead	Hinged connector rods guide canopy reversal. Locking collars enable smooth inward open/close.
Vented/Double Canopy	US5487401A	Dead	Two-layer canopy with vented lower layer.
	US4979534A	Dead	Vents placed between center post and rib tips. Elastic or channel fasteners keep top canopy taut.fig Water-channel around vents to block rain entry.
Automatic Open/Close	US6257257B1	Dead	Single-button open/close mechanism.
	US4823821	Dead	Control tube+sliding block + hooked runner for state switching. Spring-loaded ribs that toggle open/closed with one press. Push-button triggers both expansion and collapse. Foldable shaft and ribs for compact storage.
Inflatable/Bubble	US3954117A	Dead	Canopy design using a fluid-filled or inflatable sleeve. Inflatable protective enclosure for the user. Integrated housing–handle–canopy structure enabling an inflated umbrella canopy.
Telescopic/Compact	US5720311A	Dead	Telescoping shaft with nested sections for compact storage.
	US3785388A	Dead	Reinforced tube+ribs for strength and stability. Integrated spring/locking mechanism for smooth open/close.
Specialty (Lighting)	US10561210B1	Active	LED lights activate with umbrella orientation.
	US7571736B1	Dead	Sensor+processor control lighting on open/close.

Table 1: Patent Status Table

As an example shown in table 1 The team conducted a comprehensive review of existing umbrella patents and commercial products to identify opportunities for differentiation (Appendix B and F). Research revealed that most umbrella innovations focus on canopy geometry, automatic opening systems, reverse-folding mechanisms, wind resistance, or moisture sleeves.

While these solutions address individual aspects of the umbrella experience, few products specifically target the transition between outdoor and indoor environments. Reverse-folding umbrellas improve moisture containment but still require awkward closure motions and do not address vehicle-entry ergonomics. Moisture sleeves and umbrella bags provide storage solutions but introduce additional steps into the user workflow.

This analysis revealed a significant whitespace opportunity. Existing products focus on either rain protection or moisture containment, but few address the complete transition from rain protection to storage. This insight guided the development of the Liminal Umbrella toward a solution centered on vehicle-entry usability, controlled collapse, and integrated containment.

Key Research Insights

The research phase produced five primary insights that guided development of the final design simplified from Appendix G:

Research Insight	Design Response
Users struggle most during vehicle entry and indoor transitions	Focus design around vehicle-entry workflow
Existing umbrellas lack vehicle-entry adaptability	Redesign closure architecture around constrained-space operation
Excessive reaching and pulling create frustration	Develop lower-shaft actuation mechanism
One-handed operation is highly desirable	Simplify closure into a single controlled motion
Wet umbrellas create storage and containment issues	Develop integrated containment solution
Existing products fail to address the complete user journey	Design a system rather than a standalone umbrella

Collectively, these findings established the foundation for the project's differentiation strategy and directly informed the development of the final concept.

Competitive Analysis

The umbrella market contains a wide range of products that attempt to improve rain protection, portability, durability, and moisture management. To understand the current competitive

landscape, the team analyzed conventional automatic umbrellas, reverse-folding umbrellas, compact travel umbrellas, and moisture-containment accessories through product benchmarking, patent research, and product dissections. These analyses focused on how existing solutions perform during the transition between outdoor and indoor environments, particularly during vehicle entry.

Conventional automatic umbrellas prioritize convenience through push-button deployment and collapse mechanisms. While these products reduce the effort required to open the canopy, they still require users to manually compress the shaft and gather the canopy before storage. As a result, users must perform multiple steps while entering vehicles or buildings, often leading to water transfer and awkward movements.

Reverse-folding umbrellas, such as the Lanbrella, address one aspect of the problem by folding the wet canopy inward to reduce dripping. These products are frequently marketed as vehicle-friendly solutions because they contain moisture more effectively than traditional umbrellas. However, benchmarking and product dissection revealed that users must still collapse the umbrella while partially outside the vehicle and manually compress the shaft using significant force. Although moisture containment is improved, the underlying vehicle-entry and ergonomic challenges remain largely unresolved.

Compact travel umbrellas focus on portability and storage by minimizing closed length and overall size. While these products are easy to carry, they often sacrifice canopy coverage, structural stability, or ease of operation. Additionally, their closure and wrapping process remains largely unchanged from traditional umbrella designs, providing little improvement during vehicle-entry transitions.

Other products attempt to address wet umbrella storage through sleeves, bags, or standalone containment accessories. While these solutions help reduce dripping after use, they introduce additional steps into the user workflow and require users to manually wrap and store the umbrella before containment can occur. As a result, moisture management is treated as a separate task rather than an integrated part of the umbrella experience.

Benchmarking results demonstrated that all evaluated umbrella types struggled with vehicle entry. Each benchmark umbrella received a “does not fit” rating for door clearance and an “awkward” rating for entry motion flow. Closure and wrapping times remained between approximately 7.6 and 8.8 seconds regardless of umbrella type, suggesting that existing solutions have not fundamentally addressed the transition from rain protection to storage.

Overall, the competitive analysis revealed that most existing products optimize individual attributes such as automation, portability, wind resistance, or moisture containment. Few solutions address the complete user journey from rain protection through vehicle entry, closure, containment, and storage. This gap in the market identified a significant opportunity for

differentiation and guided the development of the Liminal Umbrella toward a transition-focused system that integrates ergonomics, controlled collapse, and moisture containment into a single user experience.

Area for Differentiation

The Liminal Umbrella differentiates itself by focusing on the transition between outdoor rain and indoor environments rather than solely on rain protection. Research consistently demonstrated that users experience the greatest frustration not while standing under an umbrella, but during the process of entering vehicles and managing a wet umbrella afterward. Existing products largely treat moisture management, portability, and automation as separate challenges. In contrast, the Liminal Umbrella addresses the entire transition workflow as an integrated system.

The first major differentiator is the lower-shaft actuation architecture. Conventional umbrellas require users to reach toward the upper shaft or canopy region and generate significant force to collapse the umbrella. The Liminal Umbrella relocates the primary interaction point closer to the user's natural hand position, reducing reach requirements and enabling a more ergonomic closure motion. This architecture is specifically designed to function within the constrained geometry of vehicle entry, where traditional umbrella interactions become difficult and awkward.

The second differentiator is the pleated canopy architecture. Traditional umbrellas collapse into an irregular bundle that requires users to manually gather and wrap the canopy before storage. This process often requires two hands and creates additional opportunities for water transfer during vehicle entry. The Liminal Umbrella utilizes a pleated canopy that collapses into a more organized and predictable geometry. By guiding the fabric into a compact configuration during closure, the canopy can be inserted directly into the containment system with minimal user intervention. This reduces the need for manual wrapping, simplifies the storage process, and improves the overall transition experience.

The third differentiator is integrated moisture containment. Rather than treating storage as a separate task that occurs after umbrella use, the design incorporates moisture management into the overall transition workflow. This reduces dripping, minimizes contamination of vehicle interiors, and improves the post-rain user experience.

The final differentiator is the system-level perspective taken throughout development. Existing products typically optimize a single attribute such as portability, automation, durability, or moisture containment. The Liminal Umbrella instead addresses the complete user journey, from rain protection to vehicle entry, closure, containment, and storage. By focusing on the transition moment itself, the design targets a significant gap that current products have largely overlooked.

Commonality Plan

Testing

Testing this quarter focused on validating our core use case: closing and containing a wet umbrella during vehicle entry and other tight-space transitions. Our approach combined (1) early user testing to confirm the problem and refine interaction assumptions, (2) geometry benchmarking of existing umbrellas to identify what fails in vehicles and what mechanisms enable controlled collapse, (3) structured handle evaluations to determine which interface best supports one-hand pulling in tight spaces, and (4) container lid testing to evaluate seal confidence and day-to-day storage behavior. Across all tests, we prioritized metrics and observations that translate directly into design requirements, such as reach, closure force, closure time, drip behavior, and sealed-state confidence.

Benchmarking

Benchmarking Methods

We benchmarked umbrella geometry and collapse behavior using a structured plan designed to isolate geometry-driven variables rather than materials or electronics. The benchmarking plan defines measurements across canopy geometry (open diameter, depth, panel structure), folding geometry (fold direction, folding stages, rib nesting, fold consistency), rib and strut geometry (length, curvature, pivot placement), shaft geometry (segment count, alignment during collapse), closed-state geometry (closed length and envelope), and vehicle interaction outcomes (door clearance fit, roofline interference, arm extension required). Measurements were collected using tape measurement, profile photos with scale reference, sequential video/photo capture of collapse, and a vehicle entry simulation.

Geometry Feature	What to Observe	Umbrella A: Stick Reverse	Umbrella B: Warped	Umbrella C: Automatic
Open Canopy Diameter	Wide / medium / narrow	42 in	Major: 39 in Minor: 34.5 in	40 in
Canopy Depth	Shallow / medium / deep	11.5 in	7in	9.5 in
Folding Direction	Inward / outward	Up	Down	Down

Fold Control	Clean / messy / inconsistent	Manual clean <5lb	Manual very difficult 15lb	Automatic easy 0lb (Auto)
Rib Length	Short / medium / long	24 in	Max: 29.5 in Min: 15 in	23 in
Rib Angle	Straight / curved	50 deg	30 deg	42 deg
Closed Length	Short / medium / long	31.5 in	10.5 in	12.7 in
Closed Bulk	Slim / average / bulky	Bulky when unfastened, average when fastened	Small when unfastened . average when fastened	Bulky when unfastened, average when fastened
Water Containment	None / partial / good Notes: Cup filled with water, dumped three times each on umbrella, umbrella closed and tapped on paper ten times			
Door Clearance	Fits / tight / doesn't fit	Doesn't fit	Doesn't fit	Doesn't fit
Entry Motion Flow	Smooth / awkward (reference videos)	Awkward	Awkward	Awkward

Closing + Wrapping time	Time (s) : average of three times	8.75 Notes: Strap is far from the base, have to hold it higher up to get it	8.07 Notes: have to wrap it tight because its so bulky, takes more force, probably because ribs are “huge” for wind protection, jerky motion down, really heavy	7.64 Notes: strap is at a weird angle so you have to pull it kind of up to wrap it around, shaft length is much shorter when the umbrella canopy opens up which can be a drawback because we need as much shaft length as possible to clear the car door
Notes Umbrella A has a weaker spring which makes entry motion easier/ less effort for fold control Umbrella B is extremely strenuous to lock (“at least 50 pounds to lock” - peter w.) Umbrella C fails to open up fully sometimes				

Table 2: Benchmarking Results fo Umbrella A, B, and C

Benchmarking Results

Across three benchmark umbrellas (Stick Reverse, Warped, Automatic), several patterns emerged as seen above in Table 2:

- Door clearance was a universal failure point. All three benchmarks were rated “doesn’t fit” for door clearance and “awkward” for entry motion flow, reinforcing that current closed-state geometry and required motion are not optimized for constrained vehicle entry.
- Fold control strongly depends on required user force and mechanism behavior. The stick reverse umbrella was rated “manual clean” with less than 5 pounds for fold control, while the warped umbrella was “manual very difficult” and noted at 15 pounds, indicating

higher user burden when fold geometry is less cooperative or when structural stiffness increases resistance.

- Closure and wrapping time clustered around ~7.6–8.8 seconds, but the reasons differed. The stick reverse umbrella required holding higher up due to strap position (8.75 s), the warped umbrella required tight wrapping because it was bulky and jerky (8.07 s), and the automatic umbrella had an awkward strap angle and short shaft length (7.64 s), which was flagged as a drawback because our application needs shaft length to clear the car door.
- Shaft length and bulk directly impact vehicle usability. The benchmarking notes explicitly call out shaft length as a limitation in vehicle entry, and closed bulk as a contributor to difficult wrapping and maneuvering.

What Changed

Benchmarking directly supported our decision to treat “vehicle entry performance” as a primary design requirement rather than a secondary convenience feature. It also reinforced two design priorities: (1) closing must be controlled and low effort, and (2) the closed-state envelope and shaft geometry must support door clearance and storage without awkward re-gripping or excessive repositioning.

Functional and Design Specifications

User Need	Design Requirement	Target Specification	Test Method
Smooth vehicle entry and exit	Umbrella must move through a partially open car door without awkward repositioning	Passes through car-door opening without contacting roofline or door frame	Simulated vehicle-entry test; record door clearance fit as fits / tight / does not fit
Reduced reach during closure	User should not need to reach toward upper shaft or canopy runner to close	Primary closure interaction occurs near the handle or lower shaft	Observe user interaction during prototype testing; record where users naturally reach and whether upper-shaft grabbing occurs

Low-effort closure	Closing should be achievable with one hand without excessive force	Closure force remains comfortable for one-handed use; benchmark goal below high-force manual systems	Measure pull force with luggage scale or force gauge; compare against benchmark fold-control ranges
Fast closure during transitions	User should be able to close and contain the umbrella quickly while entering a car or building	Closure and containment time should improve on benchmark wrapping times of about 7.6–8.8 seconds	Time three closure trials per user; average closure + containment time
Controlled canopy collapse	Canopy should collapse in a predictable path without bunching or spreading unevenly	Fold path should be clean and repeatable across trials	Video/photo sequence of collapse; rate fold control as clean / inconsistent / messy
Compact closed-state form	Closed umbrella must be easy to maneuver and store after use	Closed envelope should be smaller and more organized than bulky unfastened benchmarks	Measure closed length and maximum closed width; compare with benchmark closed-state measurements
Reliable water containment	System must reduce dripping after closure and during transport	No major leakage during short movement, tilt, or inversion; acceptable result is no leak or damp rim only	Tablespoon-scale water test; rate leakage from 0–3 after tilt, inversion, walking, and set-down
Clear sealed state	User should understand when the container is sealed without instruction	Users can close lid confidently and identify sealed state	User testing; record confusion, resealing attempts, and seal confidence rating from 1–5

Easy opening and closing of container	Lid should seal securely without being too hard to remove	Lid opens and closes without excessive effort or fear of breakage	User test with dry and wet hands; record open/close difficulty, comments, and number of attempts
Day-to-day storage compatibility	Product should fit into realistic storage locations after use	Users identify plausible storage locations such as backpack pocket, car cupholder area, floor, or entryway	Scenario test; ask users to demonstrate where they would store it and record objections about size, handle, or leakage

Table 3: Needs, Requirements, Spec, and Test Method

We translated testing and journey mapping into measurable specifications so that prototype performance could be evaluated objectively rather than by feel, as seen above in table 3. The benchmarking plan itself is structured to convert qualitative vehicle problems into measurable geometry and motion variables (closed length, envelope, fold consistency, arm extension, and clearance).

Core specs we are tracking (linked to test methods):

- Vehicle clearance performance: door clearance fit and entry motion flow (vehicle simulation + observation)
- Closure usability: closure time (average of multiple trials) and closure force proxies (observed difficulty and measured force where possible)
- Closed-state compactness: closed length and bulk (tape measurement + envelope observation)
- Containment performance: drip behavior and water containment observations (drip paper test, inversion risk notes)

User-centered iteration (user testing + what changed)

Early prototype user validation

We conducted early user observation with four student participants to evaluate how first-time users attempt to close the system and what interaction they naturally expect. Users initially defaulted to familiar umbrella behaviors (top slider, twisting bottom, grabbing ribs), asked for missing affordances (buttons, lock cues), and only adopted lower-shaft pulling after understanding the intended interaction. Several participants explicitly identified “reaching and

pulling” as the most difficult part of umbrella closure, and multiple participants expressed a desire for one-handed, single-motion closing with reduced finagling.

What changed: These observations validated that our core direction targets a real pain point and that improving affordances is a key requirement. The test also reinforced that the “ideal” user expectation aligns with our working solution: a more controlled, lower-shaft interaction that reduces reach and steps.

Handle testing

We ran structured handle comparisons using prompts targeted to the vehicle context (trust when rushing into a car, maneuverability near a door, storage behavior) and a scoring framework that measured ease of use, comfort/ergonomics (weighted), and tight-space maneuverability. Results show that users prioritize control and trust in tight spaces, and that certain grip types can feel restrictive if they force the hand into a fixed position. In the sample scoring tables, carabiner-style and simple rubber directions performed strongly for trust and maneuverability, while bulky handle geometries were repeatedly flagged as “too big” for tight-space movement and storage.

Handle Type	Ease of Use Average	Comfort Average	Tight-Space Average	Total Score	Key Takeaway
Hand-Molded Grip	4.0	$2.3 \times 2 = 4.7$	4.0	12.7	Intuitive, but locks the hand into one position and limits adjustability
Simple Rubber Grip	5.0	$3.3 \times 2 = 6.7$	4.0	16.0	Best simple option; flexible, familiar, and reliable, with minor wet-feel concerns

Gym Grip	2.0	$1.7 \times 2 = 3.3$	3.3	8.7	Most restrictive; difficult to switch hands or maneuver quickly
J-Shape Handle	4.3	$3.7 \times 2 = 7.3$	2.0	13.7	Comfortable and familiar, but too bulky for car-entry use
Carabiner Handle	3.3	$4.3 \times 2 = 8.7$	4.3	16.3	Highest overall score; secure, strong in tight spaces, slightly less intuitive
C-Shape Handle	3.7	$3.0 \times 2 = 6.0$	2.0	11.7	Bulky and not ideal for tight-space movement

Table 4: Handle Testing Score Summary

What changed: As seen in Table 4 and detailed in Appendix S, handle testing shifted our direction toward simpler, more maneuverable interfaces that support quick hand transitions and controlled pulling, rather than highly sculpted grips that may reduce adaptability across users and contexts.

Container lid testing (press-fit lid)

We evaluated a container prototype using a press-fit coffee lid interface against four questions: discoverability and trust in the seal, speed of opening/closing, leakage risk, and integration into daily storage and transitions. Multiple users reported that the seal felt very tight but was difficult to open, sometimes to the point of worrying about breaking it or needing a better lifting feature. The drink opening on the lid was repeatedly identified as a potential leak hazard, especially if the container is inverted, and users provided consistent storage feedback around backpack pockets and cupholder-like locations.

User	Seal / Discoverability	Opening + Closing	Leak Risk	Storage / Daily-Life Feedback	Design Implication
Yena, Electrical Engineering	Initially unsure whether to pull or turn, but figured it out	Hard to open; worried about breaking it	Coffee lid opening identified as a leak hazard	Would use in backpack, but size may limit fit in a partially full bag	Add clearer opening affordance and remove or redesign the drink opening
Kritika, Biomedical Engineering	Easy to figure out; suction felt secure	Wanted a better place to grip and pull; suggested screw-on style	Did not see major leak concern	Wanted it to fit in a backpack side pocket; handle may be obtrusive	Improve grip/lip geometry and consider cupholder or bottle-pocket proportions
Patrick, Mechanical Engineering	Seal felt very tight, but too hard to open	Opening and closing not easy; suggested twist-style cap or gasket	Did not see leaking as the main issue, but wanted a looser fit	Would like cupholder fit; asked where the lid goes when open	Reduce opening force, add gasket/seal strategy, and resolve lid storage
Jeter, Civil Engineering	Hard to figure out how to open	Could not open/close quickly	Leakage not the main concern	Would not put in bag due to size; suggested longer handle and bag clip	Improve daily carry, reduce bulk, and add external attachment option

Lailah, Sociology	Hard to open	Not quick to use	Coffee mug lid did not feel secure; concern about leakage	Would store upright when dry and upside down when wet; would swing it while walking	Improve perceived seal reliability and test upside-down carry behavior
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Table 5: Container Lid Testing Summary

What changed: As seen above in Table 5 and detailed in Appendix T Container testing clarified that “tight seal” alone is insufficient. The design must also support fast, intuitive opening, and minimize leak paths introduced by lid features not designed for this application. This feedback supports redesigning the opening interface, adding clearer grip features for lid removal, and validating container geometry against how users actually carry and store the object.

Prototype

Mockups

The team's prototyping process followed a user-centered iteration model, progressing through four mockup phases, each tied directly to identified user needs and producing documented feedback that drove the subsequent iteration. Across all phases, the core interaction architecture — lower-shaft pull-to-close actuation with integrated moisture containment — remained constant while individual subsystems were refined in response to what users and testing revealed.

1. Lower Handle Actuation Concept

The first mockup explored relocating the closure interaction point from the upper shaft to the lower handle region. A basic form was constructed to test whether users could intuitively collapse the umbrella from a lower position without reaching toward the canopy runner.



Figure 5

User needs addressed:

Reduced reach during closure; one-handed operation in tight spaces; reduced cognitive burden while multitasking.

Observations and feedback:

First-time users defaulted to familiar top-slider behaviors and initially reached for the upper shaft before understanding the intended interaction. Once the lower-shaft pull was demonstrated, users quickly recognized the ergonomic benefit. Multiple users explicitly identified reaching and pulling as the dominant pain point, directly validating the core direction. Users consistently asked for clearer affordances such as color differentiation or tactile lock signals to communicate the correct interaction point without instruction.

Takeaways:

The feedback confirmed lower-shaft actuation as the correct architecture. The team prioritized improving affordance design in subsequent iterations and began exploring cord-based force transmission as the mechanism to link the handle pull to the upper runner.

2. Container Concept

The second mockup evaluated an integrated cylindrical container that would receive the collapsed canopy after closure, addressing the post-use moisture management problem identified throughout user research as the central unmet need.



Figure 6: Mockup of Container Concept

User needs addressed:

Integrated moisture containment; clean storage immediately after use; day-to-day carry compatibility.

Observations and feedback:

Users found the concept immediately understandable as a storage solution for a wet umbrella. However, the coffee-cup-style lid opening was repeatedly flagged as a leak hazard when the container is inverted or carried at an angle. Opening force was described as too high, with users expressing concern about breaking the lid. Users wanted a clearer grip feature for removal and a more decisive sealed-state cue. Also, users mentioned not wanting an additional interaction step to seal the container, emphasizing the importance of a “smooth” motion. Storage scenarios identified across users included backpack side pockets, car cupholder areas, and the vehicle floor. Handle geometry near the base was flagged as potentially obstructive when maneuvering the umbrella.

Takeaways:

The team removed the container lid feature after considering the results of moisture testing, which concluded that water contamination came more from residue water on the canopy rather than accumulated water in the container. Container proportions were refined toward a slimmer, cupholder-compatible geometry to reduce bulk at the car-door interaction zone.

3. Locking Mechanism Iterations

The third mockup focused on exploring how to lock/unlock the umbrella shaft using a mechanism compatible with lower-handle string actuation. The goal was to identify a locking architecture that would integrate cleanly with existing umbrella manufacturing conventions, be robust enough for repeated daily use, and release reliably from the lower handle without requiring the user to interact with the upper shaft.

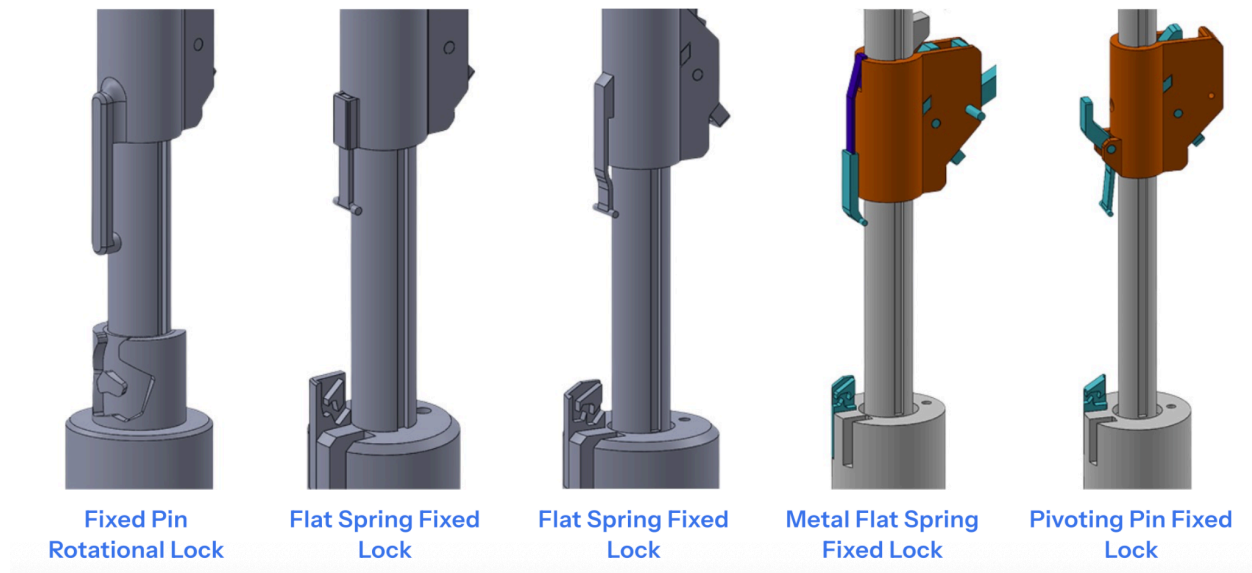


Figure 7: Locking Mechanism Iterations

User needs addressed:

Reliable locking with clean release; umbrella able to withstand strong external forces.

Observations and feedback:

The fixed-pin rotational lock provided positive engagement but required a rotational motion that conflicted with the intended linear pull-to-close interaction. Flat spring iterations showed promise for low-profile integration but exhibited inconsistent engagement and low durability. The metal flat spring version improved engagement reliability but demanded tighter manufacturing tolerances. The pivoting pin design emerged as the most promising direction, offering reliable and robust bi-directional locking with a clean release and natural integration with a cord-driven actuation path.

Takeaways: The pivoting pin direction was carried forward into the final prototype and eventually evolved into a ratchet-and-pawl system. This evolution was driven by two considerations. First, a ratchet-and-pawl architecture aligns more naturally with standard umbrella manufacturing, where the existing release button can be used directly to trigger the ratchet engagement, making the design easier to integrate and quality-control at production scale. Second, the ratchet provides a more mechanically robust and repeatable engagement than a custom-printed flat spring or pin, reducing the risk of fatigue failure over thousands of open-close cycles. The incremental closure behavior that the ratchet enables was recognized as an additional benefit following user testing feedback, and was incorporated into the system design after this mockup phase rather than being an original goal of the exploration.

Works-like Prototype

The works-like prototype validates the core mechanical actuation system: a string routed from the lower handle that interacts with both the top lock and bottom lock of the umbrella shaft without any upper-shaft interaction from the user. Rather than building on a full umbrella frame, the team constructed a focused test rig using a non-telescoping 3D-printed plastic shaft with a sliding collar on it, attached to a single rib from an actual umbrella dismantled and re-attached to the test shaft alongside the locking system. This allowed the team to isolate and validate the string actuation and locking behavior independently of the full frame complexity.



Figure 8: Works-Like Prototype

The system works as follows. A central spring pushes the collar upward along the fixed shaft, holding the rib in the extended, open position. The top lock engages when the rib is fully extended during use, holding the collar in place against the spring force. Pulling the cord downward using the pull ring from the lower handle transmits force upward, disengaging the top lock and allowing the collar to be drawn downward, collapsing the single rib inward. Pulling further brings the collar down to the handle, where it hooks onto the bottom lock, securing the collapsed state. When the user presses the button on the handle, it actuates a lever that releases

the bottom lock, allowing the spring to push the collar back upward and return the rib to the open position.



Figure 9: Pull Sting Force Measurement using Luggage Scale

The works-like prototype demonstrated that string-based lower-handle actuation of both the top and bottom locks is mechanically feasible and measurable against the force specification. Pull force at the handle measured 20.15 N, meeting the ideal target of 34 N and the marginal target of 40 N. Reach required to operate the system is 0 inches, which implies not needing any interaction above the handle. It is important to note that the prototype simplifies several real-world conditions, including the force contributions of a full multi-rib canopy, the friction between the canopy and the container, and the resistance of an actual telescoping shaft, all of which are expected to affect force and motion behavior in the final integrated system. These results nonetheless confirmed the core actuation architecture and justified carrying the string-based lower-handle system forward into the final prototype.

Looks-like Prototype

The looks-like prototype was developed independently of the mechanical system to validate the pleated canopy geometry and container form factor together as an integrated object. The two elements were evaluated as a pair because the canopy's collapsed geometry directly determines whether it seats cleanly into the container opening, making them co-dependent design variables that needed to be validated together before integration with the working mechanical system.

For the canopy, the team investigated multiple origami-inspired fold geometries before selecting a radially pleated pattern heat-set into 210T polyester fabric. Samples were folded and baked at 340°F for 30 minutes, confirming that the material retained its crease geometry after repeated collapse and deployment cycles.

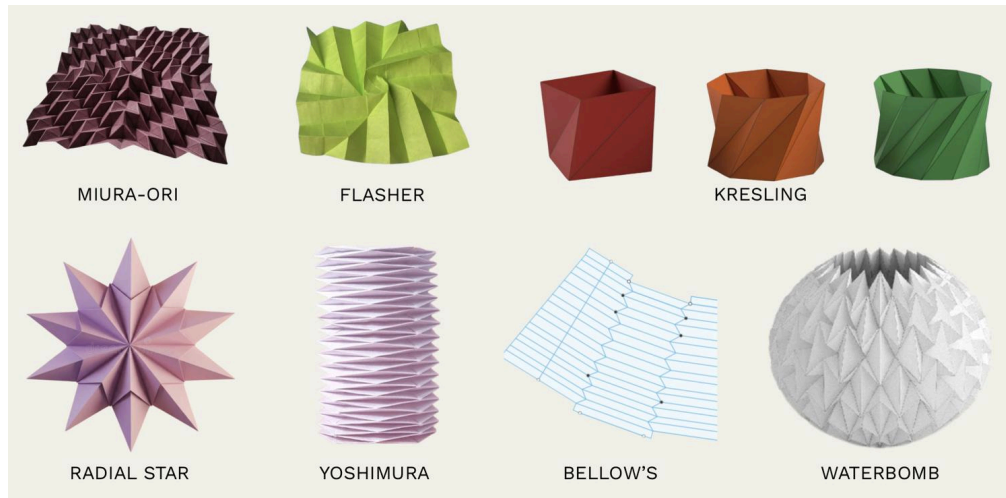


Figure 10: Origami Pattern Research

The container was refined from the earlier container mockup feedback, incorporating a slimmer cylindrical 3D-printed profile. The pleated canopy collapses into a compact radial bundle that seats into the open top of the container. The container exterior is smooth and matte, reading as a clean and professional carry object rather than a wet umbrella.



Figure 11: Looks-Like Prototype

The primary outcome of this form factor exploration was user confirmation that the object is storable in a car cupholder or backpack side pocket, and positive perception of the contained closed state as clean and professional even when wet fabric is inside. These results confirmed that the pleated canopy and container geometry were ready to be converged with the working mechanical system into the final prototype.

Final Prototype

The final prototype integrates the custom handle mechanism developed across the mockup and works-like phases onto a standard automatic umbrella platform with a custom handle casing. In addition to combining the lower-shaft string actuation and dual-lock system validated in earlier phases, the final prototype introduces three new elements. First, a ratchet-and-pawl mechanism housed within the custom handle casing enables incremental closure, so the user is never required to generate a single continuous pulling motion to fully collapse the canopy. Second, a 2:1 pulley system routed across the shaft reduces the force the user applies at the handle by half relative to the raw runner force. Third, a coordinated actuation link between the umbrella's existing release button and the ratchet pawl allows the user to disengage the pawl and reset the system in a single button press, without any secondary interaction.



Figure 12: Final Liminal Umbrella Prototype

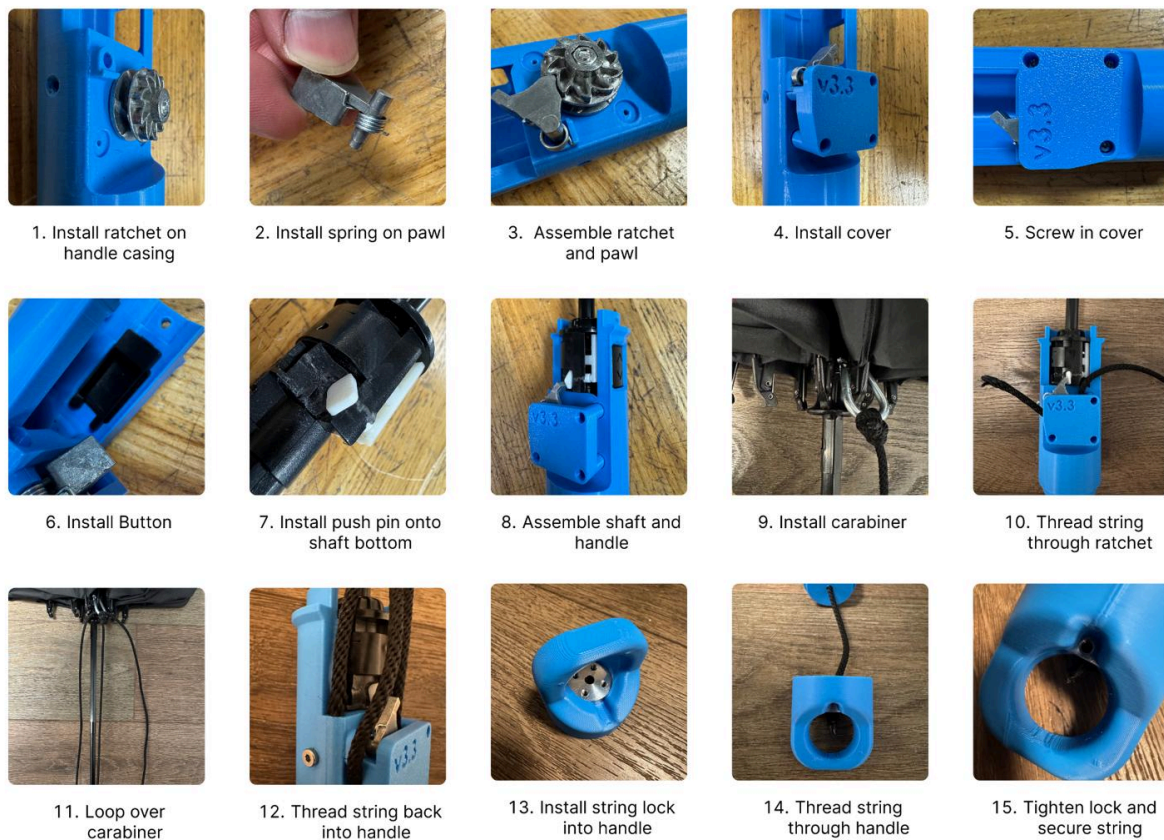


Figure 13: Final Liminal Umbrella SOP

The value-added custom components are the handle casing, which integrates the ratchet cavity, cord routing, and button linkage geometry into a single body; the ratchet-and-pawl assembly with a die-cast zinc internal mechanism and galvanized iron engagement surface; and the braided nylon pull cord terminating at the pull ring. Integrating onto an existing automatic umbrella platform rather than building a custom frame from scratch eliminates a substantial portion of the component count and manufacturing complexity, concentrating the custom value-add entirely in the handle where the user interaction differentiation lives. This is consistent with the broader lean manufacturing strategy of introducing custom integration points into an existing OEM production line rather than redesigning the full product architecture. Looking toward production, the handle casing is a clear candidate for transition from 3D-printed PLA to injection-molded ABS or nylon, which would reduce unit cost, improve surface finish, and enable tighter tolerances on the ratchet engagement geometry.

Overall, the final prototype successfully addresses the subspace of eliminating upper-shaft reach during closure. Lower-handle string actuation, incremental ratchet-controlled collapse, pulley-assisted force reduction, and coordinated button release are all validated within a handle

envelope that integrates onto a standard umbrella platform. An additional planned development is routing the pull cord internally through the umbrella shaft rather than along the exterior, which would eliminate exposed mechanisms, improve the visual cleanliness of the product, and produce a more robust and protected cord path for everyday use.

The final prototype meets the reach specification at zero inches and the pull force specification at 3.8 kg (approximately 37.3 N), within the marginal target of 40 N. The incremental ratchet-controlled closure and reduced handle force together produce a coordinated and fluid closure motion, which aligns closely with what users described as their ideal interaction throughout testing. The team looks forward to continuing to develop the handle actuation system to a production-ready level.

A significant challenge emerged, however, around the integration of the container with the umbrella's actuation sequence. Collapsing the canopy into the container generates significant friction between the fabric and container walls, which can be reduced by increasing the diameter but at the cost of a larger overall form factor and a more cumbersome object to handle. Deploying the umbrella is even more difficult, as the automatic umbrella extends the shaft and deploys the ribs simultaneously, generating active friction against the container walls and preventing the umbrella from opening while seated in the container. This outcome is in part a reflection of the team's decision to concentrate prototyping resources on the handle actuation system, which user research consistently identified as the highest-friction moment in the umbrella experience. In hindsight, the problem is most cleanly understood as two independent subproblems: eliminating upper-shaft reach through handle actuation, and enabling integrated moisture containment through a compatible container architecture. The final prototype demonstrates that the first subproblem is well addressed. The second remains an open design space, and the team's view is that tackling it independently is the right path forward, with the long-term goal of converging both solutions into a fully integrated close-and-contain system.

Design Drawings

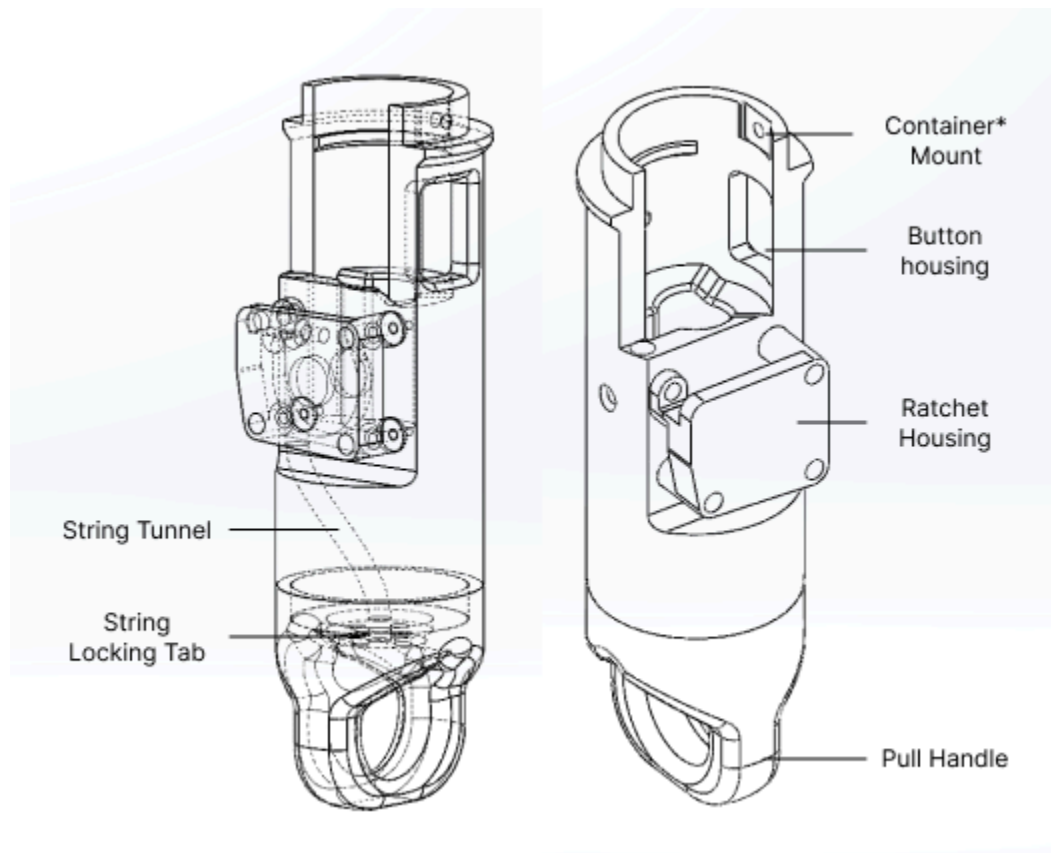


Figure 14: Ratcheting Handle Engineering Drawing

Please see Appendix R for detailed engineering drawings of the other subsections and final view of the prototype.

Patent Space Strategy

Patent Research

Appendix F shows a comprehensive patent search was conducted throughout the development of the umbrella system to understand the existing intellectual property landscape and identify opportunities for innovation. The research focused on four primary embodiment areas that are central to the final design: the bi-directional lock mechanism, pull-to-close actuation system, integrated water-containment container, and canopy folding architecture.

For each embodiment, approximately 80 relevant patents were identified and reviewed. The team performed a detailed analysis of patent claims, specifications, and figures to understand how similar systems were structured and protected. Particular attention was given to embodiment-specific patents to determine common design approaches, recurring claim language, and strategies used to broaden patent coverage. This process provided valuable insight into existing solutions while highlighting potential gaps in the market and opportunities for differentiation.

Following the initial search, the patent set was narrowed to approximately 15 highly relevant patents for each embodiment. These patents represented the closest prior art and were evaluated based on structural similarity, user interaction methods, folding mechanisms, locking systems, and handle-based actuation approaches. By focusing on the most relevant references, the team was able to establish clear boundaries between existing technologies and the proposed umbrella system.

Bi-Directional Lock Mechanism

The bi-directional lock mechanism revealed a broad range of existing prior art depending on the final locking configuration. Many of the most relevant active patents were found outside the umbrella industry, particularly in sportswear adjustment systems and window-locking applications. While the locking mechanism itself was not expected to serve as the project's primary patentable embodiment, the research indicated an opportunity to protect the umbrella-specific application and integration of the mechanism within the overall system architecture.

Structured Folded Canopy

Prior art related to canopy folding demonstrated that engineered crease networks can guide controlled fabric collapse and reduce uncontrolled draping during storage. Existing reverse-folding umbrella patents primarily focus on frame and linkage mechanisms rather than the behavior of the canopy fabric itself. This created an opportunity space around structured

canopy folding methods, including heat-set crease networks, radial fold geometries, bellows-style pleats, and their integration with reverse-folding umbrella architectures.

Pull-to-Close Actuation System

The pull-to-close subsystem presented the strongest opportunity for differentiation. Limited direct prior art was identified for pull-actuated umbrella closure systems, as most umbrella designs rely on push-based or spring-actuated mechanisms. Relevant analogs were found in adjacent industries, including cord-driven window blinds, retractable identification badge reels, pull-string lighting fixtures, and toy actuation systems. The research suggested a potential opportunity to protect the relocation of the closure actuation point from the upper shaft to the handle, improving ergonomics and user control during operation.

Integrated Self-Container System

A significant amount of prior art was identified relating to umbrella storage and water-containment solutions. One particularly relevant active patent described a bottle-style container system for umbrella storage. However, existing designs generally treated the storage container as a separate accessory rather than an integrated component of the umbrella itself. Detailed claims analysis of the most relevant references indicated that opportunities remain for protecting a self-contained system in which the umbrella and storage container function as a unified product during closure and stowage.

Overall, key themes identified in the prior art included umbrella closure mechanisms actuated from the handle, reverse-folding canopy designs, water-containment accessories, and various locking approaches for telescoping shafts. While many patents addressed individual aspects of the user experience, few integrated multiple solutions into a single cohesive system designed to improve the complete umbrella-use workflow.

Novelty and Provisional Patent

The patent search revealed that many existing umbrella patents address individual aspects of the umbrella experience, including automatic opening and closing systems, reverse-folding canopies, shaft locking mechanisms, and water-containment approaches. However, these solutions generally continue to rely on closure mechanisms actuated near the upper shaft or canopy, requiring users to reach upward during operation. This limitation is particularly problematic in constrained environments such as entering or exiting a vehicle, where maneuverability is restricted.

Additionally, the majority of prior art focuses on spring-based or push-actuated mechanisms and does not provide a practical means of relocating the actuation point to a lower, more ergonomic handle-level interface. Existing reverse-folding umbrellas improve water containment but do not fully integrate self-contained storage systems that guide the umbrella into a dedicated enclosure during closure. Furthermore, conventional canopies typically collapse in an unstructured manner, limiting their ability to be consistently stowed within a compact container.

The team's review also identified cord- and cable-driven actuation systems in adjacent industries, such as retractable devices, blinds, and pull-string mechanisms. While these systems demonstrate alternative approaches to motion transfer, they have not been widely adapted to umbrella closure systems in a manner that enables coordinated shaft retraction, guided canopy folding, and integrated water containment.

As a result, the primary area of novelty for the proposed design lies in the system-level integration of several complementary subsystems. The umbrella combines a handle-based pull-to-close actuation mechanism, a bi-directional locking system, a structured canopy-folding strategy, and an integrated containment container into a single user-centered workflow. Rather than focusing solely on a single mechanical innovation, the invention addresses the complete transition from umbrella deployment to storage, creating a more intuitive, ergonomic, and self-contained user experience.

Evolution of the Provisional Patent

One of the most significant developments during the provisional patent, detailed in Appendix R, drafting process was the incorporation of a ratcheting mechanism into the umbrella's closure and locking system. Early versions of the provisional focused primarily on the overall pull-to-close architecture and handle-based actuation concept. As prototyping progressed, however, the team identified the need for a mechanism that could provide controlled motion, positional retention, and improved user feedback during operation.

The ratcheting mechanism was introduced as a potential embodiment that could selectively engage and disengage with the umbrella shaft during closure. Rather than requiring a single continuous closing motion, the ratchet allows the umbrella to move incrementally while preventing unintended reverse movement. This approach improves controllability during closure, reduces the force required from the user at any given moment, and creates a more intuitive interaction between the user and the umbrella.

From a patent strategy perspective, the ratcheting mechanism was important because it expanded the range of embodiments covered by the provisional application. Instead of limiting protection to a single implementation of the pull-to-close system, the provisional was revised to include alternative methods for transmitting, controlling, and locking motion within the umbrella. The ratchet was therefore described as one possible embodiment among a broader family of closure

and retention mechanisms, allowing the patent to maintain flexibility while still capturing a key engineering advancement developed during the project.

The addition of the ratcheting embodiment also drove substantial updates to the patent figure set. New drawings were created to illustrate the interaction between the ratchet, locking features, handle actuation components, and umbrella shaft.

Market Strategy

Target User

The primary target user for the umbrella system is the urban professional who regularly commutes between home, work, and other destinations while maintaining a polished appearance. This user values convenience and functionality but is unwilling to sacrifice professionalism through the use of bulky rain gear such as ponchos or rain suits. Typical users include business professionals, consultants, office workers, and frequent travelers who encounter transitional scenarios such as walking between buildings, commuting on public transportation, or entering and exiting vehicles during inclement weather.

This customer segment is willing to pay a premium for products that seamlessly integrate into their daily routine and reduce friction in everyday experiences. Similar to the consumers who purchase products from brands such as Away, these users prioritize thoughtful design, durability, and convenience over the lowest possible price. The umbrella system addresses a common but often overlooked pain point—managing a wet umbrella—through a solution that feels intentional, refined, and compatible with a professional lifestyle.

Brand Positioning

The brand was developed using the positioning principles popularized by premium lifestyle brands such as Away. Rather than competing on technical specifications alone, the umbrella is framed as a carefully designed travel and commuting accessory that improves the user's experience during everyday transitions.

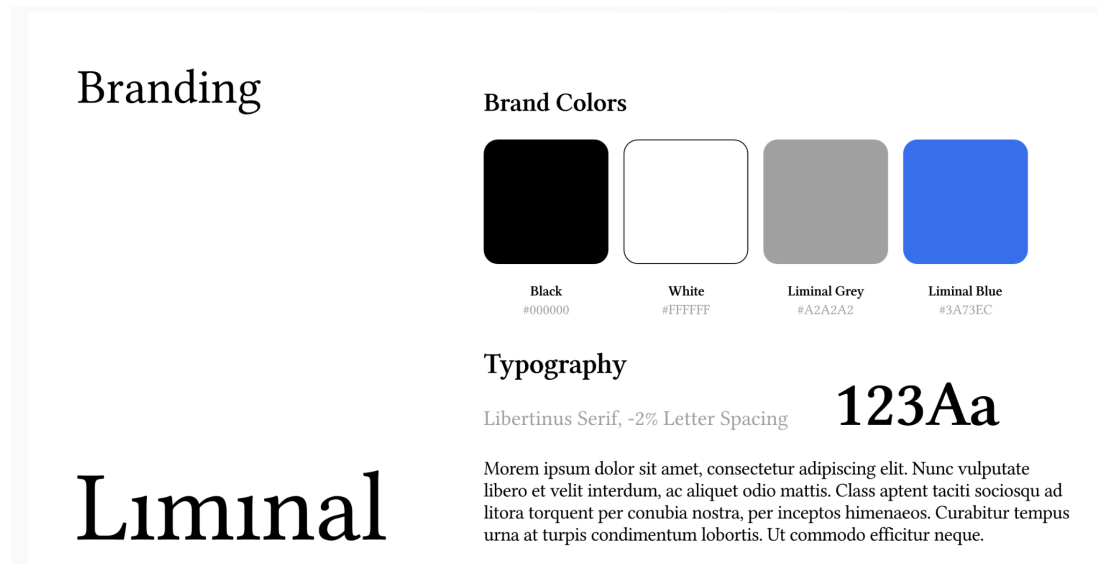


Figure 15: Liminal Umbrella Branding

The visual identity, seen above in Figure 15, emphasizes simplicity, professionalism, and confidence. Branding materials utilize a restrained color palette consisting primarily of blue, black, and neutral tones, paired with clean typography and minimal visual clutter. This aesthetic communicates reliability and sophistication while reflecting the product's focus on intentional design.

The overarching brand message centers on reducing friction in daily movement. Every design decision, from the integrated containment system to the streamlined visual language, supports the idea that professional users should be able to move through rainy environments without disrupting their routine, appearance, or workflow.

Competitive Positioning

The umbrella is positioned within the premium segment of the umbrella market while remaining more accessible than many luxury accessories. Traditional umbrellas often compete on price and are treated as disposable products, while high-end umbrella brands typically emphasize craftsmanship or material quality alone.

In contrast, the proposed umbrella differentiates itself through user experience innovation. The combination of handle-based closure, guided canopy folding, and integrated self-containment provides functional value beyond weather protection. This positions the product between mass-market umbrellas and luxury lifestyle accessories: premium enough to communicate quality and intentionality, yet accessible enough for everyday professional consumers.

The product's competitive advantage is therefore based not only on aesthetics but also on its ability to solve a meaningful user problem in a way that aligns with the expectations of modern professionals.

assembly. A third risk involves improper canopy tension after custom mechanism integration. This is controlled through standardized assembly procedures and final functional testing. By introducing inspection gates throughout the assembly process, defects can be identified early and prevented from propagating downstream.

The manufacturing journey map developed during this project revealed that umbrella production is substantially more complex than originally anticipated, involving numerous suppliers, manufacturing processes, quality checkpoints, logistics providers, and operational stakeholders. This systems-level understanding informed design decisions that support manufacturability, scalability, and long-term production feasibility.

Critical-to-Quality (CTQ) Characteristics for our umbrella include:

1. Ratchet Engagement Reliability
 - Mechanism must positively engage during closure.
 - No unintended release under normal operating loads.
2. Button Release Function
 - Release mechanism must disengage consistently.
 - User force must remain within acceptable ergonomic limits.
3. Seal and Interface Integrity
 - Mating components must maintain proper alignment.
 - Interfaces must resist wear during repeated operation.
4. Leak Resistance
 - Canopy and attachment points must prevent water intrusion.
 - Final assembly must withstand simulated rain exposure.

Incoming Inspection:

- Verification of dimensions for custom components.
- Visual inspection of umbrella frame assemblies.
- Material and finish verification.
- Supplier certification review where applicable.

In-Process Inspection:

- Ratchet installation verification.
- Alignment checks during assembly.
- Functional engagement testing after mechanism installation.
- Verification of assembly torque and fastener installation.

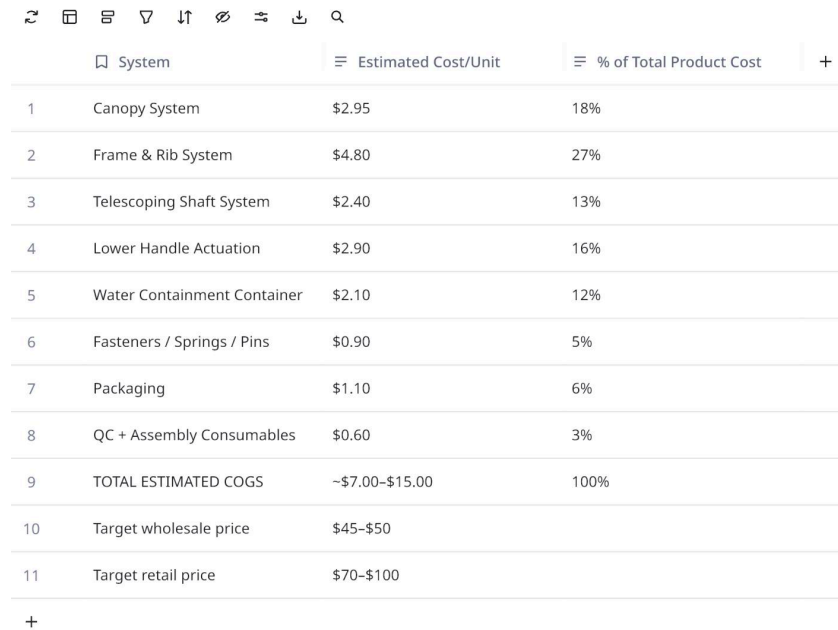
Final Functional Testing:

- Open and close cycle testing.

- Ratchet engagement validation.
- Button release validation.
- Structural integrity inspection.
- Leak testing under simulated environmental conditions.
- Final visual inspection and packaging verification.

Our manufacturing plan will continue to change as our prototype is refined. Currently, we are faced with the challenge of smoother canopy collapse into the container itself. Rib clips were designed as a potential solution to bring the ribs closer to the shaft and reduce the friction as it gets pulled into the container. However, more solutions have arisen and will be tested to identify the optimal mechanism for the umbrella. Ideation around canopy fabric pleating assembly has also begun, but is still in the early stages. Therefore, a more detailed manufacturing and assembly flow will be established once the subproblems have been thoroughly addressed.

Bill of Materials



	System	Estimated Cost/Unit	% of Total Product Cost	+
1	Canopy System	\$2.95	18%	
2	Frame & Rib System	\$4.80	27%	
3	Telescoping Shaft System	\$2.40	13%	
4	Lower Handle Actuation	\$2.90	16%	
5	Water Containment Container	\$2.10	12%	
6	Fasteners / Springs / Pins	\$0.90	5%	
7	Packaging	\$1.10	6%	
8	QC + Assembly Consumables	\$0.60	3%	
9	TOTAL ESTIMATED COGS	~\$7.00-\$15.00	100%	
10	Target wholesale price	\$45-\$50		
11	Target retail price	\$70-\$100		

Figure 17: Bill of Materials Summary and Cost Breakdown

Cost estimates are developed using supplier quotations, benchmark umbrella manufacturing costs, prototype fabrication expenses, and labor estimates. Total product cost, seen above in Figure 17, includes raw materials, purchased components, custom components, assembly labor, quality assurance, packaging, shipping, and overhead allocation.

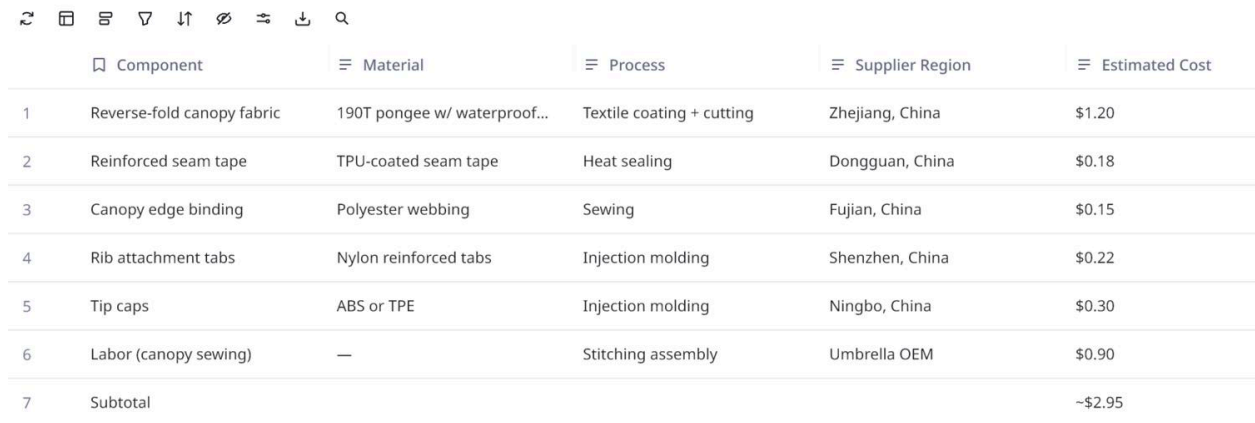
Expected Capital Expenses:

- Injection mold tooling for custom polymer components
- Machining fixtures and assembly jigs
- Inspection fixtures
- Prototype development tooling
- Production validation equipment

For low-volume production, custom machining may be more economical than dedicated tooling. As production volume increases, investment in molds and assembly fixtures becomes justified through lower unit costs.

A subsystem-based Bill of Materials (BOM) was developed to estimate the manufacturing cost of the umbrella prototype and identify key cost drivers within the design. Rather than organizing components solely by part type, the BOM was structured around the major functional systems of the product: the Canopy System, Frame & Rib System, Telescoping Shaft System, Lower Handle Actuation System, and Water Containment System. This approach provided greater visibility into how value is distributed throughout the product architecture and highlighted opportunities for design standardization and cost reduction.

Canopy System



	Component	Material	Process	Supplier Region	Estimated Cost
1	Reverse-fold canopy fabric	190T pongee w/ waterproof...	Textile coating + cutting	Zhejiang, China	\$1.20
2	Reinforced seam tape	TPU-coated seam tape	Heat sealing	Dongguan, China	\$0.18
3	Canopy edge binding	Polyester webbing	Sewing	Fujian, China	\$0.15
4	Rib attachment tabs	Nylon reinforced tabs	Injection molding	Shenzhen, China	\$0.22
5	Tip caps	ABS or TPE	Injection molding	Ningbo, China	\$0.30
6	Labor (canopy sewing)	—	Stitching assembly	Umbrella OEM	\$0.90
7	Subtotal				~\$2.95

Figure 18: Table of Cost Estimates for Canopy System

As seen in Figure 18, the canopy subsystem consists primarily of textile materials and assembly labor. Fabric and sewing operations account for the majority of the subsystem cost, making labor efficiency and fabric utilization important cost-reduction opportunities.

Frame & Rib System



	Component	Material	Process	Estimated Cost
1	Main ribs	Spring steel / aluminum alloy	Stamping + pivot hole...	\$1.60
2	Stretchers	Aluminum alloy	CNC punching	\$0.75
3	Reverse-fold guide links	Stainless steel	Precision stamping	\$0.55
4	Upper notch/crown	Zinc alloy die cast	Die casting	\$0.65
5	Runner assembly	Glass-filled nylon	Injection molding	\$0.70
6	Pivot rivets/pins	Stainless steel	Cold heading	\$0.25
7	Surface coating	Powder coat / anodizing	Finishing	\$0.30
8	Subtotal			~\$4.80

Figure 19: Table of Cost Estimates for Frame & Rib System

As seen in Figure 19, the frame and rib subsystem is the largest contributor to overall product cost and represents approximately one-third of total material cost. Structural components require multiple manufacturing processes including stamping, machining, die casting, injection molding, and finishing operations. Because of its complexity and high material content, this subsystem is expected to remain the primary cost driver in future production.

Telescoping Shaft System



	Component	Material	Process	Estimated Cost
1	Outer shaft tube	Aluminum extrusion	Tube forming	\$0.70
2	Inner shaft sections	Aluminum	Precision tube machining	\$0.60
3	Lock sleeves	POM / nylon	Injection molding	\$0.35
4	Guide rails/tracks	Stainless steel	Laser cut + formed	\$0.30
5	Spring return element	Spring steel	Coil forming	\$0.25
6	Internal cable guides	Nylon	Molded insert	\$0.20
7	Subtotal			~\$2.40

Figure 20: Table of Cost Estimates for Telescoping Shaft System

As seen in Figure 20, the telescoping shaft system provides controlled deployment and collapse of the umbrella. Costs are driven primarily by aluminum components and precision machining operations required to achieve smooth telescoping motion and reliable locking performance.

Lower Handle Actuation System

	Component	Material	Process	Estimated Cost
1	Handle shell	ABS + TPE overmold	Injection molding	\$0.95
2	Trigger mechanism	Nylon + steel insert	Molded assembly	\$0.45
3	Pull cable	Braided stainless cable	Cable forming	\$0.30
4	Spool / reel mechanism	POM engineering plastic	Precision molding	\$0.45
5	Locking pawl	Stainless steel	Fine blanking	\$0.20
6	Compression springs	Spring steel	Coil forming	\$0.15
7	Ergonomic grip coating	TPE	Overmolding	\$0.40
8	Subtotal			~\$2.90

Figure 21: Table of Cost Estimates for Lower Handle Actuation System

As seen in Figure 21, the handle actuation subsystem houses the user interface and closure mechanism. This subsystem became increasingly important as the design evolved to include a ratcheting closure system that prevents the umbrella from reopening during collapse. The locking pawl, spool mechanism, and cable system represent custom value-added components that differentiate the product from conventional umbrellas.

Water Containment System

	Component	Material	Process	Estimated Cost
1	Main containment shell	Polypropylene	Injection molding	\$0.80
2	Flexible sealing membrane	Silicone/TPE	Compression molding	\$0.35
3	Drainage reservoir	ABS	Injection molding	\$0.30
4	Water channel insert	TPU	Flexible extrusion	\$0.20
5	Gasket/seal	Silicone	Die cut	\$0.15
6	Snap-fit latch features	Integrated molding	Molded in part	\$0.10
7	Surface finishing	Texturing/coating	Finishing	\$0.20
8	Subtotal			~\$2.10

Figure 22: Table of Cost Estimates for Water Containment System

As seen in Figure 22, the water containment subsystem is the primary feature that differentiates the design from a standard umbrella. It manages water captured during closure and minimizes dripping during storage and transport. While relatively inexpensive compared to the frame system, it introduces additional sealing, molding, and assembly requirements that must be carefully controlled during production.

Consumption Chain

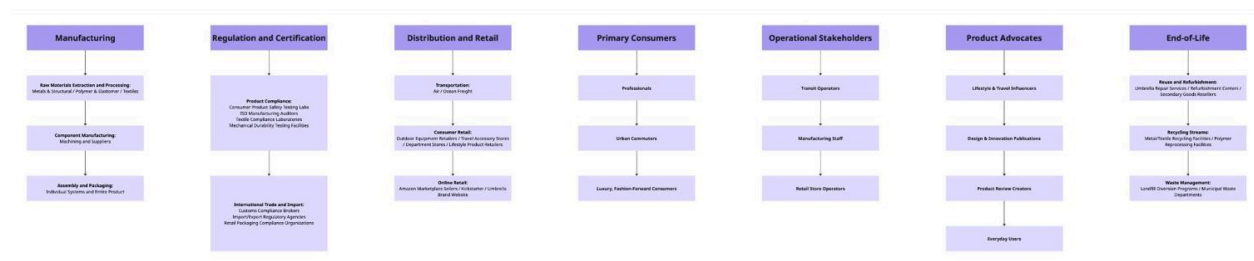


Figure 23: Detailed Liminal Umbrella Consumption Chain Overview

The consumption chain, seen above in Figure 23, maps the broader ecosystem of stakeholders involved throughout the product lifecycle. This includes raw material suppliers, manufacturers, logistics providers, regulatory bodies, distributors, operational users, maintenance personnel, advocates, waste management entities, and public or municipal partners. The section illustrates how value, materials, information, and operational responsibilities move across the ecosystem while highlighting the interconnected relationships between manufacturing, compliance, distribution, operation, and end-of-life management.

Contact

The supplier outreach strategy focused primarily on established umbrella OEM manufacturers in China due to their extensive experience, mature supply chains, and ability to integrate custom features into existing umbrella production platforms. Chinese manufacturers were identified through supplier databases, Alibaba listings, manufacturer directories, and industry references. Our team received a direct response from Xiamen Hoda Co., Ltd., a premier OEM and ODM manufacturer of the “Inverted Umbrella with Carabiner Handle”. This umbrella is of high quality and contains key features required in our own design.

A key advantage of the team was the participation of a Chinese-speaking teammate who facilitated communication with suppliers. This reduced language barriers, improved response quality, and enabled clarification of technical requirements that may have been difficult to communicate solely in English.

Supplier outreach focused on gathering information regarding manufacturing capabilities, minimum order quantities, tooling requirements, custom component integration capabilities, quality assurance processes, lead times, cost estimates, and engineering support services.

Manufacturers were specifically asked about making modifications to an existing reverse umbrella structure, general information regarding their experience with custom umbrella projects, and the approximate costs involved. These discussions provided valuable insight into the practical considerations of scaling the design from prototype to production and informed subsequent BOM, manufacturing flow, and cost analyses.

Appendices

Appendix A: Problem Exploration Research Reports

Toilets:

Interview Summary

- **Avoidance Behavior:**
People consistently avoid touching the toilet seat in shared or public restrooms.
 - **Workarounds:**
Most use folded or layered tissue to lower or cover the seat; some use a shoe to lift it.
- **Cleanliness Check:**
People often inspect or switch stalls if the seat appears wet, dirty, or suspicious.
- **Contextual Difference:**
At home, they sit directly without hesitation, showing that perceived cleanliness drives behavior more than hygiene habits.
- **Pain Points:**
 - Touching the seat (even with tissue) feels “gross” or unsafe.
 - Visible dirt or moisture triggers strong avoidance and anxiety.
 - Lining the seat with tissue feels slow and still not clean enough.
 - Few reliable hands-free options exist in shared restrooms.
- **Desired Improvements:**
 - Prefer touch-free solutions (motion sensors, foot pedals).
 - Want simple, fast, and hygienic designs that operate in seconds.
 - For home or dorms, prefer affordable, renter-friendly alternatives.

Patent Landscape Summary

62 patents– Four dominant categories of patented designs emerged:

https://miro.com/app/board/uXjVJ1Em4Bc=?share_link_id=402099843206

1. **Foot Pedal / Lever Systems:** Mechanical linkages activated by pedals; hands-free but bulky.
 - a. <https://inventionassistant.com/step-n-lift/>
2. **Hand Lever / Adhesive Tabs:** Low-cost retrofits that minimize touch but still involve hand contact.
 - a. <https://www.liftyloo.com/>

3. **Powered / Automatic Systems:** Sensor-based, motorized seats offering true touch-free use, but expensive and complex to install.
 - a. A lot of medical applications
 - i. Toilets for handicap
 - b. Japanese toilets
 - c. <https://www.totousa.com/washlet/how-it-works?>
4. **No Contact / Not Automated:** Passive or magnetic devices for indirect lifting.

The patent field is quite saturated and mature.

Insights & Opportunity Space

1. **Root Problem = Dirtiness, Not Touching**
 Disgust is driven by visible dirt and perceived contamination. Cleanliness visibility or self-cleaning materials would be more effective.
 - a. Automation doesn't solve dirtiness
 - b. Think about a diaper changing station (UV light)
2. **Crowded Market**
 Most lift mechanisms are already patented.
 - a. Every mechanism type (pedal, lever, sensor, magnetic) already exists.
3. **Contextual Behavior**
 Users' comfort at home vs. disgust in public spaces shows environmental trust outweighs technology.
4. **Practical Barriers:** Powered systems are costly; mechanical ones add bulk and cleaning issues.

Rock Climbing:

Interview Summary

1. Beginner Climbers (2)

- Climb mainly for fun and social reasons; most only boulder indoors.
- Learn from friends or YouTube, not formal instruction.
- Struggle with motivation and consistency after the initial excitement.
- Value a welcoming, community-driven environment.
- **Main Opportunity:** Service design around community and progression tools (e.g., visual progress tracker, app for climb logging).

2. Intermediate Climber (1)

- Bouldering-focused; want to improve measurable performance (strength, technique).

- Plateau in ability and lack structured training methods.
- Interested in peer mentorship but find few climbers with similar body types or styles.
- **Main Opportunity:** Accessible, guided training or mentorship tools to break plateaus.

3. Advanced Climber (Josh)

- Experienced in lead, top rope, and trad climbing, safety and equipment are priorities.
- Concerned about gear wear, belay safety, and lack of standardized inspection systems.
- Mentioned products like **Petzl Grigri** (simplified belaying) and **Avant Climbing Innovations** (lead rope multi-soloing).
- Recognized potential for innovation in safety tech and materials, but noted it's highly specialized.
- **Main Opportunity:** Smart or material-based safety tools with standardized wear indicators, if the team can invest in technical depth.

Survey Summary (8 responses)

1. Climbing Experience & Frequency

Respondents ranged from beginners (<6 months) to advanced climbers (5+ years), mostly indoor boulderers.

Regular climbers go 2–3 times a week; newer climbers attend weekly or biweekly for social and recreational reasons.

2. Training Challenges

Common struggles include performance plateaus, inconsistent routines, and low motivation. Beginners lack structure, while advanced climbers find it hard to progress beyond a certain level.

3. Learning Methods

Most learn informally through friends, mentors, or YouTube, with little formal instruction. This builds community but leads to safety and technique gaps at higher levels.

4. Welcoming Community Factors

A friendly, non-competitive atmosphere and familiar faces make gyms feel most welcoming. Community and social connection remain the biggest motivators for continued participation.

5. Desired Improvements

Respondents highlighted a need for:

- Interactive training tools (e.g., hangboard tracking, progress feedback)
- Stronger social/community connections
- More affordable gear options

Summary:

- Climbers vary in experience but share **frustrations around plateaus, inconsistent training, and lack of structure.**
- Social connection and community are key drivers of engagement.
- Opportunities lie in creating *interactive, inclusive, and affordable solutions* that blend technical progress with a strong sense of community.

Key Insights

- **Problem Variation by Experience:**
 - Beginners/intermediates: issues in *service design* (community, training consistency).
 - Advanced climbers: issues in *technical gear safety* and inspection standards.
- **Divergent Opportunity Areas:**
 - Service design and training are approachable but less tangible and already saturated with apps.
 - Gear safety innovation requires deep expertise in *materials, mechanical design, and climbing standards*.
- **Team Alignment:**
 - After discussion, the team showed less interest in pursuing a highly technical, gear-focused direction.
 - Given interest and feasibility, the climbing problem space felt fragmented and not strongly aligned with team goals.

The climbing idea presented interesting insights but split across two paths, social service design vs. technical gear innovation.

Because of existing solutions and team capacity, **the group chose to pivot toward a different problem space** with clearer user alignment and innovation potential.

Umbrellas:

Survey Results

1. Usage Frequency

Most respondents reported using umbrellas occasionally or seasonally, depending on weather and regional rainfall. A smaller group uses them daily during rainy seasons, while a few prefer raincoats or simply avoid going out in rain. This indicates that umbrella use is highly contextual and routine-dependent, influenced by local climate and transportation habits.

2. Common Scenarios of Use

Umbrellas are primarily used for commuting, walking to class or work, or short outdoor errands. Respondents often mentioned juggling bags, phones, or other items while holding an umbrella, making hand occupancy and balance a recurring challenge. Some users carry umbrellas in cars or office drawers as backups, while others forget or avoid them due to inconvenience.

3. Pain Points and Frustrations

The most frequent frustration involves managing a wet umbrella indoors—especially when entering cars, public buildings, or stores. Users expressed discomfort with dripping water, lack of proper storage, and general inconvenience. Wind-related issues were also noted: umbrellas flipping, breaking, or causing wrist strain in heavy rain.

Other cited frustrations included:

- Wet surfaces after folding (seats, floors, bags)
- Difficulty drying or storing in confined spaces
- Inconvenience when hands are occupied (e.g., carrying groceries, using phones)
- Poor durability of automatic mechanisms or ribs

4. Storage and Drying Habits

Respondents typically shake off water before entering indoors, then place the umbrella on balconies, doorways, or special stands. Some mentioned plastic bag dispensers in public places, but found them unsustainable or ineffective. A few resorted to leaving umbrellas in cars or bathrooms to dry, though this often resulted in mildew or odors.

5. Frustration Level and Replacement Willingness

Overall frustration levels ranged moderate to high (average 5–7 on a 10-point scale).

Users expressed a clear desire for a cleaner, faster, and less awkward drying process.

Nearly all respondents indicated high purchase intent (8–10) if a new umbrella design could improve ease, comfort, or moisture management.

Summary

The survey revealed that while umbrellas effectively serve their core function—keeping users dry—the post-rain experience remains incomplete. Wetness management, storage convenience, and ergonomic handling emerged as the strongest pain points. Users frequently face discomfort or social inconvenience when transitioning indoors, often improvising with inefficient solutions. While mechanical improvements (e.g., auto-open/close) have saturated the market, moisture containment and transitional usability remain under-addressed. There is strong interest in designs that offer contained, effortless, and socially seamless umbrella experiences, reducing the friction of carrying, drying, and storing.

Interview Results

1. Moisture and Drying Experience

Every interviewee described annoyance with moisture after folding umbrellas. Cars, entrances, and small spaces are especially problematic—water drips onto seats, floors, or personal items. Some users expressed a desire for instant-dry covers or automatic moisture containment systems that would let them store umbrellas anywhere immediately.

2. Hands-Occupied Scenarios

Many interviewees shared struggles with carrying bags, phones, or groceries while holding an umbrella. They expressed interest in hands-free or balanced designs—for example, wearable systems, stabilizers, or compact umbrellas that can be held with minimal grip effort.

3. Transition Between Outdoor and Indoor Use

Transition moments—such as entering cars or malls—were repeatedly cited as the most frustrating parts of the umbrella experience. Users mentioned wishing for “a way to get rid of moisture instantly” or “somewhere clean to throw it without leaking.”

4. Design and Appearance

While looks were not the top concern, respondents mentioned preferring bright colors for visibility and compact silhouettes for convenience. Aesthetic considerations were secondary to functionality but still influenced satisfaction.

5. Durability and Ergonomics

Several users mentioned umbrellas flipping in wind, handles being slippery, or wrist strain during prolonged use. The most appreciated features were sturdy frames and smooth automatic operation, but users noted that automatic folding mechanisms often wear out or jam after extended use.

6. Purchase Motivation and Likelihood

Across all interviews, users rated purchase intent for an improved umbrella between 8–10, especially if it addressed drying and storage pain points. Convenience and reliability were prioritized over novelty or visual design.

Summary

Interview data confirmed that the umbrella’s functional cycle remains incomplete—it excels in shielding rain but fails in what happens immediately after. Users improvise with makeshift drying methods, face frustration when transitioning indoors, and express openness to any innovation that reduces post-rain friction. The interviews underscored that moisture management, storage convenience, and one-handed usability are deeply interlinked pain points. Respondents envision umbrellas that are self-contained, hygienic, and effortless to manage, suggesting strong design potential in automation, material innovation, and compact containment systems.

Key Insights

1. **Moisture Management Is the Core Pain Point**
Both survey and interview data highlight wet umbrellas after use as the most universal frustration. Users dislike dripping water, damp storage areas, and the social discomfort of carrying something messy. The problem is not the umbrella's performance in rain—but what happens immediately after.
2. **Transition Moments Drive Frustration**
The transition from outdoors to indoors or into cars is the most problematic moment. Users often feel trapped between not wanting to drip water and having nowhere appropriate to store their umbrella.
3. **Hands-Free Convenience Is Underserved**
Many users find holding umbrellas difficult when carrying bags, phones, or children. This friction discourages use, especially among multitaskers or commuters.
4. **Durability and Ergonomics Still Matter**
Though secondary to moisture issues, durability and handling remain key to satisfaction. Users want umbrellas that resist flipping in wind, feel sturdy, and open smoothly. Automatic mechanisms are valued, but frequent breakage causes mistrust.
5. **Motivation for Change**
Both surveys and interviews reveal strong emotional friction. Users are open to innovation if it simplifies routine or enhances comfort—even at a higher price point.
6. **Sustainability Is Valued but Secondary**
Respondents appreciate environmentally conscious materials and reusable alternatives to plastic umbrella bags, but only if performance and durability are not compromised.
7. **Aesthetic and Social Factors Influence Perception**
Users prefer umbrellas that look neat and contained, especially in public or professional settings. Clean, compact designs signal organization and social composure.

Patent Research Summary

Patent exploration within the umbrella moisture management domain revealed two dominant innovation clusters:

1. **Umbrella Drying & Containment Devices**
Includes patents such as *US5323802A (Umbrella Bag)*, *US6655399B1 (Portable Absorbable Umbrella Stand)*, and *US20110154677A1 (Ringed Umbrella Dryer)*. These primarily focus on enclosures or sleeves that collect or absorb water. While functional, most are mechanically simple and lack true integration with umbrella form factors.
2. **Integrated or Modular Umbrella Systems**
Patents like *WO2006138600A2 (No-Drip Umbrella Sheath)* and *JP2004208735A (Portable Umbrella Holder)* explore sheath-based designs and self-contained drying cases. However, few achieve ergonomic efficiency or convenient one-handed operation.

More patent information shown in the following link:

https://miro.com/app/board/uXjVJ1P08x0=?share_link_id=477530342912

Overall Summary

Across both survey and interview findings, two main directions emerged for improving the umbrella experience: enhancing hands-free ergonomics during use and addressing moisture management after use.

While umbrellas perform reliably at protecting users from rain, they fall short in the surrounding moments of interaction—how they are handled, carried, and stored once wet.

1. Hands-Free and Ergonomic Operation

Users frequently described frustration when trying to hold umbrellas while carrying other items, such as bags or phones. This points to an opportunity to make umbrella use more comfortable and balanced, reducing the physical and cognitive effort required during rain.

Potential directions include lightweight, counterbalanced frames, wearable supports, or stabilized handles that allow freer movement without sacrificing protection. Subtle ergonomic improvements could help the umbrella feel more natural and less burdensome in daily routines.

2. Post-Use Drying and In-Car Containment

An equally prominent issue arises once the umbrella has been used—managing moisture indoors or in vehicles. Users often face dripping, lack of clean storage, and difficulty keeping personal items dry. This suggests potential for compact drying or containment systems, such as retractable shells, absorbent linings, or in-car docking mechanisms that simplify transitions and reduce clutter.

Patent research reinforces these two areas of opportunity. While several existing designs address moisture capture or structural durability, few consider hands-free ergonomics or car-specific containment as integrated features. The current landscape remains fragmented, leaving room for innovation that brings together ease of use, compactness, and practical storage.

Running Shoes:

Survey Summary:

Survey Results:

1. Running Frequency

- Majority of respondents run a few times a week (20), followed by every day (9).

- Light or occasional runners are fewer (5).

2. Shoe Lifespan

- Most shoes last 3–12 months, with 6–12 months being slightly more common (13 respondents).
- A notable number of runners experience less than 3–6 months before needing replacement (17 respondents combined), indicating frequent wear.

3. Main Reason for Replacement

- The dominant reason is soles worn out (12), sometimes combined with upper fabric tearing or discomfort.
- Other minor reasons include style changes, recommended mileage limits, or increased injury risk.
- This shows that performance and durability are the primary drivers, rather than aesthetics.

4. Spending Habits

- Most runners spend \$100–150 per pair (24), with some paying more than \$150 (5).
- Few spend under \$100, indicating a willingness to invest in quality shoes.

5. Frustration Level

- Frustration ratings skew moderate to high, with 3 being most common (11), and 4–5 (high frustration) totaling 14 respondents.
- Indicates dissatisfaction with frequent replacements and potential pain points for innovation.

The survey revealed that most respondents run several times a week, with a smaller portion running daily or occasionally. Shoe lifespan varies, though the majority report needing replacements within 6–12 months, with a notable number experiencing wear in less than six months. The primary reason for replacement is worn-out soles, often accompanied by upper fabric issues or discomfort, while style changes and recommended mileage limits are minor factors. Respondents generally spend between \$100–150 per pair, with few spending less than \$100, indicating a willingness to invest in quality footwear. Frustration levels with frequent replacements skew moderate to high, highlighting recurring pain points among runners.

Insights & Opportunity Space:

1. High-frequency replacements are common
 - A significant portion of runners replace shoes within 6–12 months or less.
 - Opportunity to develop longer-lasting soles or modular/replaceable components to extend shoe lifespan.
2. Sole wear is the major pain point
 - Almost every respondent cited soles wearing out first.
 - Opportunity to explore durable materials, tread designs, or replaceable soles.
3. Willingness to pay for solutions

- Most runners already spend over \$100, suggesting they would invest in higher-quality or more durable solutions if value is clear.
- 4. Moderate-to-high frustration indicates motivation
 - Runners experience annoyance or disruption from frequent replacements.
 - Solutions that reduce the frequency or hassle of replacement (durable shoes, recyclable soles, or subscription-based sole replacement) could gain traction.
- 5. Variation in running frequency and usage patterns
 - Daily runners may need more robust, high-end solutions, while occasional runners might prefer affordable, maintenance-free options.
 - Opportunity to segment products based on running intensity.
- 6. Secondary pain points
 - Upper fabric tearing, discomfort, and injury risk are additional concerns.
 - Opportunity exists to combine sole durability with overall shoe longevity and ergonomics.

The findings point to several clear opportunities for innovation. Frequent replacements, often within 6–12 months, suggest a demand for longer-lasting soles or modular components that extend shoe life. Sole wear emerges as the primary pain point, presenting an opportunity for durable materials, improved tread designs, or replaceable soles. The majority of runners are willing to invest in high-quality solutions, and moderate-to-high frustration levels indicate strong motivation for products that reduce replacement frequency or hassle. Variation in running frequency implies potential for product segmentation, offering robust solutions for daily runners and more affordable, low-maintenance options for occasional users. Secondary concerns, including upper fabric durability, comfort, and injury prevention, further highlight opportunities to combine sole longevity with overall shoe performance and ergonomics.

Interview Summary:

Interview Results:

1. Shoe Lifespan & Durability

- All runners experience rapid shoe wear, especially in soles and cushioning.
- A recurring desire: shoes that last longer, with some interest in modular or repairable soles.
- Users are willing to pay more for durability or a solution that reduces frequent replacements.

2. Pain & Injury Prevention as a Driver

- Pain (feet, heel, ankle, IT band, knees) signals when to replace shoes.
- Injury prevention is a key motivator, especially for marathoners and serious runners.
- Even casual runners recognize that worn shoes can impact performance or health.

3. Cost & Value Trade-offs

- Shoes are perceived as expensive, and this sometimes delays purchases.

- Users are willing to spend more if the shoe lasts longer, offers proper support, or improves experience.
- Price sensitivity varies: casual runners are more flexible if the process or fit is improved; serious runners look for longevity and performance.

4. Research & Decision Friction

- Many find it challenging to decide on new shoes: too many options, biased online info, and time/effort to research.
- Social influence (friends, social media, influencers) often impacts choices.
- Some interest in tools or consultants to simplify the selection process.

5. Fit, Comfort, & Support

- Proper fit and cushioning are critical; preferences vary individually (arch height, heel support, mobility).
- Breaking in new shoes is a pain point; immediate comfort matters.
- Users value both performance and aesthetics (“shoes still need to look good”).

6. Sustainability & Reusability

- Across casual and marathoners, there’s a clear preference for reusable and sustainable shoes.
- Users are cautious about sustainable options if they compromise durability or functionality.

7. Convenience & Accessibility

- Accessing stores, trying on shoes, and waiting for shipping are challenges.
- Fast, easy, and trustworthy solutions are highly valued.
- There’s opportunity for mobile sizing, at-home fittings, or subscription/repair services.

Runners consistently face challenges with shoe lifespan and durability, particularly regarding soles and cushioning, creating strong interest in longer-lasting or modular solutions, with many willing to pay a premium for durability. Pain and injury prevention are key drivers for replacement decisions, as worn shoes can impact performance and health for both serious and casual runners. While shoes are often perceived as expensive, users are willing to invest more if the footwear offers proper support, longevity, or improved experience, though price sensitivity varies by runner type. Decision-making can be difficult due to overwhelming options, biased online information, and time-consuming research, with social influence playing a notable role, highlighting opportunities for tools or consulting to simplify selection. Fit, comfort, and support are critical, with immediate comfort and proper cushioning being especially important, alongside aesthetics. Sustainability and reusability are valued across all runners, provided they do not compromise durability or functionality. Convenience and accessibility also influence satisfaction, with challenges in store access, trying on shoes, and shipping creating opportunities for mobile sizing, at-home fittings, or subscription and repair services.

Insights & Opportunity Space:

Insights:

- Durability matters most: Frequent shoe wear, especially in soles and cushioning, drives dissatisfaction. Users want products that last longer and maintain performance.
- Pain and injury drive decisions: Both casual and serious runners replace shoes primarily to prevent pain and injury, not just aesthetics or novelty.
- Cost is a friction point: High shoe prices cause hesitation; users are willing to pay more for longevity or improved experience, but value must be clear.
- Research is time-consuming and confusing: Overwhelming options, biased online reviews, and effort required for research reduce satisfaction.
- Fit and comfort are highly individual: Shoe fit, support, and breaking-in time strongly influence adoption; one-size-fits-all solutions are not trusted.
- Sustainability is valued but secondary to performance: Users prefer reusable, durable, and eco-friendly designs as long as performance and aesthetics aren't compromised.
- Convenience drives adoption: Easy access, online sizing, quick delivery, or in-home fitting options would increase likelihood of adoption.
- Social influence shapes decisions: Friends, social media, and influencers guide choice, particularly for casual or novelty-driven runners.

Opportunities:

1. Durable & Modular Shoes – Create shoes with replaceable or repairable soles to extend lifespan.
2. Personalized Comfort & Fit Solutions – Adjustable arch, heel, and cushioning options, or custom-fit services.
3. Cost-Efficient Longevity – Offer premium shoes that reduce long-term replacement costs or frequency.
4. Simplified Decision Tools – Platforms or apps offering unbiased reviews, comparisons, and curated recommendations.
5. Sustainable Materials with Performance – Eco-friendly shoes that maintain durability, comfort, and aesthetics.
6. Convenience-Focused Services – At-home sizing, subscription models, or easy returns to reduce friction.
7. Health & Injury Tracking Integration – Features that monitor shoe wear, suggest replacement, or prevent injuries.
8. Social/Community Engagement – Leverage influencer, peer, and social media endorsements to encourage adoption.

The research indicates that durability is the most critical factor for runners, as frequent wear, particularly in soles and cushioning, drives dissatisfaction, and users seek products that maintain performance over time. Pain and injury prevention are primary motivators for replacement decisions, with both casual and serious runners prioritizing function over aesthetics. Cost remains a friction point, as high prices can delay purchases, though runners are willing to

invest in shoes that offer longevity or improved experience. Decision-making is often time-consuming and confusing due to overwhelming options and biased information, while fit, comfort, and breaking-in time are highly individual, making one-size-fits-all solutions ineffective. Sustainability is valued but secondary to performance, with users favoring durable and eco-friendly designs that do not compromise function or aesthetics. Convenience also plays a key role, as easy access, online sizing, quick delivery, or in-home fitting options increase adoption, and social influence, from friends, social media, and influencers, shapes choices, particularly for casual or novelty-driven runners. These insights suggest opportunities for innovation, including durable and modular shoes with replaceable soles, personalized fit and comfort solutions, cost-efficient longevity, simplified decision tools, sustainable materials that do not sacrifice performance, convenience-focused services such as subscription or at-home fittings, health and injury tracking integrations, and leveraging social and community engagement to encourage adoption.

Patent Research Summary:

MiroBoard: https://miro.com/app/board/uXjVJ21LFSI=

The patent research revealed three dominant categories of existing intellectual property within the running shoe innovation space: exchangeable soles, modular shoes, and innovative sole technologies. Exchangeable sole designs primarily focus on allowing users to detach and replace worn tread components, while modular shoe patents explore interchangeable parts such as midsoles or upper attachments. Innovative sole patents, on the other hand, emphasize advancements in cushioning, material composition, and adaptive flexibility. However, most of these patents were either international filings, inactive, or both, suggesting that while the concept of modularity in footwear has been explored, few viable or currently enforceable claims exist within the U.S. market.

This landscape analysis indicates a promising white space opportunity for a new, functionally distinct locking mechanism that enables reliable and user-friendly exchangeable soles. The lack of active, enforceable patents in this area means there is room for meaningful innovation and potential protection. Developing a mechanism that improves ease of use, maintains performance integrity, and ensures durability through repeated attachment cycles could fill a clear market and intellectual property gap, positioning the design as both novel and commercially viable within the current IP environment.

Overarching Summary:

Across both survey and interview research, runners consistently identified durability, comfort, and functionality as their most significant pain points. Most participants run several

times a week and replace their shoes every 6–12 months, often due to worn-out soles or diminished cushioning. These frequent replacements create frustration and financial strain, particularly given that most runners spend \$100–150 per pair. Pain and injury prevention, rather than style or novelty, drive replacement decisions, as runners recognize that deteriorated shoes affect both comfort and performance. While users are willing to pay more for longer-lasting products, they expect clear value and lasting performance in return. Decision-making around new shoes is complicated by biased reviews and overwhelming options, while individual differences in fit, comfort, and support further challenge satisfaction. Sustainability, though valued, remains secondary to performance, and convenience factors, such as fast delivery, at-home fittings, or simplified research tools, significantly influence adoption. Social influence also plays a key role, as friends, influencers, and online communities often shape consumer preferences.

Collectively, these insights reveal substantial opportunities for innovation in the running shoe market. There is strong demand for durable and modular footwear solutions that extend product lifespan and reduce replacement frequency, particularly through designs with replaceable soles or repairable components. Personalized comfort and fit systems, eco-conscious materials that maintain high performance, and digital tools that simplify the purchase process represent additional areas of opportunity. Patent research reinforces this potential, identifying three major categories of existing intellectual property, exchangeable soles, modular shoes, and innovative sole technologies, most of which are either expired or international. This suggests a feasible white space for a novel locking mechanism that securely attaches a fresh, exchangeable sole while maintaining comfort, stability, and performance. Together, the market feedback and IP landscape point toward a clear and viable design direction: a performance-oriented, modular running shoe that prioritizes durability, personalization, and user convenience.

Makeup With Vision Loss:

Survey Summary:

Survey Results:

1. Vision Type

- The majority of respondents had normal vision, and a few had mild vision loss
- Only two respondents reported having moderate vision loss

2. Tools and Setup

- The majority of respondents use their bathroom mirror
- Two identified using a magnified and/or lighted mirror
- Only one respondent uses phone camera or app

3. Challenges

- The main challenge indicated was needing to get too close to the mirror or straining their eyes
- Other challenges included being able to clearly see their face, dim lighting, and not being able to tell if makeup is even

- No one identified mirror positioning and holding as an issue

4. Feelings as a Result of the Challenges

- The majority of respondents feel “annoyed but can manage”
- One person feels frustrated, another feels anxious

5. Products to Help

- Almost all respondents did not know products exist to help ease makeup application with vision loss
- One person identified that they have not found a product that helps them

The survey indicates that the majority of respondents do not find this issue to be frustrating enough to cause them to want to make a change. However, this is indicative of the demographic that filled out the survey. Most people have normal vision or mild vision loss, so they do not face major struggles when applying their makeup. The survey was not a strong method to validate the issue. The woman that answered with moderate vision loss is older, indicating that the target audience may be people with moderate to more severe vision loss. Another drawback of the surveys is technical difficulty for older people. Grandmothers that were contacted did not fill out the survey; they preferred to talk over the phone instead.

Insights & Opportunity Space

Insights

- Problem severity correlates with vision loss: Users with mild vision issues can adapt easily, while those with moderate or severe vision loss face real barriers.
- Visual aids dominate current behavior: Mirrors and lighting are the only assistive tools most people use; tactile or tech solutions are nearly nonexistent.
- Eye strain and lighting issues are universal: Even mild-vision users mention eye fatigue and dim environments as recurring pain points.
- Low awareness of assistive solutions: Few users know that products exist for accessible makeup application, pointing to a major market education gap.
- Emotional frustration scales with severity: While most can “manage,” those with greater impairment experience stronger negative emotions like frustration and anxiety.
- Survey limitations reveal outreach barriers: Older adults with vision loss were less likely to complete online surveys, suggesting that research on this audience must rely on interviews or in-person methods.

Opportunities

1. Tactile & Accessible Makeup Solutions: Develop tools and packaging that rely on texture or shape for product identification, especially for moderate-to-severe vision loss.
2. Improved Lighting Solutions: Create adaptive or automatic lighting systems that reduce eye strain and enhance facial visibility for users with mild vision challenges.
3. Awareness & Education Campaigns: Increase visibility of accessible beauty tools through targeted marketing or partnerships with senior centers and vision loss organizations.

4. **Inclusive Product Ecosystem:** Design solutions that work across the vision-loss spectrum, from mild to complete blindness, so users can continue using the same tools as their vision changes.
5. **Alternative Research Channels:** Use phone interviews, assisted surveys, or in-person feedback sessions to reach elderly users who can't access online tools.

The survey reveals that while most users do not yet experience severe vision loss, the pain points of visibility, lighting, and confidence begin early and worsen over time. Awareness of accessible makeup solutions is minimal, signaling an opportunity for simple, tactile, and well-marketed products that empower users to apply makeup independently as their vision declines.

Interview Summary:

Interview Results:

1. Makeup Use & Frequency

- Makeup is only worn occasionally, typically for social outings rather than daily use.
- Individuals described makeup as a routine task rather than a creative or expressive one.
- Most perform their routine in the bathroom, where their items are already organized.
- Motivation to apply makeup has decreased as vision worsened, with some users saying it feels less “worth the effort” when they can't see results.

2. Vision Loss & Routine Adaptation

- As vision declines, individuals increasingly rely on intuition and memory to complete familiar steps.
- Steps that require color judgment, blending, or precision (like foundation or eyeshadow) are often abandoned entirely.
 - Or light enough where mistakes won't be noticeable
- No visual aids, such as magnifying mirrors or lighting, are effective once vision is mostly gone.
- Users develop personal, tactile techniques for basic steps like eyeliner (feeling the lash line) and lipstick (feeling lip edges).

3. Color & Product Identification Challenges

- The biggest frustration is the inability to identify makeup colors or distinguish between similar containers.
- Current packaging lacks any tactile differentiation, forcing users to rely on family or memory.
- Even users who have adapted to touch-based applications feel limited by color uncertainty, not knowing if makeup matches or looks blended.

4. Emotional & Social Factors

- Makeup no longer provides a strong emotional reward; it's described as routine rather than enjoyable.

- Still, users value maintaining a sense of normalcy and independence in doing it themselves.
- Family members (daughters) act as visual “quality control,” which can reinforce dependence.

5. Current Aids & Technological Gaps

- Visual assistive tools (e.g., smart mirrors, lighting) are irrelevant for those with advanced vision loss.
- No existing makeup tools provide tactile or auditory feedback for application or color recognition.
- Some users use AI phone apps that describe surroundings, but these are not optimized for cosmetics or small visual details.
- There’s a lack of beauty products designed for blind or low-vision consumers, existing accessibility focuses on other daily living needs.

6. Trust, Simplicity & Product Desires

- Users prefer non-visual, tactile solutions, something intuitive, reliable, and hard to lose.
- High-tech or complex systems are viewed as intimidating or impractical.
- Trust is earned through simplicity: if a product is easy to feel, identify, and use without error, it will be adopted.

Individuals with vision loss face significant barriers in maintaining independence during makeup application. While visual-assistive tools like mirrors and lighting fail to address their needs, users have adapted through tactile methods, relying on memory, touch, and family feedback. The main challenges lie in color identification, product differentiation, and confidence in the final result. Users value simplicity, tactile cues, and reliable design over visual solutions. Emotional engagement has shifted from self-expression to maintaining autonomy, suggesting a need for products that restore independence and confidence through non-visual feedback systems. The interview results are taken from two individuals with severe and moderate vision loss. Another interview was conducted with an individual with mild vision loss, but results indicated that this problem is not very relevant for her situation.

Insights & Opportunity Space

Insights

- Color is the core barrier: Vision loss makes color differentiation impossible; no tactile or auditory system currently bridges this gap.
- Routine replaces creativity: Makeup becomes mechanical rather than expressive once sight diminishes.
- Visual aids don’t help: Magnifiers and lighting are irrelevant when vision is fully lost; new solutions must rely on touch.
- Tactility drives confidence: Users trust physical cues more than digital or spoken ones.

- Dependence on others undermines independence: Relying on family members for feedback reduces satisfaction.
- Simplicity is key: Users value straightforward, low-maintenance designs that integrate naturally into daily life.
- Market gap in accessible beauty: Few brands or devices consider blind or low-vision consumers beyond Braille labeling.

Opportunities

1. Tactile Identification System: Develop a universal tactile labeling or texture language to differentiate makeup types and shades by touch.
2. Tactile Feedback Applicator: Create applicators (brushes, wands, or pens) that use shape or vibration to confirm correct use and placement.
3. Accessible Makeup Organizer: A modular, tactilely coded storage system that helps users locate and identify products without sight.
4. Audio/Tactile Smart Companion: A simple, portable device that identifies color verbally or through vibration patterns when scanning makeup products.
5. Inclusive Packaging Standards: Encourage cosmetic brands to adopt accessible, texture-based packaging as part of inclusive design.

For individuals with vision loss, makeup routines shift from an act of expression to a challenge of independence. The largest unmet need lies not in improved vision tools but in tactile communication and product identification. Opportunities exist in creating intuitive, touch-based systems that allow users to independently select and apply makeup with confidence, bridging a major accessibility gap in the beauty industry.

Patent Research Summary:

MiroBoard: https://miro.com/app/board/uXjVJ14eZtE=

Summary

Innovation in the beauty technology field includes an array of patents focusing on improving makeup application processes through non-visual and tactile aids, smart technologies, and enhancements for ease of use. Many existing solutions assist users by employing high-tech devices like smart mirrors and auto-adjusting applicators, while others focus on tactile guidance for those who are visually impaired or prefer non-visual techniques. Despite the abundance of innovations, a common workaround observed is users employing their own fingers and eyelids as guides.

Insight 1

Smart Mirrors and Personalized Feedback: Several patents highlight the creation of smart mirrors that go beyond simple reflection. These devices provide users with personalized recommendations by assessing facial symmetry, offering detailed feedback on cosmetic applications, and even displaying selected makeup looks over the reflection. This indicates a

significant trend towards integrating augmented reality into everyday beauty routines for a more tailored and guided experience.

Insight 2

Non-Visual Makeup Application Technologies: Certain patents focus on assisting users who either cannot or prefer not to rely on visual cues. This includes tools like eyeliner stencils with handles for control, mascara shields to prevent smudging, and guides for contouring. These devices highlight the creativity in providing solutions that cater to a broader audience who might face challenges with traditional makeup application methods.

Insight 3

Tactile and Motorized Innovation: The emergence of technologies offering tactile cues, such as products providing sensory feedback during application, shows a unique approach that merges traditional makeup practices with innovative sensory aids. Additionally, motorized and robotic technologies such as computerized makeup brushes and makeup application devices further diversify available solutions, suggesting a trend towards automation and ease in cosmetic application.

Recommendations

To remain competitive, it is critical to focus on tactile and non-visual solutions to fill a gap in the market where few patents currently exist. Developing user-friendly products that offer tactile feedback can broaden accessibility and appeal to a diverse audience. Moreover, leveraging smart technologies that offer personalized guidance could provide an edge in customization, meeting the growing demand for personally tailored beauty solutions.

Kitchen Sinks:

Kitchen Drain Hygiene: Research Compilation Survey Summary

1. Cleaning Frequency & Exposure

- 80% of respondents use their kitchen sink multiple times per day.
- 65% clean or empty their food catcher daily, with others doing so weekly or “every time they cook.”
- Nearly **70% reported needing to touch wet food waste directly** when cleaning.
- Average “**grossness**” rating: 8.7/10, indicating high discomfort and aversion.
- **Odor buildup, food sticking, and direct contact with debris** were the three most common complaints, cited by over 80% of users.

2. Perceived Hygiene and Task Unpleasantness

- 90% agreed or strongly agreed that cleaning felt unsanitary or unhygienic.
- 95% agreed or strongly agreed that they would prefer a way to empty the catcher without touching debris.
- 70% rated cleaning difficulty as “8 or higher” in unpleasantness.

3. Qualitative Notes from Responses

- Users frequently mentioned “*the smell*,” “*dripping on myself*,” and “*having to poke things out with paper towels*.”
- Many noted avoiding cleaning altogether until odor became unbearable.
- Respondents indicated willingness to pay **\$10–\$25** for a “premium hygienic” or “no-touch” catcher solution.

Summary:

Cleaning the drain catcher is perceived as one of the most unpleasant kitchen chores. Frequent exposure to decomposing food and contaminated water creates high discomfort, odor, and hygiene anxiety. The combination of daily contact, visible residue, and psychological disgust reinforces avoidance behaviors, suggesting strong market potential for a *contactless*, *self-cleaning*, or *antimicrobial* design.

Insights & Opportunity Space (Survey)

1. **Strong Market Validation**

Nearly all respondents desire a more hygienic, touch-free way to clean their sink strainer.

2. **Hygiene and Smell Dominate Pain Points**

Odor, bacterial buildup, and sticky residue were prioritized over drainage performance.

3. **Avoidance Behavior Creates Compounding Problems**

Many users delay cleaning for days or weeks, worsening bacterial growth.

4. **High Purchase Willingness**

Consumers are ready to pay up to \$25 for a reliable hygienic product.

Findings from the survey highlight a highly validated problem space. Almost all respondents indicated that cleaning the drain catcher is unpleasant and unhygienic, showing that the issue is not isolated but universal among kitchen users. The leading causes of dissatisfaction were odor, bacterial buildup, and the need for direct hand contact with decomposing food. The problem extends beyond inconvenience — it evokes a strong emotional response of disgust, which drives avoidance behavior and poor hygiene maintenance.

This behavioral insight opens an opportunity for design interventions that break the negative feedback loop between disgust and neglect. A strainer that enables touch-free emptying or self-flushing behavior would significantly reduce emotional resistance and increase cleaning frequency. The high willingness to pay further strengthens this case, suggesting that consumers

view this as a comfort and hygiene upgrade, not a luxury. In short, the survey establishes both the prevalence of the issue and the potential demand for a solution that combines sanitation, usability, and emotional relief.

Opportunity Statement:

There is a clear unmet need for a **touch-free, odor-resistant, and self-cleaning sink strainer** that minimizes user contact and addresses hygiene at the material and geometric levels.

Interview Summary

1. Routine and Context

- Most cook multiple times a week and rely on a strainer daily.
- Cleaning is delayed until odor or visible buildup becomes intolerable.
- Users use gloves, paper towels, or bleach soaks to avoid touching waste.

2. Emotional and Hygiene Perception

- Participants described cleaning as “*gross*,” “*resentful*,” and “*disgusting*.”
- Touching food waste rated **9–10/10** in unpleasantness.

3. Cleaning Process and Pain Points

- Dripping water, residue buildup, and mesh cleaning cited as top frustrations.
- Odor develops rapidly even with regular washing.

4. Desired Solutions

- **Contactless disposal** requested by all participants.
- Split preference between **reusable vs. disposable** approaches.
- Ideal price range: **\$20–\$25**.

Summary:

Emotional disgust, not mechanical complexity, is the primary driver of dissatisfaction. Current solutions focus on drainage, not user hygiene.

Insights & Opportunity Space (Interviews)

Three in-depth interviews provided qualitative insight into the lived experience behind these survey findings. Across all participants, themes of disgust, inconvenience, and hygiene anxiety emerged consistently. Each user described cleaning the drain catcher as one of the most dreaded tasks in the kitchen. Interviewees reported cooking frequently, often multiple times a week, but

delayed cleaning the strainer until odors became unbearable or food visibly accumulated. When cleaning, users resorted to awkward strategies such as using paper towels, forks, or gloves to dislodge debris. All described the act of touching food waste as “gross,” “resentful,” or “disgusting,” with perceived unpleasantness scoring between 9 and 10 on a ten-point scale.

Participants also noted issues beyond contact itself, such as dripping water when carrying the strainer to the trash, food residue that stuck even after rinsing, and unpleasant odors that persisted despite regular cleaning. Some felt that reusable products quickly became “permanently gross,” while others expressed a preference for single-use, disposable alternatives to avoid recontamination. Despite differences in sustainability preferences, all participants shared a common goal: to eliminate direct contact with food debris altogether. When asked what an ideal solution would look like, each respondent independently described a product that could be emptied without touching the waste or rinsing with hands. Most agreed that a retail price between \$20 and \$25 would feel reasonable for such a product, aligning with the cost of premium sink accessories currently available on the market.

Insight	Evidence	Design Opportunity
Touch Aversion	Users describe touching food waste as intolerable.	Create no-touch release (lever, magnet, or flip system).
Odor & Residue	Foul smell appears after 1–2 days.	Add antimicrobial coating and smooth surfaces .
Inconvenient Cleaning	Cleaning avoided until odor forces it.	Enable auto-flush / self-rinsing geometry .
Drip & Splash	Strainers drip when carried to trash.	Add containment shell or base tray .
Reuse vs. Disposal Split	Half want reusable, half want disposable.	Offer modular reusable base + compostable liner .

These interviews validate and humanize the quantitative data. Users are not frustrated because cleaning takes time — they are frustrated because it feels unhygienic and emotionally uncomfortable. The strong disgust reaction causes avoidance behavior, leading to odor buildup, microbial growth, and even more aversion in a self-reinforcing cycle. This emotional element is critical: it shows that the core design challenge is not purely mechanical but psychological. To be successful, a new product must resolve both the tactile problem of hand contact and the emotional discomfort associated with perceived uncleanness.

Additional insights emerged around odor and moisture management. Users disliked handling dripping strainers and described carrying them to the trash as “a balancing act.” There was also a clear divide between those who valued reusability for environmental reasons and those who preferred single-use disposability for hygiene. This suggests an opportunity for a hybrid approach — for instance, a durable, reusable outer shell paired with a thin, compostable liner that captures waste and can be discarded without direct contact. Such a design could balance sustainability and hygiene while meeting both user priorities.

Patent Research Summary

https://miro.com/app/board/uXjVJ1F2XEs=?share_link_id=963913344433

Key Patent Clusters and Gaps

Group	Focus	Opportunity Gap
1. Quick-Release Mechanisms	Push-button or flip strainers	Manual removal still required; poor drip control.
2. Modular Inserts	Replaceable liners and cups	No touch-free option; fragile seals.
3. Material Hygiene	Antimicrobial metals & coatings	Minimal adoption; long-term performance untested.
4. Hydrodynamic Drainage	Flow optimization & self-rinse geometry	Lacks bacterial and odor control integration.
5. Odor Containment	Rubber seals & flow traps	Rare in strainers; used mainly in stoppers.
6. Historical Prior Art	Foundational “basket + holes” designs	No major innovation since early 1900s.

Patent Landscape Insight:

The drain strainer market is technically saturated but **functionally stagnant**.

Existing designs focus on *flow mechanics* and *fit*, not *user hygiene*.

This gap defines a promising space for **touch-free, antimicrobial, odor-controlling systems**.

A review of eighteen relevant patents revealed that the sink strainer market is well established yet technologically stagnant. The majority of existing patents address drainage efficiency, flow control, and ease of installation rather than hygiene or user experience. Historically, strainers have evolved from rigid metal baskets (early 1900s) to modern plastic and silicone designs (2000s), but none have fundamentally changed how users interact with food debris.

Several recent patents, such as **US 11,680,397 B2** and **US 8,763,170 B2**, explore mechanisms for easier removal, including push-button releases or scraper arms, but these still require the user to handle the contaminated component directly. Others, such as **US 7,442,296 B2** and **US 7,736,499 B2**, propose disposable or replaceable inserts, offering partial hygiene improvement but at the cost of increased waste and limited availability. A few, such as **EP 2060686 A2**, consider hydrodynamic self-cleaning through angled or spiral channels, but these remain uncommon in commercial products and do not prevent bacterial growth or odor. Meanwhile, older prior art like **US 3,308,484 A** and **US 2,890,463 A** reinforce how little functional change has occurred since the mid-twentieth century.

Across all patent categories, no design successfully integrates hygiene-first functionality, antimicrobial materials, or truly touchless operation. Most products still rely on manual inversion or rinsing, leaving the core emotional barrier — contact with decomposing food — unresolved. The patent landscape thus reveals both maturity and opportunity: while mechanical solutions are saturated, the hygienic user experience remains an open and underdeveloped frontier.

Overarching Summary

Across surveys, interviews, and patent research, the data converge clearly:

- **90%+ of users** report discomfort and hygiene concerns.
- **95%+ desire a touchless or self-cleaning** alternative.
- **No current patent or product** provides a holistic solution combining hygiene, odor prevention, and ease of cleaning.

Unmet Need:

A hygienic, touch-free, and odor-controlling kitchen drain strainer that eliminates the need for direct contact, reduces bacterial risk, and transforms one of the dirtiest kitchen chores into a seamless experience.

Conclusion

The combined survey, interview, and patent data converge on a single conclusion: the act of cleaning a kitchen drain strainer is universally disliked, emotionally unpleasant, and technologically underserved. Users experience strong feelings of disgust and hygiene anxiety, and despite incremental innovations in materials and mechanisms, no product meaningfully addresses the psychological discomfort of touching food waste. Both qualitative and quantitative findings indicate high willingness to adopt a new solution, particularly one that provides a touchless, odor-resistant, and easily maintainable experience.

The existing patent corpus confirms a century of incremental improvements but little true innovation. The absence of touch-free disposal, antimicrobial surfaces, and self-rinsing

geometries leaves clear white space in an otherwise mature market. A product that unites these elements — offering effortless cleaning without physical contact — would represent a generational advancement in kitchen hygiene.

In conclusion, there is a clear unmet need for a **hygienic, touch-free kitchen drain strainer** that redefines cleanliness and comfort in everyday use. By integrating self-rinsing geometry, antimicrobial or hydrophobic materials, and an intuitive ejection mechanism, the opportunity exists to transform one of the least pleasant household tasks into an effortless and sanitary routine. This innovation would not only solve a persistent functional problem but also deliver meaningful psychological relief to users, filling a gap that has remained open for more than a hundred years.

Appendix B: Patent Research

Umbrellas:

CLUSTER 01: ERGONOMIC / HANDS-FREE UMBRELLA				CLUSTER 02: UMBRELLA DRYING / CONTAINMENT			
IMAGE	PATENT #	STATUS	DESC.	IMAGE	PATENT #	STATUS	DESC.
	US20150000001	Active	Standard umbrella		US20150000002	Active	Umbrella with drying mechanism
	US20150000003	Expired	Umbrella with carrying bag		US20150000004	Expired	Umbrella with drying mechanism
	US20150000005	Active	Umbrella with carrying bag and drying mechanism		US20150000006	Active	Umbrella with drying mechanism
	US20150000007	Expired	Umbrella with carrying bag		US20150000008	Expired	Umbrella with drying mechanism
	US20150000009	Active	Umbrella with carrying bag and drying mechanism		US20150000010	Expired	Umbrella with drying mechanism
	US20150000011	Active	Umbrella with carrying bag and drying mechanism		US20150000012	Expired	Umbrella with drying mechanism
	US20150000013	Expired	Umbrella with carrying bag		US20150000014	Expired	Umbrella with drying mechanism
	US20150000015	Active	Umbrella with carrying bag and drying mechanism		US20150000016	Expired	Umbrella with drying mechanism
	US20150000017	Expired	Umbrella with carrying bag		US20150000018	Expired	Umbrella with drying mechanism
	US20150000019	Active	Umbrella with carrying bag and drying mechanism		US20150000020	Expired	Umbrella with drying mechanism
	US20150000021	Expired	Umbrella with carrying bag		US20150000022	Expired	Umbrella with drying mechanism
	US20150000023	Active	Umbrella with carrying bag and drying mechanism		US20150000024	Expired	Umbrella with drying mechanism
	US20150000025	Expired	Umbrella with carrying bag		US20150000026	Expired	Umbrella with drying mechanism
	US20150000027	Active	Umbrella with carrying bag and drying mechanism		US20150000028	Expired	Umbrella with drying mechanism
	US20150000029	Expired	Umbrella with carrying bag		US20150000030	Expired	Umbrella with drying mechanism
	US20150000031	Active	Umbrella with carrying bag and drying mechanism		US20150000032	Expired	Umbrella with drying mechanism
	US20150000033	Expired	Umbrella with carrying bag		US20150000034	Expired	Umbrella with drying mechanism
	US20150000035	Active	Umbrella with carrying bag and drying mechanism		US20150000036	Expired	Umbrella with drying mechanism
	US20150000037	Expired	Umbrella with carrying bag		US20150000038	Expired	Umbrella with drying mechanism
	US20150000039	Active	Umbrella with carrying bag and drying mechanism		US20150000040	Expired	Umbrella with drying mechanism
	US20150000041	Expired	Umbrella with carrying bag		US20150000042	Expired	Umbrella with drying mechanism
	US20150000043	Active	Umbrella with carrying bag and drying mechanism		US20150000044	Expired	Umbrella with drying mechanism
	US20150000045	Expired	Umbrella with carrying bag		US20150000046	Expired	Umbrella with drying mechanism
	US20150000047	Active	Umbrella with carrying bag and drying mechanism		US20150000048	Expired	Umbrella with drying mechanism
	US20150000049	Expired	Umbrella with carrying bag		US20150000050	Expired	Umbrella with drying mechanism
	US20150000051	Active	Umbrella with carrying bag and drying mechanism		US20150000052	Expired	Umbrella with drying mechanism
	US20150000053	Expired	Umbrella with carrying bag		US20150000054	Expired	Umbrella with drying mechanism
	US20150000055	Active	Umbrella with carrying bag and drying mechanism		US20150000056	Expired	Umbrella with drying mechanism
	US20150000057	Expired	Umbrella with carrying bag		US20150000058	Expired	Umbrella with drying mechanism
	US20150000059	Active	Umbrella with carrying bag and drying mechanism		US20150000060	Expired	Umbrella with drying mechanism
	US20150000061	Expired	Umbrella with carrying bag		US20150000062	Expired	Umbrella with drying mechanism
	US20150000063	Active	Umbrella with carrying bag and drying mechanism		US20150000064	Expired	Umbrella with drying mechanism
	US20150000065	Expired	Umbrella with carrying bag		US20150000066	Expired	Umbrella with drying mechanism
	US20150000067	Active	Umbrella with carrying bag and drying mechanism		US20150000068	Expired	Umbrella with drying mechanism
	US20150000069	Expired	Umbrella with carrying bag		US20150000070	Expired	Umbrella with drying mechanism
	US20150000071	Active	Umbrella with carrying bag and drying mechanism		US20150000072	Expired	Umbrella with drying mechanism
	US20150000073	Expired	Umbrella with carrying bag		US20150000074	Expired	Umbrella with drying mechanism
	US20150000075	Active	Umbrella with carrying bag and drying mechanism		US20150000076	Expired	Umbrella with drying mechanism
	US20150000077	Expired	Umbrella with carrying bag		US20150000078	Expired	Umbrella with drying mechanism
	US20150000079	Active	Umbrella with carrying bag and drying mechanism		US20150000080	Expired	Umbrella with drying mechanism
	US20150000081	Expired	Umbrella with carrying bag		US20150000082	Expired	Umbrella with drying mechanism
	US20150000083	Active	Umbrella with carrying bag and drying mechanism		US20150000084	Expired	Umbrella with drying mechanism
	US20150000085	Expired	Umbrella with carrying bag		US20150000086	Expired	Umbrella with drying mechanism
	US20150000087	Active	Umbrella with carrying bag and drying mechanism		US20150000088	Expired	Umbrella with drying mechanism
	US20150000089	Expired	Umbrella with carrying bag		US20150000090	Expired	Umbrella with drying mechanism
	US20150000091	Active	Umbrella with carrying bag and drying mechanism		US20150000092	Expired	Umbrella with drying mechanism
	US20150000093	Expired	Umbrella with carrying bag		US20150000094	Expired	Umbrella with drying mechanism
	US20150000095	Active	Umbrella with carrying bag and drying mechanism		US20150000096	Expired	Umbrella with drying mechanism
	US20150000097	Expired	Umbrella with carrying bag		US20150000098	Expired	Umbrella with drying mechanism
	US20150000099	Active	Umbrella with carrying bag and drying mechanism		US20150000100	Expired	Umbrella with drying mechanism

Running Shoes:

Innovative Soles

Diagram illustrating various innovative sole structures for running shoes, including different patterns, materials, and configurations.

Modular Soles

Diagram illustrating modular sole structures for running shoes, showing how different components can be combined to create a custom sole.

Modular Shoes

Diagram illustrating modular shoe structures for running shoes, showing how different components can be combined to create a custom shoe.

Problem - Runner's Shoes Wearing Down Too Quickly
Whisper - Exchangeable Sole With New Technical Locking Function To Shoe

Makeup With Vision Loss:

Patents Grouped by Theme

Minors/Touch

These patents are relevant, but do not address the direct need. Most inventors focus on workarounds, using their own eyes, fingers as primary guide.

Transfer/Strapping

These patents are more relevant, as they do not rely on visual cues to apply makeup. Functions do exist within the space although they are not as easy to use to produce a competitive alternative solution.

Transfer/Strapping

These patents are relevant. Few patents exist within the specific space, which is one we intend to pursue. Many solutions are workarounds, using their own eyes, fingers as primary guide.

Kitchen Sinks:

Group 1: Flip / Invert / Flexible Strainers (Manual Ejection)

Core theme: Allow user to invert or flip the basket to release debris quickly — partial hygiene improvement, but still manual contact.

Group 2: Push-Button / Ejection & Stopper Mechanisms

Core theme: Integrate mechanical buttons, toggles, or collapsing geometries to open or eject the strainer.

Group 3: Disposable / Detachable Strainers & Liners

Core theme: Avoid touching debris by discarding or replacing the dirty portion.

Group 4: Flow / Geometry Optimization & Anti-Clog Concepts

Core theme: Maintain drainage even when debris accumulates — indirectly improves hygiene by reducing standing waste.

Group 5: Scraper / Wiper / Cleaning-Assist Mechanisms

Core theme: Integrate wiping or scraping features to remove residue or hair — cross-applicable from bathroom drains.

Group 6: Foundational / Historical Prior Art

Core theme: Establishes baseline "strain + removable basket" concepts for novelty analysis.

Appendix C: Umbrella Use Interview Guide

Problem Space: Challenges with umbrellas when hands are occupied or items get wet

Section 1: Context & Routine (2–3 min)

Tell me about a typical rainy day — what are you usually carrying or doing?

How do you decide whether to bring an umbrella?

Where do you usually keep it when not in use?

Section 2: Experience & Pain Points (5–6 min)

Can you recall a time when using an umbrella was frustrating or inconvenient?

What happens when you go indoors or get into a car with a wet umbrella?

Have you ever avoided bringing one because it felt like too much of a hassle?

Section 3: Emotions & Perceptions (3–4 min)

How do you feel about carrying an umbrella — prepared, burdened, or somewhere in between?

How much does its appearance or design matter to you?

When it rains, do you ever prefer to just get a little wet rather than use one?

Section 4: Design Discovery (5 min)

What would make using or carrying an umbrella feel simpler or more natural?

What parts of the experience (opening, closing, storing, drying) would you most want to improve?

If you could change one thing about how umbrellas fit into your routine, what would it be?

Quick Ratings

How inconvenient is using an umbrella for you? (1 = not inconvenient, 10 = extremely inconvenient)

How likely are you to purchase a new umbrella if it clearly improves ease or comfort? (1 = not likely, 10 = extremely likely)

Appendix D: Umbrella Use Interview Notes

Interviewee 1

Section 1: Context & Routine (2–3 min)

Tell me about a typical rainy day — what are you usually carrying or doing?

- Wear waterproof shoes
- Want big umbrella, don't want to get wet, small umbrella affected by wind easily
- Bring plastic bags for umbrella contain, or bag has wet umbrella compartment
- Like unfoldable, cane-like umbrella because big. Also act as cane.

How do you decide whether to bring an umbrella?

- Weather forecast
- Bring light foldable umbrella when light rain

Where do you usually keep it when not in use?

- Balcony to dry / special umbrella basket for big, drawer for small.

Section 2: Experience & Pain Points (5–6 min)

Can you recall a time when using an umbrella was frustrating or inconvenient?

- When carrying a lot of stuff, hands occupied
- When wind extremely big, wrist force not enough
- When carrying small child

What happens when you go indoors or get into a car with a wet umbrella?

- Fold, there will be lots of moisture.
- When car, put in passenger seat or backseat
- Shopping malls have automatic plastic bag sealers
- When at home, shake before going in and bring to balcony as fast as possible

Have you ever avoided bringing one because it felt like too much of a hassle?

- No, don't want to get wet.

Section 3: Emotions & Perceptions (3–4 min)

How do you feel about carrying an umbrella — prepared, burdened, or somewhere in between?

- Feel safe, willing to carry

How much does its appearance or design matter to you?

- UV umbrella care about appearance, needs to go together with fit
- Rain umbrella not matter as much, want bright color for safety. More recognizable by cars, traffic, etc. in rainy / dark environment

When it rains, do you ever prefer to just get a little wet rather than use one?

- No. Acid rain, bad for health.

Section 4: Design Discovery (5 min)

What would make using or carrying an umbrella feel simpler or more natural?

- Biggest concern is when going into car, nowhere to put umbrella, always leave moisture. Wish there was instant dry cover

What parts of the experience (opening, closing, storing, drying) would you most want to improve?

- Drying

If you could change one thing about how umbrellas fit into your routine, what would it be?

- When going from outside to inside, immediately get rid of moisture

Quick Ratings

How inconvenient is using an umbrella for you? (1 = not inconvenient, 10 = extremely inconvenient)

- 6

How likely are you to purchase a new umbrella if it clearly improves ease or comfort? (1 = not likely, 10 = extremely likely)

- 10

Interviewee 2

Section 1: Context & Routine (2–3 min)

Tell me about a typical rainy day — what are you usually carrying or doing?

- only go out in rain for meetings, try not to go out

How do you decide whether to bring an umbrella?

- when no car

Where do you usually keep it when not in use?

- in car or in home basket

Section 2: Experience & Pain Points (5–6 min)

Can you recall a time when using an umbrella was frustrating or inconvenient?

- windy
- hands occupied when buying things

What happens when you go indoors or get into a car with a wet umbrella?

- just retract
- lower body still wet often
- Annoying not elegant

Have you ever avoided bringing one because it felt like too much of a hassle?

- yes

Section 3: Emotions & Perceptions (3–4 min)

How do you feel about carrying an umbrella — prepared, burdened, or somewhere in between?

- uncomfortable
- don't have a choice

How much does its appearance or design matter to you?

- not important

When it rains, do you ever prefer to just get a little wet rather than use one?

- afraid of getting sick, dirty rain
- still use umbrella

Section 4: Design Discovery (5 min)

What would make using or carrying an umbrella feel simpler or more natural?

- don't use hands
- coverage
- springs auto retract and open

What parts of the experience (opening, closing, storing, drying) would you most want to improve?

- closing
- wet

If you could change one thing about how umbrellas fit into your routine, what would it be?

- open and close is inevitable
- but storing and drying can improve

Quick Ratings

How inconvenient is using an umbrella for you? (1 = not inconvenient, 10 = extremely inconvenient)

- 5

How likely are you to purchase a new umbrella if it clearly improves ease or comfort? (1 = not likely, 10 = extremely likely)

- 10

Interviewee 3

Section 1: Context & Routine (2–3 min)

Tell me about a typical rainy day — what are you usually carrying or doing?

- Very lazy person, won't go out in rain if not necessary
- Sometimes walk out of class and there is unexpected rain. Don't have long enough to wait till rain stops
- Playing golf, rain

How do you decide whether to bring an umbrella?

- Almost never bring umbrella, honestly don't have one
- Have one on car that never bring out

Where do you usually keep it when not in use?

- Car
- Don't wanna keep it just never took out

Section 2: Experience & Pain Points (5–6 min)

Can you recall a time when using an umbrella was frustrating or inconvenient?

- Get out of car, have to open up in weird way, have to get a little wet
- Sharing umbrella is hard.

What happens when you go indoors or get into a car with a wet umbrella?

- Put beneath legs on car.
- Indoor, if going to public area dont mind
- Own place dont wanna bring in, keep outside until dry

Have you ever avoided bringing one because it felt like too much of a hassle?

- Yes

Section 3: Emotions & Perceptions (3–4 min)

How do you feel about carrying an umbrella — prepared, burdened, or somewhere in between?

- burdensome

How much does its appearance or design matter to you?

- Decent amount. Dont wanna look immature. In public area.

When it rains, do you ever prefer to just get a little wet rather than use one?

- Yes

Section 4: Design Discovery (5 min)

What would make using or carrying an umbrella feel simpler or more natural?

- If it can be dry when decided to store away, majority of time have to hold because wet, can't be put in bag with computers etc. If can be smaller too.

What parts of the experience (opening, closing, storing, drying) would you most want to improve?

- Drying, Storing.

If you could change one thing about how umbrellas fit into your routine, what would it be?

- Not part of routine now
- Would use when going out on date

Quick Ratings

How inconvenient is using an umbrella for you? (1 = not inconvenient, 10 = extremely inconvenient)

- 8

How likely are you to purchase a new umbrella if it clearly improves ease or comfort? (1 = not likely, 10 = extremely likely)

- 5

Appendix E: Umbrella Survey Results

Timestamp	Email Address	How often do you use your umbrella when it rains?	What do you usually use instead of an umbrella when it rains?	In what situations do you find umbrellas most difficult or inconvenient to use?	When it's raining and you need both hands free, what do you usually do?	On a scale of 1-5, how satisfied are you with your umbrella in general? (1 - I hate it, 5 - Very satisfied)
10/16/2025 10:29:52	xinyiwang2027@u.northwestern.edu	Always	Hoodie, I just get a little wet / don't mind	When carrying other items (e.g. coffee, phone), When walking in crowds	Look for cover (e.g. building)	3
10/16/2025 10:32:46	ericji2027@u.northwestern.edu	Always	Raincoat or waterproof jacket, Hoodie, I just get a little wet / don't mind	When carrying other items (e.g. coffee, phone), When entering and exiting a car, When walking in crowds, When it's windy.	Look for cover (e.g. building)	4
10/16/2025 10:35:24	fsy.joshua0@gmail.com	Always	Raincoat or waterproof jacket	When carrying other items (e.g. coffee, phone)	Look for cover (e.g. building)	4
10/16/2025 11:13:04	jerryxiang2027@u.northwestern.edu	I don't go out in the rain	I just get a little wet / don't mind	When carrying other items (e.g. coffee, phone), When entering and exiting a car, When walking in crowds, When storing it indoors after use	Try to juggle everything while holding the umbrella	4
10/16/2025 11:34:13	krystalxu2027@u.northwestern.edu	Always	Hoodie	When carrying other items (e.g. coffee, phone), When entering and	Look for cover (e.g. building)	4

				exiting a car, When storing it indoors after use		
10/16/2025 11:46:17	chelseytao 2027@u.n orthweste n.edu	Always	Hoodie, I just get a little wet / don't mind	When carrying other items (e.g. coffee, phone), When storing it indoors after use	Try to juggle everything while holding the umbrella	3
10/16/2025 13:14:27	amychen2 028@u.no rthwestern. edu	Always	Raincoat or waterproof jacket, Hoodie, Hat or cap	When carrying other items (e.g. coffee, phone), When entering and exiting a car, When storing it indoors after use	Try to juggle everything while holding the umbrella	3
10/16/2025 14:29:33	anthonyhu ang2028@ u.northwes tern.edu	Never	Raincoat or waterproof jacket, Hoodie, I just get a little wet / don't mind	When carrying other items (e.g. coffee, phone), When entering and exiting a car, When walking in crowds	Look for cover (e.g. building)	2
10/16/2025 15:01:04	kathyzhou 2026@u.n orthweste n.edu	Only when the rain is heavy	Raincoat or waterproof jacket, Hoodie, Hat or cap, I just get a little wet / don't mind	When carrying other items (e.g. coffee, phone), When entering and exiting a car, When storing it indoors after use	Get wet and skip the umbrella	2
10/16/2025 16:34:14	amberlueth 2026@u.n orthweste n.edu	Only when the rain is heavy	Hoodie, I just get a little wet / don't mind	When carrying other items (e.g. coffee, phone), When entering and exiting a car	Try to juggle everything while holding the umbrella	4
10/16/2025 19:27:34	caihe2027 @u.north western.ed	Always	Raincoat or waterproof jacket, Hoodie	When carrying other items (e.g. coffee,	Try to juggle everything while holding the	3

	u			phone), When entering and exiting a car, When storing it indoors after use, heavy in bag	umbrella	
10/18/2025 22:26:45	jiangm@u. northweste rn.edu	Always	Hoodie	When carrying other items (e.g. coffee, phone), When entering and exiting a car, When storing it indoors after use	Try to juggle everything while holding the umbrella	5

Appendix F: AI Generated Patent Research

1) Expanded US patent table — mechanical / physical umbrella moisture-management (representative)

Notes: I filtered for mechanical, drainage, inverted/reverse folding, handle reservoirs, dryers and closure kinematics — excluding coatings/chemistry patents. For each entry I include patent number, short title, short description of the mechanical feature, and a link reference to the source I used.

Patent (US)	Short title / focus	Key mechanical feature (why relevant)	Source
US 9,486,044 B1	Reverse folding umbrella	Reverse/inverted closing kinematics — wet surface folded inward to contain water. Useful baseline for reverse-fold mechanisms.	Google Patents
US 9,322,593 B2	Umbrella draining assembly	Dedicated draining/containment cistern + centrifugal/fling removal concept for umbrellas carried indoors — shows device-level draining approaches.	Google Patents
US 7,484,518 B2	Reversible umbrella	Slidable canopy assembly on pole enabling reversible/inward folding — early utility on mechanical reverse canopy.	Google Patents
US 2011/015467 7 A1	Ringed umbrella dryer (published app)	A ring/bristle dryer that removes water from an unfurled umbrella (building/stand device) — useful to compare public-infrastructure solutions.	Unified Patents
US 2013/009220 6 A1	Umbrella mechanism and methods of use	Kinematic rib arrangements where ribs open vertically before horizontally — reduces contact during opening/closing (kinematic control).	Google Patents

US 2008/016390 7 A1	Inverted collapsible umbrella	Collapsible umbrella configurable to close opposite direction — demonstrates claim language for inverted closure.	freepatentsonline.com
US 2010/007805 1 A1	Liquid-squirting umbrella apparatus	Hollow shaft with handle reservoir and conduit — example of integrating a handle reservoir and internal plumbing (mechanical routing).	google.com
US 2008/000631 3 A1	Stick umbrella structure for retracting rib units in a reverse direction	Structural folding path for reverse-direction folding — shows mechanical rib/runner approaches.	Justia Patents
US 5,143,107	Raining umbrella (reservoir + flow)	Early example where a reservoir at top and perforated tubes were used (reverse idea: routing liquids inside umbrella structure).	Justia Patents
US 2014/028388 8 A1	Umbrella anti-inversion apparatus	Removable cuff and straps to prevent inversion; useful for ensuring structural integrity under wind while adding drainage features.	Justia Patents

(These are representative — there are many additional US families that modify ribs, runners, and handle shapes. I can expand this to 30–50 filings and export CSV if you want.)

Five most load-bearing sources cited above: US9486044B1; US9322593B2; US7484518B2; US20110154677A1; US20130092206A1. [Google Patents+4Google Patents+4Google Patents+4](#)

2) Example independent + dependent claim drafts (mechanical focus only)

Below are *draft* claim templates (engineering language, mechanical-only). Use them as starting points for provisional drafting — they aim to claim structure + measurable performance features.

Claim 1 (independent — reverse closure + reservoir)

1. An umbrella comprising:
 - a central shaft;
 - a canopy supported by a plurality of ribs connected to a runner slidably mounted on the shaft;
 - a sliding outer sleeve operable to move the canopy from an open state to a closed state, wherein motion of the sleeve causes the canopy to invert such that an outer canopy surface faces a sleeve-interior when in the closed state; and
 - a water collection reservoir disposed within a handle of the shaft, the reservoir in fluid communication with at least one channel formed in one or more ribs such that water routed from the canopy during closure flows into the reservoir; wherein, when the canopy is moved from the open state to the closed state by actuation of the sleeve, total water collected in the reservoir or otherwise removed from the canopy is ≤ 5 mL as measured by the residual drip volume test.

Dependent examples

2. The umbrella of claim 1, wherein the reservoir has a capacity of ≥ 50 mL and a removable plug for drainage.
3. The umbrella of claim 1, wherein the runner motion forces the ribs to follow a fold path that maintains an outer canopy angle toward the handle of $\leq 30^\circ$ during the entire closure.
4. The umbrella of claim 1, wherein channels in the ribs are integrally formed as hollow rib members and terminate at the handle reservoir opening.
5. The umbrella of claim 1, wherein the canopy and sleeve reach the closed state in ≤ 1.5 s from initiation of closure.

Note: these examples aim to combine mechanical structure and measurable performance targets — useful for establishing inventive step (improvement) over earlier reverse umbrella filings. Don't copy verbatim into filing — have your patent counsel formalize.

3) White-space & patentable opportunities (mechanical only)

Concrete ideas I see as under-covered or defensible mechanically (subject to claim searches):

1. **Sequential fold runner that enforces containment angle** — a runner with cam surfaces that ensure the outer canopy never exceeds X° toward the user during closure; claimable as kinematic constraint + performance metric.
2. **Hinged gutter ribs that open upon closure to route water into handle** — ribs that articulate open channels only during closure, then collapse for compactness (hard to design-around).

3. **Modular handle reservoir with one-way valve and removable cartridge** — mechanical-only (no chemistry) replaceable cartridge that traps water and is washable.
4. **Single-actuation entry closure that combines runner + handle latch for car-entry** — mechanical latch that prevents closure unless handle is inside car threshold, ensuring contained closure in small spaces.
5. **User-operated squeeze drainage (mechanical pump)** — squeeze the handle to push reservoir water out into a detachable cup — a mechanical dispensing/draining mechanism.

4) Risk assessment & mitigation

- **Risk:** Existing US patents on reverse closure/runners may have broad claims.
 - *Mitigation:* Draft claims on combination features + measurable performance (e.g., handle reservoir + channel routing + residual drip metric) and design minor mechanical differences (fold path, channel termination) to avoid direct overlap. Perform claim-by-claim charting.
- **Risk:** Chinese/foreign utility models may block manufacturing in China.
 - *Mitigation:* Use alternate manufacturing locations or negotiate manufacture rights; do CN model searches early if you plan China fab.
- **Risk:** User acceptance / perceived complexity or weight.
 - *Mitigation:* Limit added mass, use ergonomic handles, ensure one-hand operation.

Patent (Year)	Name	Prior Art / Context (plain English)	Key Claims (moisture angle)	Status*
US 8,667,975 (2014)	Umbrella for dripless closure	Cites earlier “inverted”/containment ideas (e.g., Weiss et al.) to keep water inboard while closing. (Justia Patents)	Two-canopy structure + closure mechanism that inverts fold so wet surface drains inward into a handle cavity ; water retained until user empties. (Justia Patents)	Likely active (filed 2012; est. expiry ~2032 unless lapsed).

US 9,486,044 B1 (2016)	Inverted close umbrella	Continuation of “reverse/inverted” family improving driveline and folding geometry for water containment.	Reverse-folding canopy so wet side stays inside upon closure, minimizing drips on user/car/interiors.	Likely active (filed ~2014–2016 ; est. expiry ~2034–2036 unless lapsed).
US 7,484,518 B2 (2009)	Reversible umbrella (inward folding)	Builds on prior dripless/inverted patterns; earlier take on inward fold to mitigate dripping.	Inward-fold mechanism that encloses wet canopy surfaces to reduce external dripping after use.	Possibly active (priority circa 2006 → est. expiry ~2026 if fees paid).
US 8,910,392 (2014)	Dewatering machine for umbrella	Notes problems with sleeves/bags; improves centrifugal drying machines. (Justia Patents)	Centrifugal “spinner” appliance with water-receiving box to rapidly expel water from wet umbrellas in public venues. (Justia Patents)	Likely active (filed 2010; est. expiry ~2030 unless lapsed).
US 5,495,866 (1996)	Umbrella drip water collector	References older ferrule-mounted collectors; generalizes to retrofit devices. (Google Patents)	Ferrule-mounted receptacle + sponge that collects drips and can be squeezed to expel water safely. (Google Patents)	Expired (1996 issue).
US 5,381,812 (1995)	Rainwater collecting device for umbrella	Improves on earlier collars/sleeves by guiding water into a vessel. (Google Patents)	Collector near canopy tip routes runoff into a container to stop floor dripping after closure. (Google Patents)	Expired (1995 issue).

US 5,323,802 (1994)	Umbrella bag	Prior sleeves leaked or were flimsy; proposes more robust containment. (Google Patents)	Sleeve/bag for wet umbrellas designed to retain water and vent to dry; improves carry comfort. (Google Patents)	Expired (1994 issue).
US 5,625,960 (1997)	Umbrella dryer	Precedes today's countertop/public units; addresses slippery floors. (Google Patents)	Drying apparatus for umbrellas to remove moisture before entering interiors. (Google Patents)	Expired (1997 issue).
US 2013/012657 1 A1 (2013)	Bag for umbrella (application)	Acknowledges flimsy sleeves; proposes fashion-grade sleeve with drain seal. Status: Abandoned. (Google Patents)	Sleeve with end-cap drain aperture that seals around the umbrella tip ; selective draining; multi-ply water-resistant construction. (Google Patents)	Abandoned (per Google Patents). (Google Patents)
US 8,667,975 & US 9,486,044 — Market tie-ins	(Not patents—commercial references)	Modern reverse-close umbrellas (Totes “InBrella”, Stromberg “Inversa/Reversa”) implement wet-side-inside containment described in above patents. (Totes.com USA)	Show commercial embodiment of the inverted/containment concept: keep drips inside during closure/transport. (Totes.com USA)	—

Appendix G: Umbrella Moisture Management Research Summary

Executive Overview

Umbrellas perform their main function well—shielding users from rain—but **fail at every moment before and after use**, especially during transitions into cars, buildings, and indoor spaces. User research shows consistent frustration with:

- Wet umbrellas dripping onto floors, seats, bags, and clothing
- Juggling bags, phones, and laptops while holding an umbrella
- No clean or convenient place to store a wet umbrella
- Slow drying, mildew, and odor
- Poor durability and wind stability

Patent research confirms that while many inventions address *moisture containment* or *mechanical structure*, almost **none solve the complete post-rain experience** or integrate *hands-free usability* into everyday routines.

Across surveys, interviews, and global patent analysis, two dominant opportunity spaces emerge:

1. Moisture Management

Contain, collect, or remove water so the umbrella can be stored instantly without dripping.

2. Hands-Free Convenience

Enable users to carry or operate the umbrella while handling other objects.

A next-generation umbrella ecosystem should address **both**.

1. User Research Insights (Survey + Interviews)

1.1 Usage Patterns

Umbrella use is highly situational—commuting, walking to class, running errands. Many users keep umbrellas in cars or offices but forget them due to inconvenience.

1.2 Pain Points

The strongest and most universal frustrations:

- **Dripping water indoors**, especially when entering cars
- **No clean storage** for a wet umbrella in bags, hallways, or classrooms
- **Awkward one-handed use** while juggling items

- **Wind-induced flipping** and fragile automatic mechanisms
- **Mildew or odor** from slow drying
- **Social discomfort**—wet umbrellas appear messy or unprofessional

Users repeatedly emphasized the **transition moment** (outdoor → indoor) as the worst part of the experience.

1.3 Storage + Drying Habits

Users typically:

- Shake umbrellas, but water still drips
- Leave them in bathrooms, balconies, or cars to dry
- Use plastic bags in public places, though these are disliked for sustainability reasons

Most users lack a reliable, hygienic drying method.

1.4 Emotions, Motivation, and Purchase Intent

Frustration scores averaged **5–7/10**. Purchase intent for an improved umbrella was **8–10/10**, especially for:

- Quick, contained drying
Cleaner storage solutions
- Hands-free usability

Users are open to paying more if the umbrella simplifies everyday routines and reduces hassle.

2. Patent Landscape Summary

Patent exploration revealed two major innovation clusters:

A. Moisture Capture & Drying Devices

Patents such as:

- **US5323802A** — absorbent bag
- **US6655399B1** — sponge-lined portable stand
- **US20110154677A1** — bristle ring for mechanical drying

These focus on sleeves, cases, or external devices that absorb or collect water. They address dripping but often require **two hands**, cleaning, or bulky accessories.

B. Integrated or Modular Umbrella Systems

Examples include:

- **WO2006138600A2** — airtight no-drip sheath
- **KAZbrella (Reverse Fold)** — canopy folds inward to trap water
- **Kjaro Umbrella** — drainable case

These designs bring moisture management closer to the umbrella itself, but few optimize for **one-handed operation**, car entry, or seamless transition.

C. Hands-Free Devices

- **Huriia Holder** — wearable clip
- **Nubrella** — backpack-mounted canopy

These solve hands-free use but sacrifice portability, comfort, or aesthetic appeal.

Conclusion:

Existing designs solve *pieces* of the problem. No single product meaningfully integrates **containment + quick drying + hands-free functionality** into a compact, commuter-friendly umbrella.

3. Key Cross-Research Insights

Insight 1 — Moisture Management Is the Central Unmet Need

Users struggle most with wet umbrellas after use. Patent solutions exist but lack ergonomic integration.

Insight 2 — Transition Moments Create the Most Friction

Car entry, building entry, and crowded indoor spaces are the biggest sources of frustration.

Insight 3 — Hands-Free Convenience Is Underserved

People often have full hands—phones, backpacks, groceries—and want partial or full hands-free operation.

Insight 4 — Durability and Ergonomics Still Drive Satisfaction

Wind resistance, smooth operation, and robust frames remain essential.

Insight 5 — High User Motivation

Users are eager for a premium solution that reduces mess, effort, and time.

4. Existing Solution Typologies (Across Patents & Market)

1. **Containment + Absorption Sleeves**
 - Pros: Prevent drips
 - Cons: Get dirty, require cleaning
2. **Reservoir / Drainage Cases**
 - Pros: Fully drip-proof indoors
 - Cons: Bulky, require two hands to use
3. **Mechanical Drying Systems**
 - Pros: Fast water removal
 - Cons: Typically non-portable
4. **Material Innovations (Superhydrophobic Fabrics)**
 - Pros: Repels most water
 - Cons: Coatings wear off
5. **Hands-Free Mounts and Harnesses**
 - Pros: Allow multitasking
 - Cons: Bulky, awkward appearance

This typology demonstrates a gap for **minimal, integrated, ergonomic mechanisms**.

5. Design Opportunity Areas

1. Indoor / Car Transition Containment

- Auto-sealing case
- Reverse-fold trap + reservoir
- Magnetic or mechanical “snap-dry” system

2. Compact Drying System

- Absorbent cores
- Ventilated shells
- Wiper ring integrated on umbrella shaft

3. Hybrid Repel + Contain Approach

- Hydrophobic canopy + small sealed reservoir
- Cuts reliance on large absorbent surfaces

4. Lightweight Hands-Free Support

- Integrated clip or strap in the handle
- Balanced or counterweighted frame

5. Smart UX Indicators

- Moisture sensors
- Color-change or mechanical signals
- Auto-lock mechanisms

6. Synthesis: Where the Market Fails

Current umbrellas are optimized for **rainfall**, not for:

- The transition indoors
- Drip containment
- Car entry
- Hands-free convenience
- Quick drying
- Social professionalism (clean, contained look)

Existing patents suggest interest in these areas but do not combine them into a user-centered system.

7. Overall Summary

Survey, interview, and patent research collectively identify a clear design gap: umbrellas succeed outdoors but fail in the everyday routines surrounding their use. Users want:

1. **Hands-free, low-effort operation**
2. **Contained, instant moisture management**

Patent benchmarks show partial solutions, yet no product integrates them seamlessly. This opens a strong opportunity for a “**Post-Rain Umbrella System**”—an ecosystem combining:

- A hydrophobic or quick-dry canopy
- A compact drip-proof case or reservoir
- A one-hand stowage mechanism
- Optional hands-free carry or mount
- Clean, professional aesthetic

Such a design would address the full umbrella lifecycle—**use** → **fold** → **carry** → **store**—solving the friction points that users have considered “normal” for decades.

Appendix H: AEIOU Observation Video Library

All videos supporting this research are available in the project repository:

https://drive.google.com/drive/folders/1QyY1ZDm_YPD97jOFFRyr9xoeAik-XFPi?usp=drive_link

Screenshot of key frame from video ReverseTest2(in).MOV



Appendix I: AEIOU Observation Findings

AEIOU Field Observation Report: Umbrella Use in Vehicle Entry and Exit

1. Stick Umbrella

Getting In

- Opens car door with one hand while holding umbrella in the other.
- Performs quick “hand transfer” to close door.
- Struggles to bring umbrella inside due to limited door opening.
- Steps back out to fold umbrella before entering.
- Holds umbrella slightly away from body to avoid wet contact.

Getting Out

- Partially opens door, extends umbrella outward, then steps out.
- Long shaft makes maneuvering difficult in tight spaces.
Requires coordinated hand-door movement to prevent dripping inside.
- Minor drips on seat or clothing occur if timing is off.

2. Reverse-Folding Umbrella

Getting In

- Collapses umbrella inward while entering car, trapping water inside canopy.
- Allows door to remain more closed; interior stays dry.
- Requires two-hand operation but feels natural and contained.

Getting Out

- Opens umbrella through narrow door gap before stepping out.
- Canopy unfolds outward, keeping car dry.
- Smoothest transition among all types; not held away from body.

3. Auto-Folding (Non-Reverse) Umbrella

Getting In

- Presses button multiple times to fold and lock umbrella.
- Stands outside getting wet during locking process.
- Holds umbrella at arm’s length to avoid touching wet canopy.

Getting Out

- Opens door slightly, raises umbrella above head, then steps out (“hands-up, then body-out”).
- Movement shows learned muscle memory and spatial awareness.

4. Manual Folding Umbrella

Getting In

- Manually slides ring up to fold canopy while balancing door and bag.
- Reaches arm outward to shake off water before entering.
- Holds umbrella away from body when bringing it inside.

Getting Out

- Grabs umbrella from side pocket or floor, reaches outward through narrow opening to open.
- Occasionally struggles to find enough clearance for full expansion.
- Movement is slower but more predictable.

5. Overall AEIOU Synthesis (Cross-Type Summary)

Category	Detailed Observations and Insights Across All Umbrella Types
Activities	• Transitional phases—getting in and out of vehicles—reveal the highest friction points. • Users develop personalized micro-routines (“fold first,” “extend first,” “raise first,” “shake first”) to minimize water exposure. • Frequent gestures include twisting the torso, reaching outward, or holding the umbrella at arm’s length to maintain distance from wet surfaces. • Manual and stick umbrellas demand more coordination and effort; auto and reverse types reduce some steps but introduce mechanical delays. • All users strive for the same goal: shorten exposure time and keep the car interior dry.

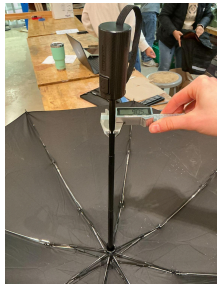
Environments	• Common settings include parking lots, curbsides, and tight urban spaces under active rainfall. • Limited door clearance and surrounding vehicles constrain movement, forcing awkward postures. • Wet ground and puddles create instability and hesitation, slowing down the sequence. • Dim lighting or wind intensifies difficulty, particularly when users need to see button mechanisms or align handles. • The environment amplifies urgency—every second spent outside equals visible discomfort and dripping.
Interactions	• For every type except the reverse-folding umbrella, users instinctively keep the wet canopy away from the body when entering the car. • Reaching outward to shake water off or extending the arm to separate “wet” and “dry” zones are nearly universal behaviors. • The umbrella shifts psychologically from a protector to a contaminant once wet, prompting caution and spatial awareness. • Mechanical feedback (buttons, locks, sliders) is often unclear, creating hesitation and repetitive testing motions. • Users improvise one-handed or mid-motion adjustments, revealing design gaps in tactile feedback and ergonomic flow.
Objects	• Stick umbrellas: strong coverage but poor maneuverability in confined spaces; frequently bump doors or nearby cars. • Manual folding: compact yet demanding; relies on fine motor control under stress. • Auto folding: faster closing but still exposes user while waiting for mechanical action.

	• Reverse folding: best containment of water but bulkier and needs two-hand coordination. • None provide effortless containment and storage during transition—design tension persists between size, speed, and cleanliness.
Users	• Users develop adaptive “muscle memory” routines that balance dryness, time, and limited space. • Actions evolve from trial-and-error to learned choreography after repeated use. • Emotional tone ranges from mild frustration (manual, stick) to cautious satisfaction (reverse-folding). • Behavior reflects pragmatic problem-solving rather than intuitive product interaction. • Across contexts, users prioritize personal dryness first, then car cleanliness, then convenience—clear hierarchy of needs.
Design Opportunities	• Create umbrellas that enable seamless transition behavior within car environments. • Support one-handed folding/unfolding and minimal arm extension. • Integrate built-in containment—absorptive sleeves, drain channels, or internal catch layers. • Provide clear mechanical feedback to confirm lock or close state without visual check. • Optimize form factor for door-arc geometry, allowing clean entry and exit in tight spaces. • Reduce perceived “wet object” contamination by guiding natural, dry-side handling points. • Design for cognitive simplicity under time pressure—intuitive gestures, fewer steps, faster closure.

Appendix J: Product Dissection Measurements Table

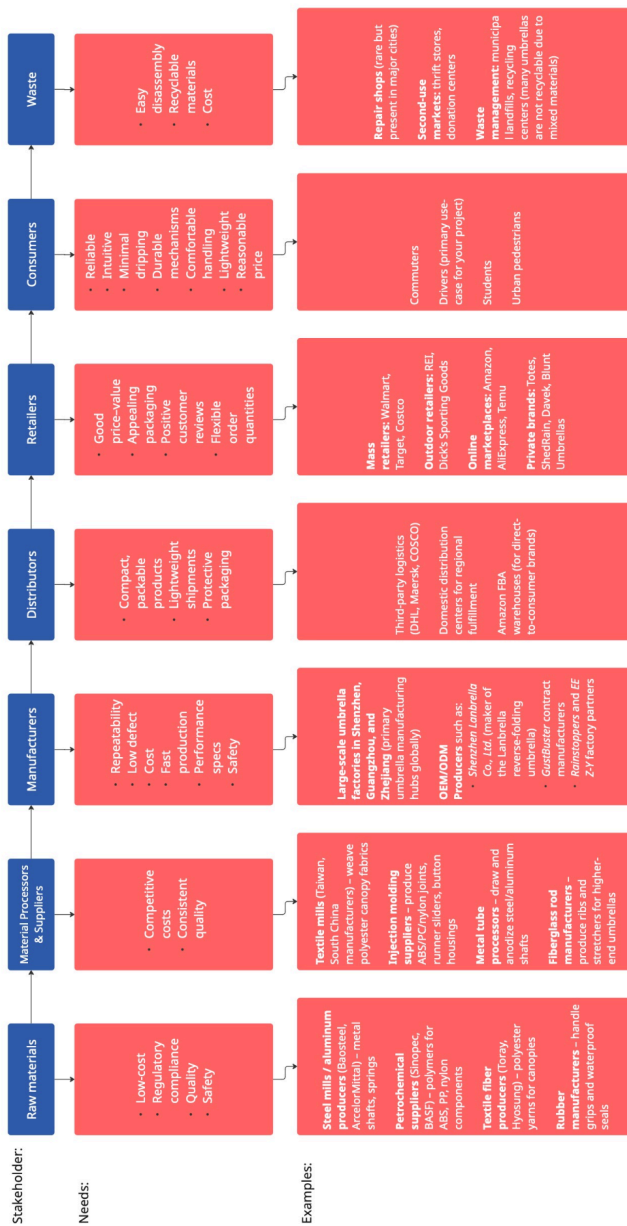
Part	Measurements
Handle	Diameter: 35.98mm Length: 80.63mm
Button	Width: 13.89mm Length: 19.88mm
Shaft 1	Diameter: 9.05mm
Shaft 2	Diameter a(circular): 10.22mm Base-Height b(hexagonal): 10.35mm
Shaft 3	Base-Height a (hexagonal): 11.32mm Diameter b (circular): 12.46
Slider	Maximum diameter: 28.94 Minimum diameter: 17.54

Appendix K: Product Dissection Function Table

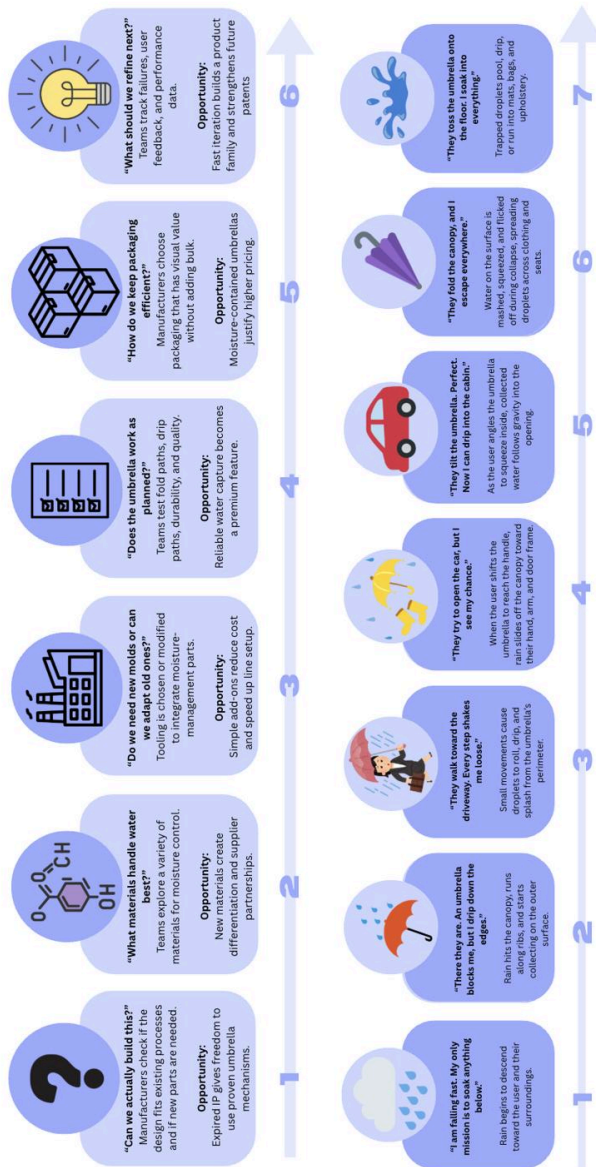
Part Name and Image	Function	Material	Production Method
 Outer casing, button, and handle	Houses internal locking mechanism; provides user grip; button actuates the auto-open feature.	Injection-molded ABS plastic (button + handle)	Injection molding
 Shaft 1 (Upper + Lower Section)	Structural backbone of umbrella; transfers force during opening/closing; telescopes.	Aluminum alloy with black coating	Drawn metal tube extrusion Coating process (powder coat or black oxide)
 Shaft 2 (Upper + Lower Section)	Allow collapsible telescoping motion; connect to runner mechanism.	Aluminum alloy with black coating	Drawn metal tube extrusion Coating process (powder coat or black oxide)
Runner (Sliding Lock Mechanism)	Moves along shaft to open/close canopy; interfaces with button-release system.	Injection-molded ABS plastic	Injection molding

 Reverse-folding canopy frame	Enables reverse folding motion; controls canopy shape; supports umbrella fabric	Fiberglass ribs (resin coated); Nylon pivot joints (not ABS because need to be fatigue-resistant and undergo repeated flexing)	Pultrusion of fiberglass ribs (pulling fibers through a resin bath and then through a heated die) Injection molded nylon pivots
 Canopy fabric	Double-layered; main rain-protective surface; provides water-repellency.	High-density 210T polyester (waterproof)	Weaving, heat cutting, seam stitching
 Strap + Velcro Closure	Wraps around umbrella to keep fabric closed.	High-density 210T polyester (waterproof) + Velcro	Weaving and stitching
Main Spring	Provides stored mechanical energy to open umbrella quickly.	High-carbon spring steel	Wire forming, heat treatment

Appendix L: Stakeholder Analysis



Appendix M: Stakeholder Journey Maps



- 1 user grabs umbrella from home.
- 2 user deploys umbrella after opening door to outside
- 3 user walks approx 20 ft to car parked on the street
- 4 user opens car door, steps halfway in, then presses button to retract umbrella
- 5 user sits halfway on seat, reaches up to grab and retract umbrella
- 6 user fully retracts umbrella and places it on passenger seat floor before closing car door.

who are the stakeholders?

→ include the passenger

PAIN POINTS

u m b r e l l a

- 1) umbrella does not fit into the space between car and car door
- 4) umbrellas are not easy to fully close w/o full range of motion

W a t e r

- 2) water is collecting on the outer portion of the umbrella
- 4) water rolls off the sides of the umbrella, creating a brief moment of concentrated water-fall
- 5) umbrella shaft becomes slippery making it unviable to grab.
- 5) water gets onto the hands of the user despite efforts to stay dry.
- 6) water gets inside the car — onto the seats and floor.

C a r

- 4) car door and handle are wet making it difficult to open while maneuvering umbrella
- 4) while car door is open, water gets on the inner driver door and seat.
- 5) car door design does not allow for an umbrella to easily close
- 6) car interiors are generally hydrophobic and do not respond well to the introduction of a small pool of water.

OPPORTUNITIES

u m b r e l l a

- 2) umbrella deployment can be violent, decreasing users from opening the umbrella inside — superstition — gentler umbrellas? can we get around superstition?
- 3) umbrella's sometimes flip or get swept away in the wind — wind proof design, ventilated hood?
- 4) the shape of the umbrella is not ideal for usage when entering a vehicle — reshape the umbrella hood?
- 5) a fully retracting umbrella can be dangerous because of the speed and force required — is a slower, fully automated retraction system possible?
- 6) umbrellas are hydrophobic, but water does not bead up on the outer surface to remain "dry" — how might we create an umbrella canopy that does not retain surface water when retracted?

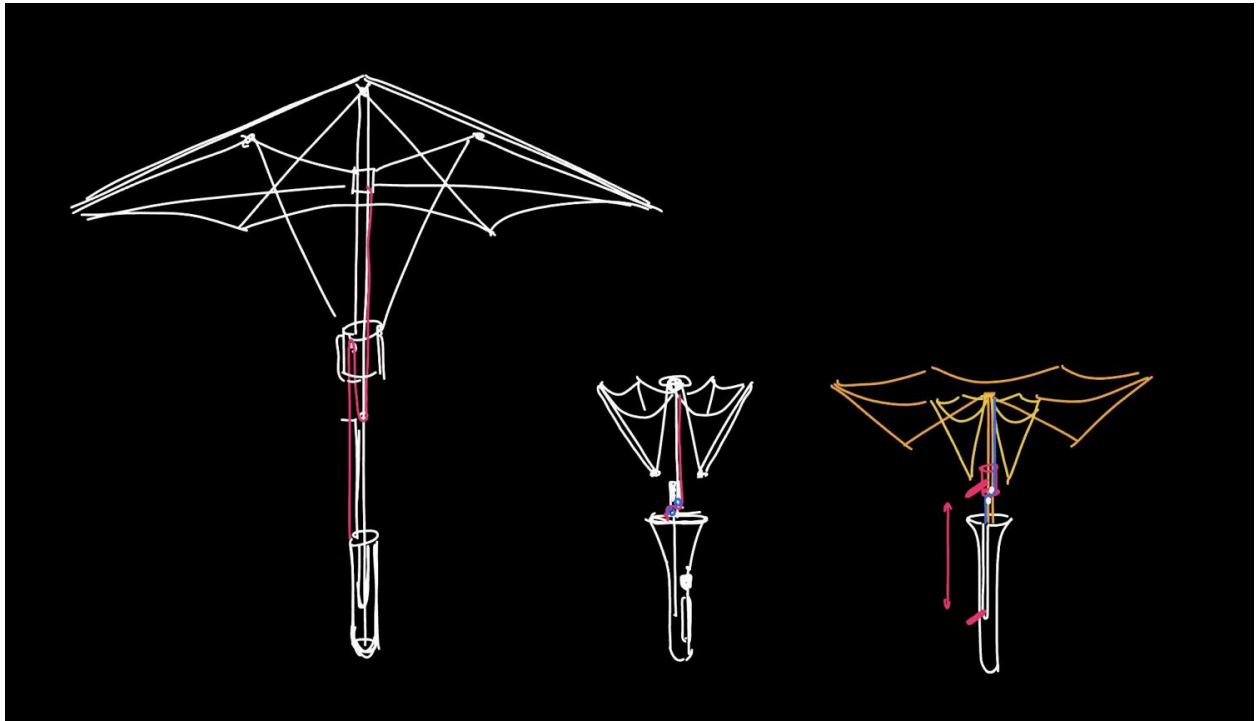
W a t e r

X

C a r

- 3) car is wet — can we keep driver door dry w/o garage?
- 4) range of motion of door feels limited with an umbrella in hand — can we change the range of motion for the door? buy a Tesla
water gets onto the inner driver door while user fumbles with umbrella — can we create a way to hang the umbrella on the car door for hands-free use?
- 5) car interiors should be kept dry. — how can we ensure the interior of the car (and user) stay dry during this process?
- 6) there is no designated place for an umbrella. — can we create a "cup holder" or "glove box" for umbrellas? will it be an external accessory for the vehicle or will it be implemented during manufacturing?

Appendix N: Preliminary Design Concepts

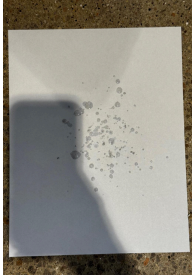


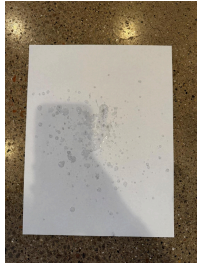
Appendix O: Umbrella Geometry Benchmarking

One column per umbrella

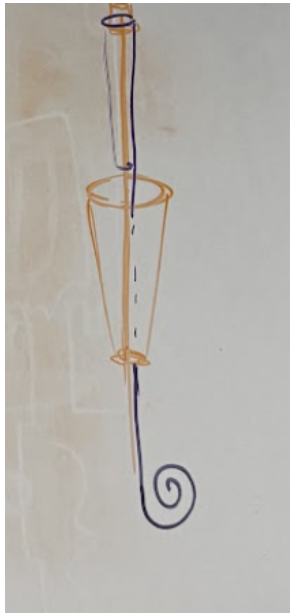
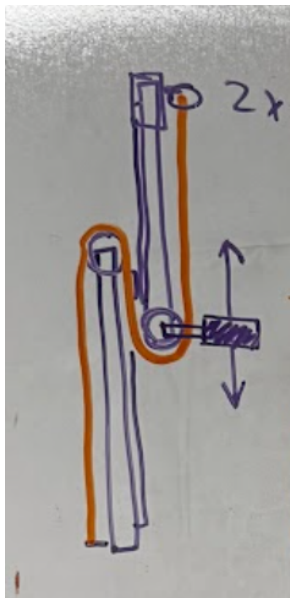
Rows = what you observe

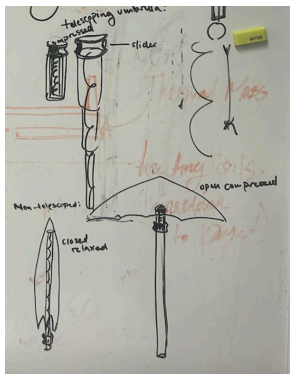
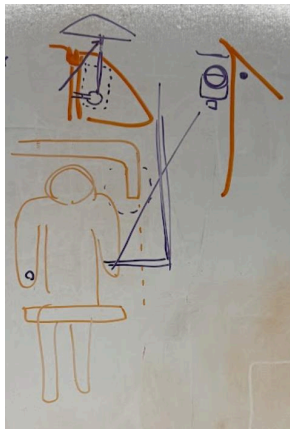
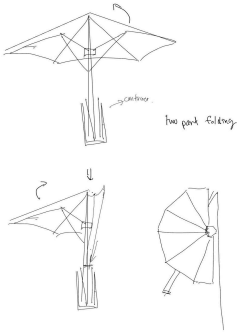
Cells = where you write notes, sketches, or quick ratings

Geometry Feature	What to Observe	Umbrella A: Stick Reverse	Umbrella B: Warped	Umbrella C: Automatic
Open Canopy Diameter	Wide / medium / narrow	42 in	Major: 39 in Minor: 34.5 in	40 in
Canopy Depth	Shallow / medium / deep	11.5 in	7in	9.5 in
Folding Direction	Inward / outward	Up	Down	Down
Fold Control	Clean / messy / inconsistent	Manual clean <5lb	Manual very difficult 15lb	Automatic easy 0lb (Auto)
Rib Length	Short / medium / long	24 in	Max: 29.5 in Min: 15 in	23 in
Rib Angle	Straight / curved	50 deg	30 deg	42 deg
Closed Length	Short / medium / long	31.5 in	10.5 in	12.7 in
Closed Bulk	Slim / average / bulky	Bulky when unfastened, average when fastened	Small when unfastened . average when fastened	Bulky when unfastened, average when fastened
Water Containment	None / partial / good Notes: Cup filled with water, dumped three times each on umbrella, umbrella closed and tapped on paper ten times		 	 

		 		
Door Clearance	Fits / tight / doesn't fit	Doesn't fit	Doesn't fit	Doesn't fit
Entry Motion Flow	Smooth / awkward (reference videos)	Awkward	Awkward	Awkward
Closing + Wrapping time	Time (s) : average of three times	8.75 Notes: Strap is far from the base, have to hold it higher up to get it	8.07 Notes: have to wrap it tight because its so bulky, takes more force, probably because ribs are "huge" for wind protection, jerky motion down, really heavy	7.64 Notes: strap is at a weird angle so you have to pull it kind of up to wrap it around, shaft length is much shorter when the umbrella canopy opens up which can be a drawback because we need as much shaft length as possible to clear the car door
Notes Umbrella A has a weaker spring which makes entry motion easier/ less effort for fold control Umbrella B is extremely strenuous to lock ("at least 50 pounds to lock" - peter w.) Umbrella C fails to open up fully sometimes				

Appendix P: Needs-Function-Metrics Table

Need/Pain Point	Item	Mechanical Function	Sketch	Metrics
Reaching to pull down umbrella makes you wet	String	Minimize reach in pull		Reach(cm)
	Pulley 1	Minimize reach in pull		Reach (cm)

	Reverse Spring	Spring in compression when open at the top to balance out force required to close		Maximum Force required to pull down (N)
	Pulley 2	Redirect pull force direction		Direction of pull (degrees)
Moisture contamination after retracting umbrella	Canopy	Geometry (close in/out)/ Multiple Step folding		Water retained after closure (mL)

Umbrella gets stuck on door during closure	Magnet	Magnet to clear ribs from car door		Clearance between ribs and door edge (mm)
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Appendix Q: Metrics and Specs Table

Metric	Target Specification	How it was measured
Reach (in)	21"	User can close umbrella without leaning outside the car (Measure max hand reach during closure when seated)
Open Length (in)	34" - 17" = 17"	Minimum length of open umbrella that clears car ceiling (distance from handle to car roof) Mid Handle to bottom of tesla: 17" Mid Handle to top of tesla: 34"
		● Pocket umbrella open = 20.5" ● Reverse Telescoping umbrella: open = 24.8" ● Risetime automatic umbrella: open = 26.3"
Maximum pull force (N)	*test next week when luggage scale arrives	Test with new umbrellas + luggage scale to get numbers. Then test seeing which umbrellas can be closed with one hand to determine the threshold
Direction of pull (degrees relative to vertical) (Pulley 2)	90 degrees	Most convenient pull motion/angle

Metric	Target Specification	How it was measured
Water retained after closure (mL) (Canopy folding)	*test once we move into the canopy stage.	Qualitative Observation (No water escapes onto user or vehicle interior)
Closed bulk of umbrella (in) (Container)	< 2” in diameter	An average car glove compartment: < 12” length < 6” depth < 4” width Average car door compartment (door pocket/storage bin) : < 12” length < 3” depth < 2” width
		● Pocket umbrella diameter = 1” ● Reverse telescoping umbrella diameter = 3” ● Risetime automatic umbrella diameter = 5” *measured after “messy” closing
Retracted length (in) (Telescoping shaft)	< 12” in length	An average car glove compartment: < 12” length < 6” depth < 4” width

Metric	Target Specification	How it was measured
		Average car door compartment (door pocket/storage bin) : < 12" length < 3" depth < 2" width
		● Pocket umbrella length compact = 5.9" ● Reverse Telescoping Umbrella: compact = 13" ● Risettime automatic umbrella: compact = 13.4"

Appendix R: Provisional Patent

TITLE

Improved Umbrella with Advanced Retraction, Canopy Retraction, and Containment

INVENTORS

Isaac Braunfeld, Nick Visuthikosol, Symphony Smith, Peter Wang, Juliana Gonzalez

FIELD OF THE INVENTION

[001] The present invention relates generally to portable weather protection devices, and more particularly to umbrellas. Specifically, the invention pertains to an umbrella system incorporating an improved handle-actuated retracting mechanism, canopy retraction, and containment features to operate an umbrella in a restricted area.

BACKGROUND

[002] When umbrellas are used in inclement weather, particularly during transitions between environments, users often encounter difficulties in closing and storing the umbrella in a controlled and convenient manner. Conventional umbrellas require the user to have an extended reach upward along the shaft to actuate the retraction mechanism, which can be awkward when entering a car, for example. This motion frequently forces the user to extend the umbrella outside the vehicle, exposing the interior to rain and increasing the likelihood of water entering the vehicle and coming into contact with the user.

[003] In one instance, when entering a space, the user must partially open the door while holding the umbrella overhead. To close the umbrella, the user is required to reach upward toward the canopy or runner, often beyond comfortable reach, especially while seated or in the process of sitting down. This results in an inefficient and unergonomic motion that can delay entry into the vehicle and increase exposure to rain. Additionally, the lack of control during this process may cause water collected on the canopy to drip into the vehicle interior.

[004] In another instance, once the umbrella is closed, the canopy fabric tends to retract in an uncontrolled manner, with portions of the material draping outward and holding water. The fabric retains and exposes water, contributing to dripping and wetting of surrounding surfaces.

[005] Furthermore, after closing, wet umbrella canopies are typically brought directly into the vehicle or indoor space without effective containment of residual water. This leads to accumulation of water on seats, floors, and other surfaces, reducing user comfort and requiring additional cleanup. While some reverse-folding umbrella designs attempt to address water containment, they do not sufficiently resolve issues related to compactness of the retracted canopy or ease of operation during the closing process.

[006] Users rely on umbrellas to provide convenient and effective protection from rain during daily activities. However, the limitations associated with current umbrella designs, specifically the need for upward reaching to close the umbrella, the lack of controlled canopy retraction, and inadequate containment of water, reduce their usability in common scenarios such as entering vehicles. Accordingly,

there is a need for an improved umbrella design that enables more ergonomic operation, controlled and compact canopy folding, and enhanced containment of water during and after use.

SUMMARY

[007] The primary object of the invention is to provide an umbrella system that enables closure of a canopy through actuation at a lower portion of a shaft, thereby reducing the need for a user to reach toward an upper portion of the umbrella during operation.

[008] Another object of the invention is to provide a bi-directional locking and actuation mechanism configured to selectively engage and disengage a canopy locking state, enabling controlled opening and closing of the umbrella.

[009] Another object of the invention is to provide a telescoping shaft mechanism consisting of a plurality of spring elements distributed between adjacent shaft segments and a plurality of intermediate locking features, thereby enabling automatic extension, reduced retraction force, and reliable locking in an extended configuration.

[010] Another object of the invention is to provide a pull-to-close mechanism that allows a user to initiate and guide canopy retraction from a lower portion of the shaft in a smooth and controlled manner.

[011] Another object of the invention is to provide a canopy structure configured to fold in a controlled, compact, and predictable closed configuration, thereby reducing outward draping of canopy material during closure.

[012] Another object of the invention is to improve usability of an umbrella during transitions into confined or restricted spaces, including but not limited to vehicles, by enabling more ergonomic operation and reducing interference interactions with surrounding surfaces.

[013] Another object of the invention is to provide a water containment system configured to receive at least a portion of the umbrella in a closed configuration, thereby reducing dripping and exposure of surrounding surfaces to residual water.

[014] Another object of the invention is to enhance user comfort and cleanliness by combining controlled canopy folding with water containment to limit water dispersion during and after use.

DETAILED DESCRIPTION

[015] As shown in Sections 1-3 of the next section and the corresponding figures, in various embodiments an umbrella system comprises a canopy assembly (200), a shaft extending between an upper portion and a lower portion, a bi-directional locking and actuation system configured for user actuation from the lower portion of the shaft, a controlled canopy folding mechanism, and a water containment system. The system further includes a lower actuation interface (100), a force transmission

element (110), a storing mechanism (120), a lower locking mechanism (130), a containment structure (140), and an upper locking mechanism (160).

[016] In another embodiment, the umbrella system includes the upper locking mechanism (160) configured to maintain the canopy (200) in an open position and the lower locking mechanism (130) and lower actuation interface (100) to facilitate retraction, disposed at or near the lower portion of the shaft.

[017] The lower actuation interface (100) is configured for ease of user engagement and for translating force to the upper locking mechanism (160) via the force transmission element (110) extending along the shaft. The force transmission element (110) may include a linkage, cable, spring element, or similar force-transmission component. Improved actuation access is accomplished from the lower portion of the shaft as opposed to traditionally reaching up the shaft to release the upper locking mechanism (160) for canopy retraction.

[018] In another embodiment, the system is bi-directional, enabling both disengagement and re-engagement of the locking state via the lower actuation interface (100) and lower locking mechanism (130) at the lower end of the umbrella shaft, and allowing operation without requiring the user to reach toward the upper portion of the shaft.

[019] In another embodiment, the system incorporates an alternative ratchet-based engagement mechanism forming part of the lower locking mechanism (130), configured to selectively permit incremental movement in one direction while restricting reverse motion, thereby enabling controlled retraction and re-extension of the umbrella shaft through actuation at the lower actuation interface (100) without requiring user interaction with the upper portion of the shaft.

[020] In another embodiment, the shaft (150) comprises a multi-stage telescoping structure including a plurality of nested shaft segments configured to extend and retract relative to one another. The telescoping structure may include a first fixed shaft segment (151), a second shaft segment (152), and a third shaft segment (153), each configured to translate relative to an adjacent segment. The telescoping structure may further include at least two spring elements disposed between adjacent shaft segments, each spring element configured to apply an extension force between corresponding segments. In one embodiment, a first spring element (155) is disposed between the first shaft segment (151) and the second shaft segment (152), and a second spring element (156) is disposed between the second shaft segment (152) and the third shaft segment (153). A collar (154) may be disposed at an upper interface of one or more shaft segments.

[021] In another embodiment, the use of multiple spring elements (155, 156) distributed between shaft segments (151, 152, 153) enables reduced individual spring stiffness while maintaining sufficient extension force for automatic deployment of the umbrella. This configuration reduces the force required for user-initiated retraction of the shaft (150) as compared to systems utilizing a single spring extending along a majority of the shaft length.

[022] In another embodiment, one or more locking elements are disposed between adjacent shaft segments, the locking elements configured to selectively engage at a fully extended position to maintain the shaft (150) in an extended state. The locking elements may include detent-based mechanisms,

protrusions, or engagement features that interact with corresponding recesses or apertures in adjacent shaft segments. In one embodiment, the locking elements include a lower detent lock (157) and an upper detent lock (158).

[023] In another embodiment, the locking elements (157, 158) are positioned between spring elements (155, 156) within the shaft (150), enabling mechanical locking without interference from the spring components.

[024] In another embodiment, at least one locking element comprises a resilient member mounted along the inner surface of a shaft segment. The resilient member may include a flat spring element (159) configured to deflect laterally to engage and disengage with a corresponding feature on an adjacent shaft segment.

[025] In another embodiment, the use of a resilient member (159) disposed along a wall of the shaft (150) enables preservation of a central channel (161) within the shaft. The central channel (161) may be configured to accommodate the force transmission element (110), such as a cable or pull line, extending along the length of the shaft (150).

[026] In another embodiment, the combination of distributed spring elements (155, 156) and intermediate locking elements (157, 158) enables automatic extension of the umbrella while maintaining a reduced retraction force and reliable locking at full extension.

[027] In another embodiment, the canopy assembly (200) is configured to transition from an open configuration to a closed configuration via the lower-shaft actuation mechanism initiated at the lower actuation interface (100) and transmitted through the force transmission element (110).

[028] The pull-to-close mechanism enables a user to initiate and guide closure from the lower portion of the shaft via the lower actuation interface (100), promoting smooth and controlled retraction.

[029] In another embodiment, the canopy (200) folds into a compact configuration through segmented folding rib geometry and guided folding paths, reducing outward draping and improving usability in confined spaces. The storing mechanism (120) is configured to manage excess length of the force transmission element (110) during operation, thereby reducing slack and enabling controlled actuation.

[030] In another embodiment, the umbrella system includes a containment structure (140) configured to receive at least a portion of the umbrella in a closed configuration.

[031] The containment structure (140) is configured to at least partially enclose the canopy (200) to retain residual water and reduce dripping. The containment structure (140) may be flexible or semi-rigid and may include absorbent, drainage, or sealing features.

[032] In another embodiment, the combination of controlled canopy folding (200) and the containment structure (140) improves water containment and reduces exposure to surrounding surfaces.

[033] Various modifications may be made without departing from the scope of the present disclosure. For example, the actuation interface (100), force transmission element (110), storing mechanism (120),

lower locking mechanism (130), upper locking mechanism (160), canopy structure (200), and containment structure (140) may take different forms while achieving similar functionality.

[034] A method of retracting an umbrella includes actuating the lower actuation interface (100) to transmit force through the force transmission element (110) to disengage the upper locking mechanism (160) and control retraction via the lower locking mechanism (130), thereby enabling improved reach, controlled retraction, and containment within the containment structure (140).

BRIEF DESCRIPTION OF THE DRAWINGS

Section 1: Bi-Directional Locking and Actuation Mechanism

[035] Figure 1.1: Exploded view of an embodiment of the umbrella assembly illustrating components associated with the locking and actuation system.

[036] Figure 1.2: Assembled view of the umbrella system highlighting interaction between upper and lower locking mechanisms.

[037] Figure 1.3: Detailed view of an upper locking mechanism configured to maintain the canopy in an open position.

[038] Figure 1.4: Detailed view of a lower locking and actuation mechanism disposed at a lower portion of the shaft.

[039] Figure 1.4: Detailed view of the handle interaction with release mechanism.

[040] Figure 1.5: Telescoping shaft assembly illustrating multi-stage spring configuration and intermediate locking mechanisms.

Section 2: Pull-to-Close Mechanism and Controlled Canopy Folding

[041] Figure 2.1: Perspective view of the umbrella in an open canopy configuration.

[042] Figure 2.2: Perspective view of the umbrella in a partially closed configuration illustrating initiation of canopy retraction.

[043] Figure 2.3: View of the canopy in a folded configuration demonstrating controlled and compact retraction of the canopy material.

Section 3: Water Containment System

[044] Figure 3.1: Illustration of a container configured to receive at least a portion of the umbrella in a closed configuration.

[045] Figure 3.2: View of the umbrella in a closed and contained configuration within the container, illustrating water retention and reduced external dripping.

DRAWINGS

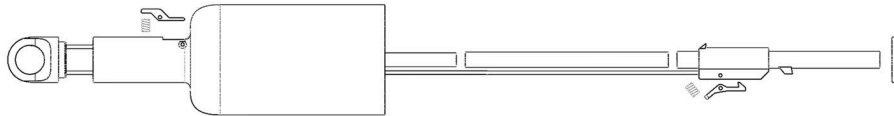


Figure 1.1: Exploded View

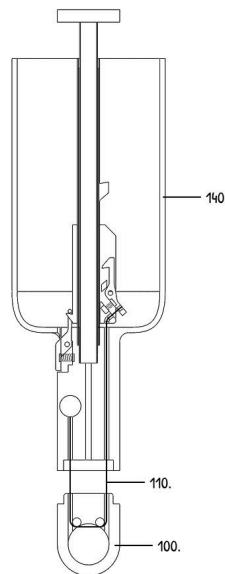


Figure 1.2: Retracted Shaft Assembly

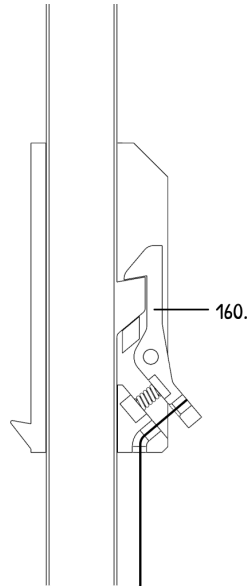


Figure 1.3: Upper Lock Mechanism

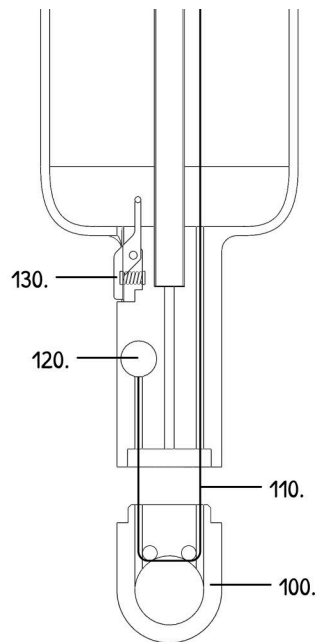


Figure 1.4: Handle & Lower Lock Mechanism

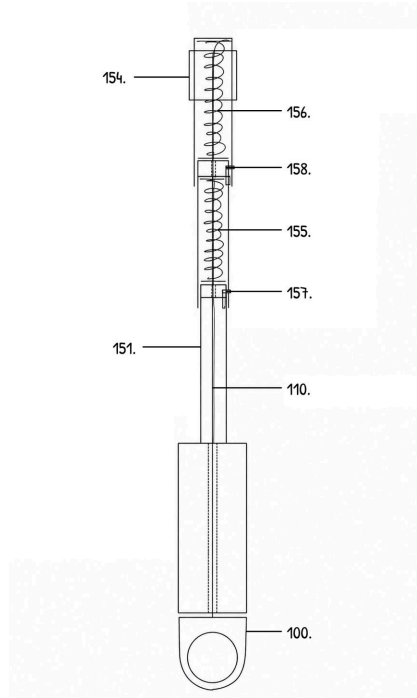


Figure 1.5: Telescoping Shaft Mechanism

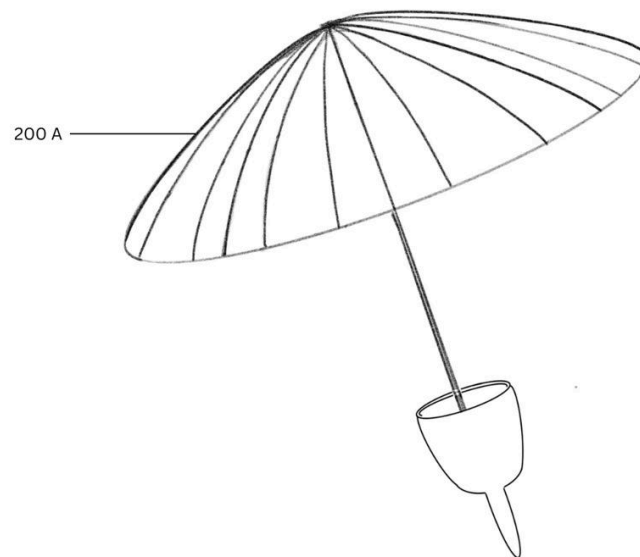


Figure 2.1: Open Canopy

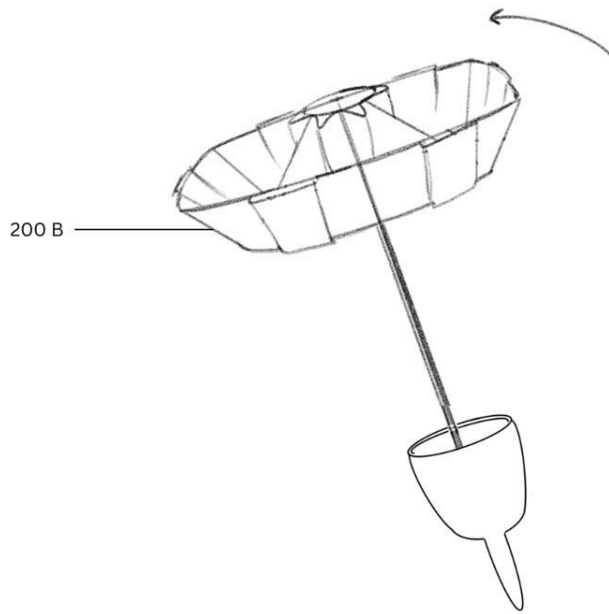


Figure 2.2: Partially Closed Canopy

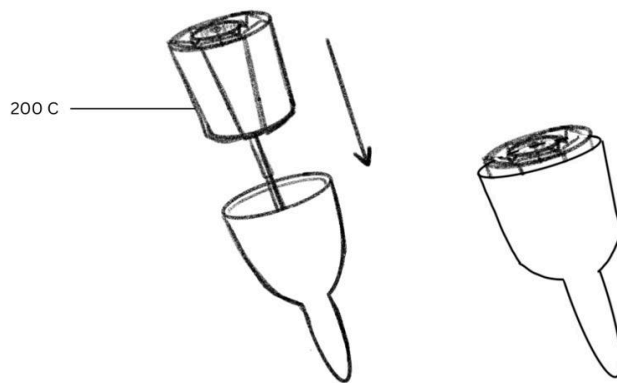


Figure 2.3: Folded Canopy

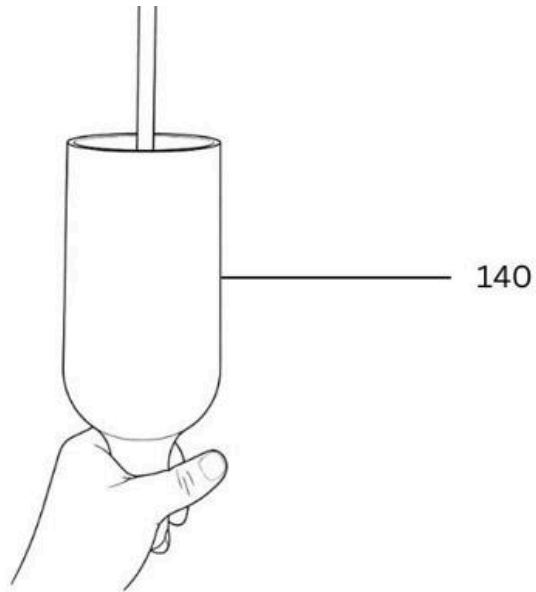


Figure 3.1: Container

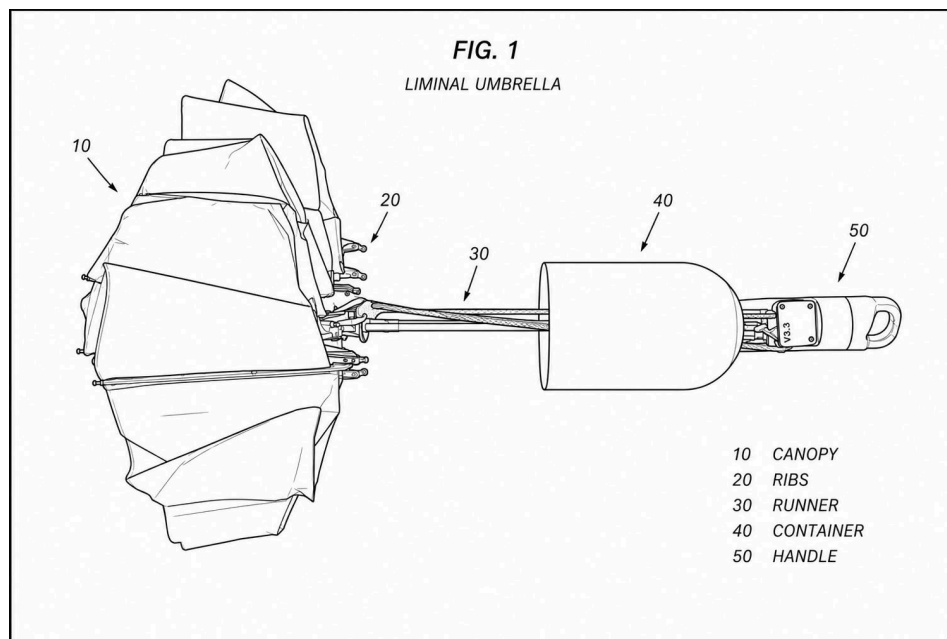


Figure 3.2: Container with Umbrella

PRIOR ART REFERENCES

Table 1: Bi-Directional Lock Mechanism

Patent Number	Status	What is Claimed
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US17325163A	Alive	Collapsible cone-shaped holder with removable water reservoir, vehicle-mountable, using absorbent/wicking materials to capture and retain runoff
US16741855A	Alive	Multi-component sealed case that physically separates wet canopy and collected water using threaded modular compartments
WO2015CN91331A	Dead	Telescoping umbrella case with detachable shell, locking features, and extendable protective housing
US14941379A	Dead	Hybrid rigid-flexible expandable case that changes internal volume via sliding flexible segments
US14886277A	Dead	Pivoting sheath-handle system that rotates around a joint to enclose umbrella and act as handle
WO2012KR7039A	Dead	Adjustable-length sheath with integrated water storage base, drainage control, and leak-prevention sealing features
US13330193A	Dead	Flexible sheath with sealed tip receiver and gasket system to prevent leakage while allowing controlled drainage
US13195024A	Dead	Umbrella with deployable waterproof bag stored inside hollow handle for on-demand enclosure
WO2012IB53828A	Dead	Waterproof container with engineered drainage system, including tank, valves, and controlled outflow mechanisms
US2011983307A	Dead	Integrated umbrella-storage device combining cylindrical case with automatic umbrella deployment mechanism
US2009322865A	Dead	Two-piece cylindrical enclosure with locking/sealing interface to create watertight umbrella storage
US2005155846A	Dead	No-drip cylindrical sheath with internal reservoir, sealing interface at umbrella tip, and optional collapsibility
US2004770499A	Dead	Rigid receiver case with internal alignment features to lock umbrella into place during storage
US2003704734A	Dead	Box-style enclosure mechanically interfacing with umbrella handle to enable storage and deployment
WO2002KR557A	Dead	Adjustable umbrella holder with multi-stage drainage system and anti-backflow fluid control design

US1999435952A	Dead	Dual-function housing that alternates between sealed storage case and structural handle via reversible threading
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Table 2: Pull-to-Close Mechanism

Patent Number	Status	Claims
EP4627914A1	Alive	Pull line winds on spool with dual constant-force springs, enabling staged tension and controlled retraction
WO2024061993A1	Indeterminate	Closure uses force threshold + locking elements, requiring manual force to transition states
JP2022167726A	Alive	Pulley system multiplies pull force, translating user pull into runner motion to close umbrella
US11378218B2	Alive	Cyclic cable pulling with gripper + pulley, converting repeated linear pull into continuous cable motion
US20210113865A1	Alive	Retractable line with user-controlled descent, regulating tension during pull/release
US10322305B2	Alive	Lifeline wraps around spring-loaded rotating hub, automatically retracting cable via stored tension
WO2017218650A1	Dead	Dual pull cords synchronize motion, translating pull into vertical displacement (blinds mechanism)
TWM468266U	Dead	Volute spring rewinds cord automatically, storing energy during pull and releasing for retraction
TWI418317B	Alive	Dual cable routing system pulls multiple internal components (runner, tubes) to open/close umbrella
US20050279977A1	Dead	Cable tension device stores compressive energy to maintain line tension, preventing slack during pull

GB2345482A	Dead	Reel + ratchet system controls cable payout vs pull-in, enabling directional tension control
(Blinds system – extended claims)	Dead	Bound dual cords + tensioning system coordinate movement and leveling via shared pull input
(Curtain cord system – extended claims)	Dead	Cord routed through guide + spring housing, enabling controlled pull + automatic rewind
(Umbrella cable variant – extended)	Dead	Primary + secondary cable coupling, distributing pull force across multiple structural elements
(Cable + spring hybrid systems – general group)	Mixed	Pull input stored in spring + released via controlled retraction, enabling smooth bidirectional motion

Table 3: Folded Canopy

Patent Number	Status	What is Claimed
CN209573523U	Alive	Claims a folding umbrella with a built-in containing structure formed by semi-surrounding parts that create a storage space. Focuses on post-use containment rather than controlled canopy folding or lower-shaft actuation.
CN223262449U	Alive	Claims a quick-folding umbrella using folding sheets and crease structures to accelerate canopy collapse. Emphasizes speed of folding rather than controlled, compact folding or ergonomic user actuation.
CN221044424U	Alive	Claims an umbrella with torsion elastic elements that drive automatic directional folding. Focuses on automated motion rather than user-controlled closure or lower-shaft actuation.
CN205902936U	Dead	Claims a fully automatic reverse-folding umbrella with elastic and linkage-driven opening and closing. Relies

		on conventional actuation and does not address controlled manual closure from the lower shaft.
TWM536490U	Dead	Claims a multi-section reverse-folding umbrella with automatic storage behavior. Focuses on reverse folding but does not integrate ergonomic actuation or controlled canopy management.
CN206182541U	Dead	Claims a reverse-folding umbrella structure with a specialized closing mechanism. Focuses on structural configuration rather than user ergonomics or controlled folding.
CN212971980U	Dead	Claims a multi-fold umbrella with automatic opening/closing and magnetic or structural folding aids. Focuses on automation rather than user-guided pull-to-close mechanisms or water containment.
US20150230569A1	Alive	Claims a portable canopy deployment system with wearable or supported structures. Focuses on deployment and support, not umbrella closure, folding control, or water containment.
US4652039A	Dead	Claims an umbrella with a manually operated collapsing mechanism using a runner and shaft system. Relies on conventional mid- or upper-shaft actuation and does not address ergonomic lower-shaft operation or controlled canopy folding.
CN212325686U	Dead	Claims an umbrella with an improved folding and storage structure. Focuses on structural arrangement for compactness, but does not provide user-controlled actuation from the lower shaft or integrated water containment.
WO2015157408A2	Dead	Claims a collapsible umbrella with improved frame and folding mechanics. Emphasizes structural folding efficiency and durability, without addressing ergonomic actuation or controlled user-guided closure.

CN211431338U	Alive	Claims a folding umbrella with enhanced rib linkage and support features. Focuses on mechanical structure rather than lower-shaft actuation or coordinated locking systems.
CN119828331B	Alive	Claims an umbrella with an improved folding and closing mechanism. Focuses on structural and mechanical improvements, but does not integrate user-actuated closure from the lower shaft or water containment features.
GB2475572A	Dead	Claims an umbrella with a modified opening and closing mechanism to improve usability. Still relies on conventional actuation locations and does not address controlled canopy folding or containment of water.
US5455098A	Dead	Claims an umbrella with a collapsible frame and improved rib structure. Focuses on durability and compact storage, but lacks ergonomic actuation from the lower shaft and controlled folding behavior.
CN205106652U	Dead	Claims a folding umbrella with a simplified structural mechanism. Emphasizes manufacturability and structure rather than user-controlled closure or integrated water containment.

Table 4: Water Containment System

Patent Number	Status	What is Claimed
US20210362657A1	Alive	Claims a separate umbrella holder with a conical body and removable water collection compartment. Focuses on capturing and disposing water after storage, not integration with the umbrella itself. Operates as an external accessory rather than a built-in system.

(Umbrella Case – Janczewski)	Alive	Claims a multi-part container system that separates umbrella storage from water collection using segregated compartments. Emphasizes fluid isolation and drainage, but requires manual placement of the umbrella into the case. No integration with umbrella actuation or folding process.
US10986905B1 / WO2017054206A1	Dead	Claims a telescoping umbrella case with detachable housing for adjustability and cleaning. Focuses on modularity and ease of assembly rather than real-time containment during closure. Still requires post-use insertion of umbrella.
(Collapsible Umbrella Case – Gorouvein)	Dead	Claims a volume-adjustable case with flexible sections that expand and collapse. Protects the concept of variable storage capacity, but does not guide or integrate with canopy folding mechanics.
US9320331B1 / US20160135557A1	Dead	Claims an umbrella with a sheathing loop structure surrounding the umbrella in the closed state. Focuses on enclosing the umbrella externally rather than directing canopy into a chamber. Limited integration with actuation or ergonomic closure.
(Can-Shaped Portable Umbrella Sheath)	Dead	Claims a rigid sheath with water retention and leak prevention features. Emphasizes storage and transport of wet umbrellas, but requires manual insertion. Does not integrate with umbrella structure or folding sequence.
WO2013032307A2 / US20130048693A1	Dead	Claims a flexible umbrella carrier with sealing mechanism at the tip. Focuses on preventing water leakage during transport. Operates independently from umbrella mechanics and requires user placement.
(Umbrella Bag Within Handle – Emerson)	Dead	Claims a deployable bag stored inside the handle that can be pulled out to cover the umbrella. Important precedent: storage integrated into handle, but requires manual deployment and wrapping. Does not provide automatic containment during closure.

US20130032184A1 / WO2013014642A1	Dead	Claims a waterproof container with drainage features for storing umbrellas. Focuses on post-use containment and water removal, not dynamic folding or guided insertion.
US8297295B2 / US20100200029A1	Dead	Claims a two-piece cylindrical tube system that seals around a wet umbrella. Emphasizes watertight sealing but requires manual assembly around umbrella. No integration with umbrella structure.
(No-Drip Umbrella Sheath)	Dead	Claims a reservoir-based sheath that captures water from a stored umbrella. Focus is on passive water collection, not on how the umbrella enters the system.
US20060283905A1 / US20050166949A1	Dead	Claims an umbrella with a dedicated receiver/case that fully encloses the folded umbrella. Focuses on storage integration but still requires manual placement into receiver.
(Box Umbrella – Acer)	Dead	Claims a box-like container integrated with the umbrella handle. Stores the umbrella fully when not in use. However, lacks guided folding or seamless transition during closure.
US20050098199A1 / WO2003047382A1	Dead	Claims an adjustable umbrella holder with drainage and backflow prevention. Focuses on versatility across umbrella sizes. Still functions as an external storage device.
(Collapsible Umbrella with Sheathing Handle – IGT)	Dead	Claims a handle that doubles as a storage sheath for the umbrella. Important precedent for integration, but relies on telescoping insertion, not guided canopy folding.
US6334454B1 / WO1996036251A1	Dead	Claims a rigid container that doubles as a handle and storage unit. Focuses on enclosing umbrella within a case-like structure. Does not address water management during transition.
(Umbrella Bag – Kiedrowski)	Dead	Claims a dual-layer absorbent bag system for drying and containing umbrellas. Focus is on material properties, not structural integration.

US5323802A / US5111835A	Dead	Claims a handle with integrated water collection and deployable sheath. Key idea: water collection in handle. However, still requires manual extension of sheath and does not automate containment.
(Emergency Umbrella – Bexel)	Dead	Claims an umbrella where the handle acts as a storage container for internal components. Focuses on compact storage, not wet containment or guided folding.
US4842003A / US4456023A	Dead	Claims a water-tight sheathing handle where the umbrella retracts into the handle. Strong structural precedent, but relies on axial retraction, not canopy folding into a chamber.
US3844301A	Dead	Claims a frame-based enclosure system that compacts canopy into a rigid structure. Early concept of integrated containment but lacks ergonomic actuation and water management features.

Appendix S: Handle Testing Results

Prompts to Ask During Testing

Direct Comparison

- “Which feels easiest to use and why?”

- “Which feels the most natural without thinking?”
- “Which would you trust the most when rushing into a car?”

Tight Space / Car Scenario (IMPORTANT for your project)

- “Which handle is easiest to maneuver when opening/closing near a car door?”
- “Which gets in the way the least?”
- “Which would be easiest to store quickly after closing?”
- “Does this feel awkward when sitting down or reaching inside a car?”

Grip & Control

- “Do you feel like you have enough control with one hand?”
- “Does the handle slip or feel secure when wet?”

Traditional J-shaped handles provide a familiar grip and enable hanging/storage, while C-shaped and carabiner-style handles introduce hands-free carrying and alternative wrist-based interactions. Hand-molded grips are designed to conform to the natural contours of the hand, improving comfort and control during force application, whereas simple rubber grips offer a lightweight, low-cost baseline with reliable friction. Gym grips, typically used for lifting, were included for their high-friction materials and ability to support stronger, more secure pulling forces with reduced slippage.

User 1: Sarah Thornton

Handle Type	Ease of Use / Intuitiveness (1–5)	Comfort & Ergonomics (1–5) (x2 Weight)	Tight Space Maneuverabil ity / storage (1–5)	Total	Key Comments (Why / Observations)
Hand-Molded	5	3	4	15	Hands don't fit
Simple Rubber	5	3	4	16	• Everything is good but when its wet, rubber feels weird. • Shorter the better because most intuitive (comfortable to hold something small)

					and can hold top if umbrella small)
Gym Grip	4	2	4	12	- Hands don't fit: "Don't like built in grip because it makes you feel like you have adjust your hands in a certain position. - Hard to switch hands when opening door or maneuvering
J-Shape	5	4	1	14	Too big
Carabiner	4	5	5	19	- Like this one best because it feels like it's holding your hand and secure. - Like it when you are just holding it - Trust most when rushing
C-Shape	5	4	1	14	Too big

User 2: Kanan Nozaki

Handle Type	Ease of Use / Intuitiveness (1–5)	Comfort & Ergonomics (1–5) (x2 Weight)	Tight Space Maneuverability / storage (1–5)	Total	Key Comments (Why / Observations)
Hand-Molded	3	2	4	11	Hands feel locked in a particular position and hard to adjust.

Simple Rubber	5	4	4	17	- Trust this the most when rushing into car - Sticky when wet
Gym Grip	1	1	2	5	When rushing into car, it will get in the way because you have to switch hands and hand is locked in particular position (hard to maneuver)
J-Shape	4	4	3	15	Like the most for daily use because can adjust hand placement and angle (especially when there is wind)
Carabiner	2	3	4	12	I like material the most. Not intuitive (don't want to think when grabbing)
C-Shape	3	2	3	10	Too big, not intuitive

User 3: Wynn Chavapokin

Handle Type	Ease of Use / Intuitiveness (1–5)	Comfort & Ergonomics (1–5) (x2 Weight)	Tight Space Maneuverability / storage (1–5)	Total	Key Comments (Why / Observations)
Hand-Molded	4	2	4	12	Fit well but hard to adjust so cant hold for long
Simple Rubber	5	3	4	15	Nice and simple

Gym Grip	1	2	4	9	Doesn't fit in hand
J-Shape	4	3	2	12	Too big, feels too open and will slip out
Carabiner	4	5	4	17	Likes the feeling of hand being secure on both sides
C-Shape	3	3	2	11	Too big

Appendix T: Handle Testing: Handle types

Name	Photo	Description
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Healeved Umbrella Handle		J-shape
LALAFINA Umbrella Handle		Hand molded grip
Bike Handlebar		Simple Rubber grip
Rubber Gym Grips		Gym grip

<u>QWB TUMAAOTU Inverted Umbrella with Carabiner</u>		Caribiner
<u>GANAZONO Replacement Umbrella</u>		C- Shape

Appendix U: Container Testing Results

User Testing - Container Lid

Questions

1. Can they close without instruction, do they trust it's sealed
2. Can users close and open quickly
3. Does the lid leak?
4. How is the object carried, stored, used during transition?

User 1: Yena, Electrical Engineer

1. Do I pull or turn? Figured it out.
2. It is hard to open, but storing a wet umbrella makes sense because the seal is tight. I was worried about breaking it. I would rather not have the coffee lid opening, turning the umbrella upside down with the lid open is just as useful.
3. The coffee lid opening is a leak hazard
4. I would mostly use this in my backpack since I do not put umbrellas in there because they're often wet, but it is large and might not fit into a bag that is already partially full.

User 2: Kritika, Biomedical

1. Easy to figure out, wish there was more of a place to hold it to pull, the suction is good so it doesn't fall off.
2. I like the place to shake out the water, not sure about the coffee cup lid, maybe something screw-on. Like a Stanley, it's hard to remove.
3. not a leak hazard
4. I would want to put it into the water bottle side pocket of my backpack but the handle is obtrusive. maybe a wider base and a handle to hold it.

User 3: Patrick, Mechanical

1. The seal is incredibly tight, too hard to open. only one tab and can only open one way. there needs to be a grip to lift. extended lip on all sides?
2. Opening and closing is not easy at all. could use a medicine type cap that only needs a little bit of a twist to open and seal, a gasket can be added to the top.
3. I don't see any leaking issues, but it needs a slightly looser fit. ridges to twist and pull the cap off maybe. or loose threaded cap.
4. It would be nice if it would fit in a coffee cup holder, since the lid has good dimensioning. i would put it on the floor upside down but the coffee cup lid would definitely make it leak. I would be comfortable putting it in my bag. The grip is ergonomic. Where does the lid go once it's open?

User 4: Jeter, Civil

1. It's really hard to open, it was hard to figure out how to open
2. This user cannot open and close it quickly
3. Leaking is not the concern
4. The handle needs to be longer. I think a bag clip should be included. I wouldn't put it in my bag because it is big. i'm using it in the rain, i know somehow the outside will get wet.

User 5: lailah, sociology

1. It is hard to open
2. No
3. Ideally not, but the coffee mug lid does not seem secure. They leak.
4. I would store it upright if it's dry, upside down if it's wet. I would swing it by my side as I walked. It should be spiked for self defense, and have pepper spray in the handle.

Appendix V: Manufacturing Plan

Likely OEM / ODM Umbrella Manufacturers

These are manufacturers already producing reverse-folding and telescoping umbrellas very similar to your reference product:

- [Xiamen Hoda Co., Ltd.](#) — reverse and inverted umbrella OEM/ODM supplier
- [Zhangzhou Dongtai Rain Gear Industrial Co., Ltd.](#) — produces reverse and double-layer umbrellas
- Shaoxing Shangyu Ping An Umbrella Co., Ltd. — reverse folding umbrella supplier
- Shenzhen Blueprint Umbrella Co., Limited — automatic reverse umbrella manufacturing
- Shaoxing Shangyu Runtian Tourist Products Co., Ltd. — custom umbrella manufacturing with reverse-fold options

Recommended Next Step

Your professor's advice is exactly how many consumer hardware startups begin sourcing in China:

1. Buy 3–5 benchmark umbrellas
2. Fully dissect each subsystem
3. Identify:
 - stamped part markings
 - fabric tags
 - molding marks
 - spring dimensions
 - rivet/rib construction
4. Match those subsystems to Alibaba/OEM suppliers
5. Contact umbrella OEMs with:
 - CAD renders
 - mechanism sketches
 - target BOM
 - desired MOQ
6. Ask whether they already manufacture similar subassemblies

The most important realization is that umbrella manufacturing is usually *assembly-driven*, not invention-driven. Most OEMs already have:

- rib suppliers
- spring suppliers
- shaft suppliers
- sewing vendors
- molding vendors

Your innovation is likely going to be in:

- shaft actuation architecture
- pull-to-close routing

- locking integration
- canopy folding behavior
- containment integration

—not in reinventing every component from scratch.

what should i include in a slide about our manufacturing progress? i think we also have a general bill of materials for each of the components.

For a manufacturing progress slide, you want to show that the team has moved from “conceptual design” into a realistic production pathway. Since your professors specifically pushed you toward identifying real umbrella manufacturing ecosystems in China, the slide should communicate three things:

1. You understand how the product is actually manufactured
2. You have begun identifying suppliers/processes
3. You have enough technical detail to estimate feasibility and cost

A strong slide structure would look something like this:

Manufacturing Progress

1. Manufacturing Strategy

Briefly explain the pivot your team made after professor feedback.

Example wording:

After discussions with faculty advisors, the team shifted toward a manufacturing-first benchmarking strategy. Because most umbrella manufacturing is concentrated in China, the team began reverse engineering existing high-quality telescoping reverse umbrellas to identify likely manufacturing methods, component suppliers, and assembly processes relevant to our patented design.

2. Competitive Benchmarking

Include:

- image of the reference umbrella
- why you selected it
- what features overlap with your concept

Example bullets:

- Selected a commercially successful telescoping reverse umbrella as baseline reference
- Analyzed:
 - telescoping shaft architecture
 - rib linkage geometry
 - inverted canopy construction
 - handle integration
 - locking mechanisms
- Compared benchmark product to provisional patent claims

You can include:

- estimated materials
- estimated manufacturing methods
- estimated sourcing regions

3. Preliminary Bill of Materials (BOM)

This is probably the most important technical section.

Make a clean table like:

Component	Material	Likely Manufacturing Process	Notes
Telescoping shaft	Aluminum / stainless steel	Tube extrusion + CNC/notching	Multi-stage locking shaft
Ribs/stretchers	Spring steel / fiberglass	Stamping + forming	Corrosion resistant
Handle	ABS or TPE overmold	Injection molding	Potential electronics cavity
Canopy	Polyester pongee	Cut + sewn assembly	Reverse-fold geometry
Springs	Stainless steel	Coil forming	Auto-open/assist mechanism
Locking buttons	Stainless steel + ABS	Stamping + injection molding	Telescoping retention

This demonstrates:

- DFM thinking
- supplier awareness

- realistic production understanding

4. Supplier / Manufacturing Research

This section shows traction.

Possible bullets:

- Identified umbrella manufacturing clusters in China
- Began mapping likely OEM/ODM suppliers for:
 - shafts
 - rib systems
 - canopy sewing
 - molded handles
- Researching manufacturers capable of:
 - reverse umbrellas
 - telescoping compact systems
 - custom tooling

You can also mention:

- Alibaba
- Made-in-China
- OEM umbrella factories
- sourcing platforms
- RFQ preparation

5. Key Manufacturing Challenges

Professors LOVE this section because it shows engineering realism.

Potential points:

- Integrating reverse-fold geometry with compact telescoping architecture
- Maintaining durability while minimizing weight
- Waterproofing around moving shaft interfaces
- Custom tooling costs for novel mechanisms
- Balancing spring force with user safety and ergonomics

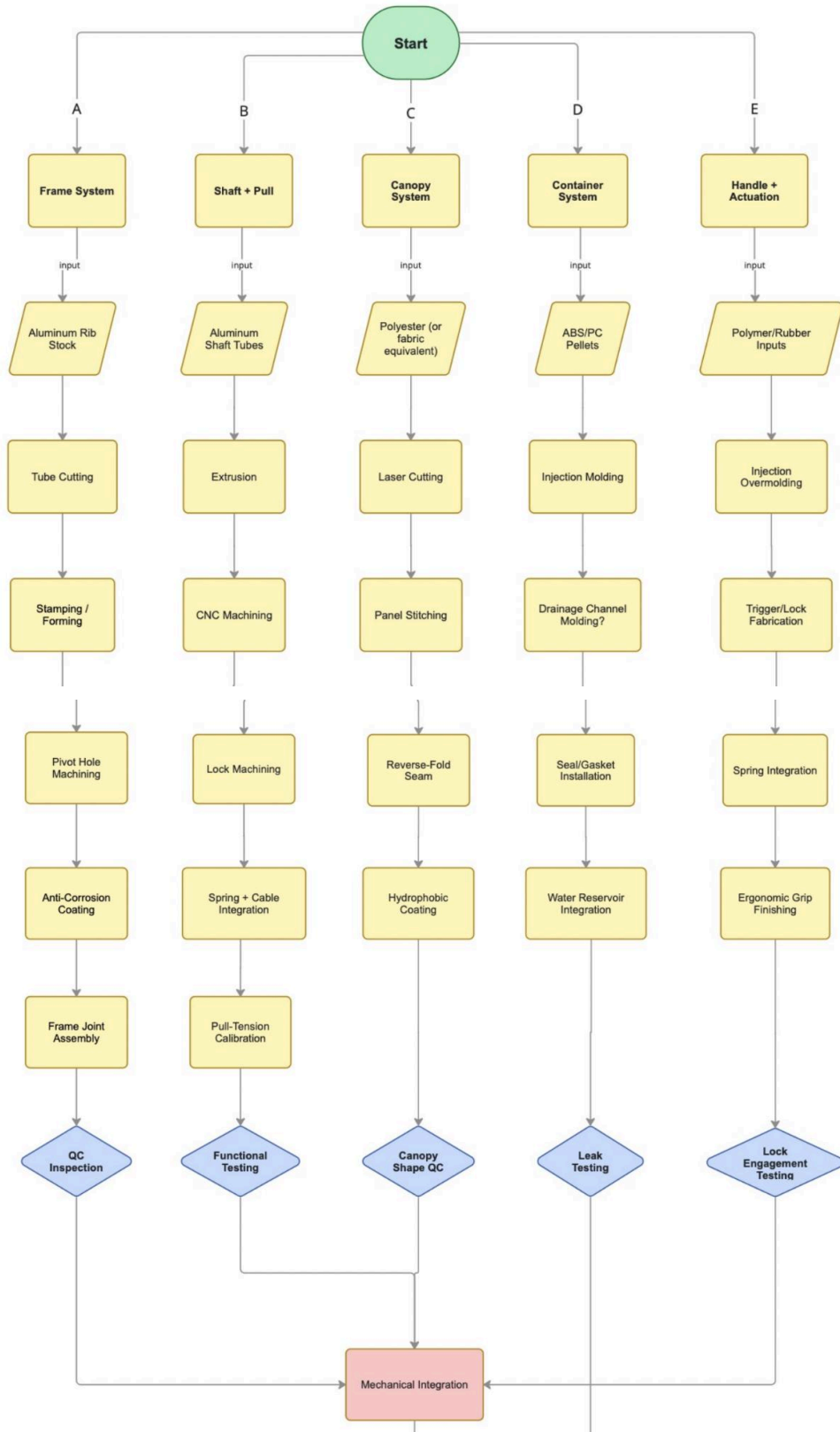
6. Next Steps

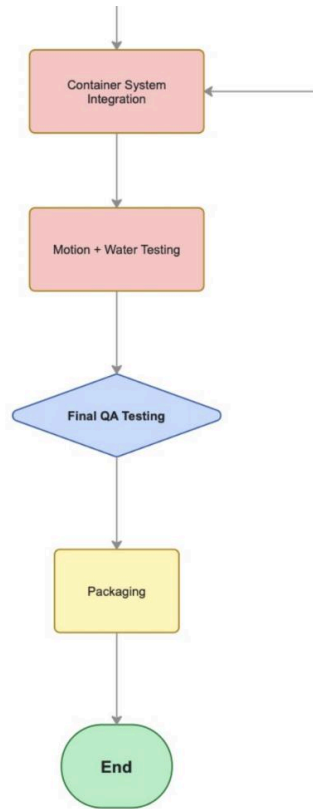
This makes the slide feel actionable.

Example:

- Finalize component-level BOM
- Contact umbrella OEM manufacturers for RFQs
- Validate spring rates and shaft tolerances
- Prototype manufacturable locking mechanism
- Estimate tooling and MOQ costs

Appendix W: Manufacturing Overview





Appendix X: Full Prototype User Testing Plan

Key Performance Metrics

Metric	Description	Measurement Method
Closing Force	Force required to fully close the umbrella	Luggage scale / force gauge
Closure Time	Time required to close and secure umbrella	Stopwatch
Water Leakage Rate	Amount of water escaping during closure/storage	Visual inspection + absorbent paper
One-Handed Usability	Ability to operate system using one hand	Observation + user rating
Ergonomics & Comfort	User comfort while gripping and operating	User survey (1–5 scale)
Ease of Use	Perceived intuitiveness and simplicity	User survey + observation
Tight-Space Maneuverability	Ability to operate inside vehicle entry space	Scenario testing
Overall User Confidence	User perception of reliability/security	Post-test interview

Participants

The testing will include approximately 5–10 participants with varying umbrella usage habits and hand sizes. Participants will include students and peers unfamiliar with the detailed mechanism to better simulate first-time user interaction.

Testing Environment

Testing will take place in Vehicle entry and exit scenarios

- Outdoor wet conditions or simulated rainfall
- Tight-space maneuverability testing
- A parked vehicle will be used to simulate entering and exiting a car while managing the umbrella system.

Testing Scenarios

Scenario 1 — Vehicle Entry

Participants will approach a vehicle while holding the umbrella in an open position. They will attempt to close and store the umbrella while entering the car using one hand.

Observations:

- Amount of body exposure to rain
- Ease of closure motion
- Dripping/leakage into vehicle
- User hand positioning and natural interaction
- Any confusion or hesitation during operation

Scenario 2 — Vehicle Exit

Participants will retrieve and deploy the umbrella while exiting the vehicle.

Observations:

- Ease of deployment
- Time required to fully open umbrella
- Handle accessibility and maneuverability
- Comfort during transition

Testing Procedure

1. Introduce participant to the prototype without detailed instruction
2. Explain testing scenario and safety considerations
3. Allow participant to interact naturally with the system
4. Record observations, timing, and force measurements
5. Conduct short follow-up interview and usability survey
6. Repeat across multiple scenarios and participants

Data Collection

Quantitative Data

- Closure force values
- Closure times
- Leakage/drip observations
- User ratings (1–5 scales)

Qualitative Data

- User comments and frustrations
- Observed behaviors and workarounds
- Preferred interaction methods
- Suggestions for improvement

User Feedback Questions

Participants will be asked:

1. How intuitive was the system to use?
2. Did the umbrella feel comfortable to operate?
3. Did you feel in control while entering the vehicle?
4. Was the closure motion smooth and natural?
5. Did the system reduce the inconvenience of getting wet?
6. What aspects felt awkward or difficult?
7. What would you improve about the design?

Success Criteria

The prototype will be considered successful if:

- Users can consistently operate the system with one hand
- Water leakage into the vehicle is minimized
- Closing force remains manageable and comfortable
- Closure time is fast and efficient
- Users report high usability and confidence ratings
- Participants naturally understand system interaction without extensive instruction

Expected Outcomes

The testing is expected to provide insight into how effectively the prototype addresses the core problem of umbrella management during vehicle entry. The results will guide final refinements related to ergonomics, mechanism performance, water containment, and overall user interaction before final presentation and documentation.

Participant	Scenario	Closure Time (s)	Closing Force	Leakage/Drip	Comfort (1–5)	Ease of Use (1–5)	Tight-Space Maneuverability (1–5)	Key Observations / Comments
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P1	Vehicle Entry							
P1	Vehicle Exit							
P2	Vehicle Entry							
P2	Vehicle Exit							
P3	Vehicle Entry							
P3	Vehicle Exit							
P4	Vehicle Entry							
P4	Vehicle Exit							
P5	Vehicle Entry							
P5	Vehicle Exit							

Participant	Observed Behaviors	Confusion / Hesitation	Preferred Interaction	Suggested Improvements
P1				
P2				
P3				
P4				
P5				

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